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Masterarbeit

There may be migrants in-between: Support Vector Machines (SVMs) and cluster in-betweenness

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Abstract

The analysis of isotopic signatures has emerged as a powerful tool in archaeology for understanding ancient individuals' diet, mobility, and provenance while offering insights beyond the archaeological contexts, such as grave goods. The introduction of multi-isotope analysis and the associated challenges have opened the field for machine learning applications within this field of research. This work presents a machine-learning approach using Support Vector Machines to identify migrants within a multi-isotope dataset. We combine the novel concept of cluster in-betweenness with the real-world use case of provenance research through multi-isotope analysis.

This thesis proposes an approach for the identification of in-between instances, which are samples exhibiting characteristics of both origin and destination locations. We developed a method leveraging the capability of Support Vector Machines to capture high-dimensional and complex patterns. This method was evaluated on a synthetic dataset. The evaluation proved its capability to detect in-between instances under different circumstances, demonstrating its effectiveness under varying conditions, such as differing cluster densities and spread. Having established the effectiveness of this technique, we applied it to a real-world multi-isotope dataset, extending the work of Grupe et al. (2020) on the genesis and spread of the Fritzens-Sanzeno culture. Our analysis identified 34 candidates with a possible migratory background, including most of the migrant candidates identified in earlier research. This supports earlier research findings while providing additional migrant candidates.

Our research highlights the value of integrating machine learning techniques in domains such as archaeological provenance research. It offers a new approach to detecting migrants based on their isotopic fingerprints. We proved that applying the concept of in-betweenness can enhance our understanding of the underlying patterns within an isotopic dataset, providing deeper insight into past humans' provenance.

Statement of Originality

I hereby declare that I have independently completed the present work without any outside help and have not used any sources or aids other than those indicated.

Furthermore, I assure that this work has not yet been submitted as a thesis elsewhere.

Kiel, July 15, 2024

Jan Robert Bensien

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1 Introduction

1.1 Motivation

Isotope ratio analysis is a powerful tool in archaeology for analyzing the diet, mobility, and provenance of specimens (Price, 2014). Through the food they eat, the water they drink, and the air they breathe, individuals continuously incorporate locally available isotopes. Our research focuses on the application of isotope analysis to provenance research. In provenance research, an isoscape defines the range of locally occurring isotope ratios. Based on the comparison of the isoscape and the isotope ratios, samples from the archaeological remains of an individual can be identified as non-local.
Traditionally, single isotope ratios have been used to analyze the provenance of an individual. However, with the widespread adoption of machine learning algorithms in various data science fields, new opportunities are emerging to gain further insight by analyzing multi-isotope fingerprints (Tschetsch et al., 2020). One such application is to search not only for non-local individuals, but also for the origin of their migration by applying machine learning methods such as Gaussian Mixture Models (Grupe et al., 2020).
However, detecting migrants and their origins within a dataset of isotope signals poses new challenges for data scientists. Traditional applications for machine learning algorithms include classification, where samples are grouped according to their inherent characteristics, and outlier detection, where samples that deviate from the rest of the dataset are identified. Since the isotopic signal of migrants still shares some characteristics of the place of origin, but also begins to adapt to the new location, the goal is to find individuals that share characteristics of two known regions.
In this thesis, we will try to overcome this challenge by applying the novel concept of cluster in-betweenness to develop a machine learning algorithm for migrant detection in a multi-isotope dataset.

1.2 Document Structure

This thesis is structured as follows: Section 1 introduces the core concepts and technologies that will be used throughout the thesis. These include isotope analysis, support vector machines (SVMs), cluster in-betweenness, and Gaussian distributions. Section 1 also contains the goals we aim to achieve. Afterward, we will present related work in Section 2. In Section 3, we will explain our approach to using SVMs to find points between two clusters. We will also explain which experiments we conducted to evaluate the performance of our approach and identify migrants within a real-world isotope dataset. The results and discussion of the

experiments and the application to the real-world dataset are presented in Section 4. Finally, Section 5 contains our conclusion and future work.

1.3 Foundations

In this chapter, we will explain the foundations of our thesis and introduce the technologies we used. Initially, we will introduce the analysis of isotopic signatures within an archaeological context, focusing on the utility of multi-isotope analysis for provenance research. This will provide a rationale for a brief introduction to introduce machine learning, classification, and outlier detection. As the isotopic datasets are assumed to be Gaussian, we will explain what defines a Gaussian distribution. The concept of cluster in-betweenness plays a central role in our thesis and will be introduced within this chapter. Finally, we will focus on Support Vector Machines and introduce them and their mathematical foundations in detail.

1.3.1 Isotope Analysis for Provenance Research

The chemical properties of an element are defined by the number of protons in its atomic nucleus, which is called the atomic number and is equal to the number of electrons in neutral, non-ionized atoms. The mass number defines the number of nucleons, which are protons and neutrons. The mass number for the same element can differ due to different numbers of neutrons. These variants of an element are called isotopes. Different isotopes of the same element have the same atomic number, thus the number of electrons, while they have a different mass number due to different amounts of neutrons. The different isotopes of a given element have similar chemical properties but can, due to differences in mass, have different physical properties, such as density and stability.

Isotopes are classified as either stable or radioactive. The nuclei of stable isotopes do not change over time and maintain a constant number of protons and neutrons. Radioactive isotopes, on the other hand, have unstable nuclei as their composition of protons and neutrons changes over time. Thus, radioactive isotopes may decay over time and become other isotopes or elements. The isotopes that are formed by the radioactive decay of another isotope are called radiogenic.

Isotope analysis emerged as a tool in archaeology and forensic research in the 1960s (Brill and Wampler, 1967). An individual absorbs isotopes through the food they eat, the water they drink, and the air they breathe. The incorporation of isotopes into an individual's body occurs via different pathways, with strontium, for instance, following the pathway of calcium intake and being incorporated, among other places, in bones. In the past, humans mostly sourced their food locally, and the ratios of isotopes within their bodies reflected the locally occurring isotope ratios. Thus, analyzing the ratio of different isotopes in human remains can give insight into their diet, mobility, and migration patterns. In this context, isotope ratio analysis can aid traditional identification methods, such as objects found at the burial scene and genetic analysis. Together they might supplement the shortcomings of the individual approaches. Isotope analysis of human skeletal remains became more popular

in the 1970s and 1980s (Van der Merwe and Vogel, 1978; Bender et al., 1981). Strontium isotope ratios were examined, and regional differences became evident (Ericson, 1985). The strontium isotope ratios of individuals were examined and compared with strontium isotope ratios from other locations. This paved the way for further research in place-of-origin studies of human remains.

The Application of Strontium and Lead Isotopes in Provenance Research

As different elements and their isotopes follow different pathways before being incorporated into an individual's body, the insights that can be gained vary widely. For example, oxygen is uptaken through drinking water (Kohn, 1996). It reflects the local rainfall. However, the variation of oxygen in certain regions of the world is very low, and unexplained variation exists in the oxygen isotope ratios of individuals from the same location (Price, 2014). Elements such as nitrogen are absorbed through the food and can give insight into an individual's very recent diet. The analysis of the nitrogen isotope ratios in hair and nail collagen can give insight into whether a human consumed marine foods, grains, or fruits (Price, 2014).

As we focus on provenance research, strontium is our most interesting element. Of the four naturally occurring strontium isotopes, ^{84}Sr, ^{86}Sr, ^{87}Sr, ^{88}Sr, all are stable and only ^{87}Sr is radiogenic and generated through radioactive decay of ^{87}Rb. With a half-life of 4.92×10^{10}, the decay of ^{87}Rb does, in realistic scenarios, not impact the strontium signatures of humans. Strontium behaves similarly to calcium and follows the same pathway into human tissue. It is absorbed through food together with calcium, and due to its chemical similarity, it can replace calcium in the hydroxyapatite crystals of bones and teeth, albeit in smaller amounts (Price, 2014).

Originating from rocks, it is absorbed by plants and through the food chain by humans (Price, 1989). The analysis of $^{87}Sr/^{86}Sr$ in human teeth to assess human mobility patterns has first been proposed by Ericson (1985). The widespread adoption of this method has since shown the efficacy of this approach.

If the goal is to find out if a human spent the first years of his life at the same place he spent the last years, comparing bone $^{87}Sr/^{86}Sr$ with its enamel counterpart is an evident approach (Ericson, 1985). Dental enamel mineralizes early in life while bone remodels and incorporates new isotopes continuously. Thus, a difference between the bone and dental enamel strontium ratios indicates that the individual migrated to this place. Depending on the teeth, dental enamel forms between the ages of 3 and 16. Thus, their composition reflects the nutrients and the contained strontium isotopes eaten by the mother and the child during early childhood (Ericson, 1985). Even if an individual dies many years later, the isotope signature of the birthplace can still be reconstructed using dental enamel. Instead of comparing the $^{87}Sr/^{86}Sr$ from dental enamel with the $^{87}Sr/^{86}Sr$ from bones, many researchers opt to compare the $^{87}Sr/^{86}Sr$ of dental enamel with the strontium ratios found at multiple locations. The premise of this approach is that there is an isoscape, a geographic region defined through a specific range of $^{87}Sr/^{86}Sr$ ratios, which is reflected in human

dental enamel through human diet and absorption during the specific infancy period in which the dental enamel is formed (Burton and Hahn, 2016). If the $^{87}Sr/^{86}Sr$ does not match the isoscape of the burial location, the individual is assumed to have spent his childhood in another place and is assumed to be a migrant.
Instead of dental enamel as $^{87}Sr/^{86}Sr$ source, this method can also be applied when using bone as $^{87}Sr/^{86}Sr$ source. Bones within human bodies remodel at a rate of approximately 4% each year. If an individual arrives at a new location, their bones will still contain the $^{87}Sr/^{86}Sr$ of the origin location. It will then slowly begin to remodel, and after 20 years, it will have fully remodeled and will contain the $^{87}Sr/^{86}Sr$ of the new location. Thus, comparing the $^{87}Sr/^{86}Sr$ with the local isoscape for strontium can give insight into whether this individual migrated to this place within the last 20 years. In the following, we will refer to this as the linear isotopic remodeling process.

After death, the human body, and thus its bones and teeth, are subject to diagenetic changes. Diagenetic are alterations due to chemical, physical, and biological processes in the burial environment. They can change the original isotopic composition of the individual's body. As dental enamel is much denser and mineralized than bone, it is more resistant to diagenetic changes and preserves the original strontium isotope ratio better than, for example, bone (Budd et al., 2000). Many human provenience studies measure the isotopic signature on dental enamel because of its resilience against diagenetic changes and its ability to reflect the local strontium isotope ratios during childhood. In addition to strontium, lead isotopes in dental enamel are examined and can give further insight into the birthplace of a human (Price, 2014; Bowen et al., 2005).

Lead is used similarly to strontium within provenance studies, albeit to a lesser extent. Both elements are incorporated as substitutes for calcium in human bones and teeth. Lead isotope ratios in human bodies reflect the ratio of lead isotopes in the place where the diet was sourced. Lead is mainly incorporated into the human body by inhalation of aerosols and dust particles (Grupe et al., 2018). Animals also incorporate lead by soil ingestion. Lead isotope analysis has multiple problems. On the one hand, the geographic lead distribution is poorly known (Price, 2014). On the other hand, global lead contamination from the combustion products of leaded gasoline makes it even harder to define a lead isoscape (Albarède et al., 2012). Lead is bone-seeking, and lead sources in the burial environment will contaminate archaeological bone and alter its lead isotope ratios. While strontium also exhibits this behavior, it's of greater importance in the case of lead due to the worldwide contamination with industrial Lead (Reimann et al., 2012). The bone of a buried individual can contain a mix of diet-based and post-mortem-absorbed geological strontium and lead. Research on modern and fossil marine animals showed that fossil marine animals showed an $^{87}Sr/^{86}Sr$ closer to the geological ratio than that of seawater due to the post-mortem-absorbed geological strontium (Nelson et al., 1986). As the modern marine animals showed a $^{87}Sr/^{86}Sr$ close to that of seawater, this highlights the impact of post-mortem contamination. Thus, when measuring $^{87}Sr/^{86}Sr$ in bone, it is important to consider the potential for bias due to local

geological $^{87}Sr/^{86}Sr$ contamination. To overcome this, one could use samples of recently deceased animals, as animals accumulate local lead within their bones and pose as a suitable source for finding local lead isotope signatures and defining a local lead isoscape (Price et al., 2017). In humans, the nearest natural lead isotope may be replaced by anthropogenic lead sources, such as kitchenware or lead processing in the vicinity, which may introduce new lead sources into the local food. An advantage of lead and strontium is that it is not altered through cremation (Grupe et al., 2018). Furthermore, cremated bone behaves similarly to dental enamel and is less contaminated through post-mortem-absorbed geological strontium and lead.

When defining the local strontium isoscape, it is important to note that even in the same place, rocks might have different strontium ratios (Burton and Price, 2013a). The $^{87}Sr/^{86}Sr$ ratio is changing due to radioactive decay of 87^{Rb}. However, as noted before, the half-life of 87^{Rb} is orders of magnitude greater than the age of the earth. In practice, the strontium ratio of rocks depends mostly on their composition (Grupe and McGlynn, 2016). There can be widely different strontium ratios at a single location depending on the exact material from which it was sourced. Furthermore, variation of strontium ratios in local geological material is orders of magnitude higher than that in biological materials, and thus also humans (Burton and Price, 2013a). In contrast to that observation, 4885 samples of human dental enamel from individuals found on six continents have shown that 95% of $^{87}Sr/^{86}Sr$ ratios fall within a range of less than 0.015 (Burton and Price, 2013b). The range is even smaller if only individuals from one location are examined. Geological material exhibits ranges that are many orders of magnitude higher. Thus, in practice, a direct comparison might lead to false conclusions. Grupe and McGlynn (2016) argues that multiple factors play a role in the compression of $^{87}Sr/^{86}Sr$ in humans in comparison to bedrock. Bedrock with high strontium ratios often occurs in small quantities and thus contributes less to the total strontium in humans. Additionally, a large part of biologically accumulated strontium stems from non-geological sources such as sea salt, dust, smoke from fires, and combustion of fossil fuels (Grupe and McGlynn, 2016). Lastly, strontium is not evenly distributed among food sources. Certain foods in a diet dominate the total strontium intake of a human. Grupe and McGlynn (2016) shows that in a diet consisting of chicken, 90% of the total volume consumed, and kale, 10% of the total volume consumed, the isotope ratio of the diet is essentially that of kale. Strontium also competes with calcium to be absorbed into bone and teeth. When a lot of calcium is present in a meal, less strontium might get absorbed. However, calcium-rich food sources often also contain more strontium than food sources with less strontium.

Instead of using local rocks, local fauna might be better suited to sample a local isotope signature. Animals have, like humans, a low variation in $^{87}Sr/^{86}Sr$. They should absorb the same strontium sources through plants and other animals. Modern animals can be used, but a shift in the $^{87}Sr/^{86}Sr$ ratio could have happened in the past. This could be due to changes, such as shifting river courses, human usage of fertilizers, the use of concrete and

cement in construction, and the consumption of imported foods by humans and animals (Burton and Price, 2013a).

To avoid this problem, archaeological teeth and bone samples from animals are preferred. Differences in diet between the animals and humans could still lead to problems when interpreting the animal strontium isotope ratio as a proxy for the human local strontium isotope ratio. For example, snails are strongly biased to surficial strontium from precipitation and atmospheric dust (Burton and Price, 2013a). In summary, archaeological samples from a species with a similar diet or as locally identified archaeologic human material are best suited to serve as a local human isotope signature source.

It can be concluded that defining an isoscape for strontium involves many factors that must be considered. Most of these points also apply to lead, which is further complicated by the additional issue of global industrial lead pollution.

Multi-Isotope Analysis

We have seen that different isotope systems can contain different information and provide unique insights. Until now, we have only examined individual isotope systems separately. However, combining multiple isotope systems can provide a more comprehensive picture of an individual's mobility and provenance.

Tschetsch et al. (2020) argues that multi-isotope analysis of human remains is better suited for provenance studies than single-isotope analysis. In isolation, the $^{87}Sr/^{86}Sr$ of an individual can be used to identify if this individual is non-local. However, due to the spatial redundancy of $^{87}Sr/^{86}Sr$ isoscapes, it can not alone be used to infer the origin of a non-local individual Tschetsch et al. (2020). Combining the $^{87}Sr/^{86}Sr$ with multiple lead isotope ratios can help overcome this redundancy and differentiate multiple isoscapes which beforehand could not be separated. The resulting multi-isotope system contains more complex patterns and information. While this can provide unique insights, it introduces new challenges and is harder to interpret. Machine learning models can identify these complex patterns and correlations that may not be immediately apparent to human researchers.

A more detailed description of Tschetsch et al. (2020)'s and other multi-isotope research using machine learning methods can be found in Section 2.

1.3.2 Gaussian Distributions

A Gaussian distribution, also referred to as a normal distribution, is a continuous probability distribution with the following probability density function

$$f(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} \tag{1.1}$$

where μ is the mean and σ^2 is the variance of the distribution. The standard, or unit, normal distribution is the special case where $\mu = 0$ and $\sigma^2 = 1$. In this case, the probability density function has the following form:

$$\varphi(x) = \frac{e^{-z^2/2}}{\sqrt{2\pi}} \tag{1.2}$$

The standard normal distribution has a mean of 0 and a standard deviation of 1. This distribution has its peak at $z = 0$. Every Gaussian can be described as a standard normal distribution whose domain has been stretched by a factor σ and then translated by the mean μ. Thus, every Gaussian deviation can be scaled by a factor σ, shifted by μ, and still obtain a Gaussian distribution.

$$f(x) = \frac{1}{\sigma}\varphi(\frac{x-\mu}{\sigma}) \tag{1.3}$$

If a random variable X is Gaussian distributed with mean μ and variance σ^2 we will write $X \sim \mathcal{N}(\mu, \sigma^2)$.

One of the most fundamental theories in modern statistics is the central limit theorem (Kwak and Kim, 2017). Collecting all elements from a population to obtain the underlying distribution within real-world scenarios is often not feasible. This would pose an insurmountable problem. Instead, researchers use a sample, which is a subset of the full population, to make inferences about the population parameters. The Central Limit Theorem is the basis for this idea and states that if the sample size is sufficiently large, the means of samples obtained using random sampling with replacement are distributed normally with the mean, μ, and the variance, $\frac{\sigma^2}{nm}$ regardless of the population distribution (Kwak and Kim, 2017). The larger the sample size, the closer the distribution will be to a normal distribution. Kwak and Kim (2017) notes that a sample size of 30 is often large enough to apply the Central Limit Theorem. Islam (2018) investigated the role of appropriate sample sizes in research and its impact on applying the Central Limit Theorem. They found that increasing the sample size from 15 to 50 led to more tight clusters around the true mean and approximated the real distribution better. The sample means become better estimators of the true population mean, and the shape of the distribution becomes more Gaussian as the sample size grows. Their simulation results were consistent with the Central Limit Theorem. The primary importance of the Gaussian distribution lies in its ability to approximately describe many real-world phenomena (Gaines and Denny, 1993). The Central Limit Theorem explains why so many variables in nature appear to have a probability density function similar to a Gaussian distribution. Random biological or physical processes are affected by a large number of random processes, and the sum of them creates the random variable, which, due to the Central Limit Theorem, converges to a normal distribution regardless of the underlying distribution of the individual small processes (Gaines and Denny, 1993). Some voices within research allude that this view is too simplistic and too often applied. Lyon

(2014) argues that the qualifying conditions might not be satisfied, the variables might be chosen in a way to create Gaussian distributions, and the Central Limit Theorem does not address these problems. However, the research on multi-isotope signature analysis often assumes an underlying Gaussian distribution. Thus, within our thesis, we will apply this idea and assume a Gaussian distribution of certain distributions. This should hold true as long as the sample size is large enough.

One important observation on Gaussian distributions is the three-sigma rule of thumb, also known as the 68-95-99.7 rule. It states that for a Gaussian distribution, approximately 68% of the data points lie within one standard deviation of the mean, 95% lie within two standard deviations of the mean, and 99.7% lie within three standard deviations of the mean (Huber, 2018). We will apply this rule within our thesis to set the bounds of the area where we expect that an isotopic signal belongs to a certain reference location.

1.3.3 Cluster In-Betweenness

Detecting in-between instances is novel, and no extensive research has been done in this field. Our definition of in-between instances is based on a vision paper (Kazempour et al., 2024).



Figure 1.1: Visualization of Classes A,B, and In-Between Instance

In the following, we will explain the proposed concept of in-betweenness and the provided definition. In-between instances serve as connectors between multiple clusters. These instances exhibit characteristics that are shared by both clusters. However, as the two clusters are distinct, they also exhibit differences. Otherwise, these clusters would not have discernible characteristics and would fuse. Figure 1.1 illustrates an instance that is in-between clusters A and B. In the two-dimensional feature space, the instance is situated

between clusters A and B. Consequently, its characteristics are a mixture of both clusters but not similar enough to be recognized as an instance of either cluster.

The task of finding in-between instances is related to the tasks of clustering and outlier detection. Clustering tries to differentiate objects into multiple groups based on their shared characteristics. In contrast, outlier detection aims to find objects whose characteristics deviate so much from the rest of the dataset that they should not be considered part of this group.

The task of finding in-between instances contains some of these ideas. On the one hand, there needs to be a way to differentiate an in-between instance from the instances of a cluster, similar to outlier detection. On the other hand, a notion of similarity is needed to identify the characteristics of an in-between point that are similar to, or a mixture of, the characteristics of two clusters. Thus, the task of finding in-between instances differs from that of outlier detection in that it attempts to infer how similar the deviating instance is to a mixture of characteristics of two clusters rather than merely attempting to distinguish the instance from the regular pattern.

Kazempour et al. (2024) define an in-between instance, referred to as IBI, as follows.
Given a dataset $X \in \mathbb{R}^{n \times d}$, a partitioning $C = \{C_1, ..., C_k\} \subseteq X$ and a set of outliers $O \subseteq X$ where $C \cap O = \emptyset$ and $\{X\} = C \cup O$. An object $o \in O$ is an in-between instance if it is (a) deviating mostly from the characteristics of clusters and (b) still exhibits some characteristics of at least two or more clusters.
Thus, an IBI is located between two or more clusters and potentially shares the characteristics of the clusters it is located between. This further delimits the concept of IBIs from outliers. A point can be an outlier if it is near a single cluster, while an IBI needs to be in the proximity of at least two clusters. To address the question of when we consider a point to be in the proximity of multiple clusters, Kazempour et al. (2024) introduced a predicate-driven definition of in-between instances as follows.
Given a dataset $X \in \mathbb{R}^{n \times d}$, a partitioning $C = \{C_1, ..., C_k\} \subseteq X$ and a set of outliers $O \subseteq X$ where $C \cap O = \emptyset$ and $\{X\} = C \cup O$. An object $O \in o$ is an in-between instance if it satisfies a predicate $\theta(o, C_i, C_j) \in \{true, false\}$ with $i \neq j$.
The choice of this predicate is non-trivial. It needs to capture the semantics of a point in-between multiple clusters. Three possible concepts and predicates based on these concepts were proposed (Kazempour et al., 2024).
The first predicate is based on the concept of neighborhood. For two objects $x \in C_i$ and $y \in C_j$ with $i \neq j$, an object $o \in O$, and the neighborhood $\mathcal{N}(x)$ for each object.

$$\theta(o, x, y) = o \in \mathcal{N}(x) \wedge o \in \mathcal{N}(y) \tag{1.4}$$

This predicate describes an in-between point as an outlier that lies within the neighborhood of points of at least two clusters. Thus, the similarity to multiple clusters is interpreted as being in the neighborhood and thus in the proximity of multiple clusters.

The second predicate uses probabilistic models M_i and M_j for given clusters C_i and C_j with $i \neq j$. For an object $o \in O$ the predicate is defined as follows:

$$\theta(o, M_i, M_j) = p(o|M_i) \approx p(o|M_j) \tag{1.5}$$

Here $p(o|M)$ is the probability of an object o belonging to a model M. Using this predicate, we define points whose probability of belonging to cluster C_i and their probability of belonging to another cluster C_j are similar, as in-between instances.
The last predicate leverages geometric properties of in-between points. We use two objects, $x \in C_i$ and $x \in C_j$ with $i \neq j$, as angle references for the possible in-between instance $o \in O$. Let $d_{cos}(x, y) = 1 - \frac{\langle x,y \rangle}{\|x\|\|y\|}$ be the cosine distance between two objects. The predicate is defined as follows:

$$\theta(o, x, y) = d_{cos}(x, o) \leq d_{cos}(x, y) \wedge d_{cos}(y, o) \leq d_{cos}(x, y) \tag{1.6}$$

An in-between point is located such that the angle between the object o and the reference objects in both clusters is smaller than the angle between the two reference objects.

1.3.4 Support Vector Machines

Support Vector Machines, hereafter referred to as SVMs, were originally introduced by Vladimir Vapnik (Boser et al., 1992) in the early 1990s. They were introduced as a tool for binary classification and nonparametric regression. Binary classification is the problem of dividing a set into two disjoint classes based on their inherent characteristics. In nonparametric regression, the structure and patterns of the dataset are captured to predict discrete values. We will focus on the use case of applying SVMs to classification problems. Their widespread adoption in different research fields can be attributed to their robust performance, their solid theoretical foundations, the inherent ability to avoid overfitting, fast and effective training even on small sample sizes, and the ability to handle high-dimensional data (Noble, 2006; Meng and Zhao, 2015; Ben-Hur et al., 2008).

SVMs belong to the class of supervised machine learning algorithms. As a supervised machine learning algorithm, SVMs are trained on a dataset containing annotations indicating the class to which each datapoint belongs; these annotations are called the label. As a tool for binary classification, SVMs aim to separate two classes from each other. This is done using a decision boundary where one class lies on each side of the boundary. SVMs search for the optimal hyperplane, which fulfills the aforementioned criterion and best differentiates the two classes. Additionally, the hyperplane is optimized to maximize the distance between the hyperplane and the nearest data points of each class, which is referred to as the margin. This characteristic of SVMs avoids overfitting while still obtaining high-accuracy classifying models. The maximum margin hyperplane is based on the principle that a larger distance from the separating hyperplane to each class reduces the generalization error, while smaller distances lead to overfitting. While classifiers that minimize the mean squared error are

susceptible to outliers, SVMs are able to detect these points and eliminate their influence on the separating hyperplane (Boser et al., 1992). In an n-dimensional feature space, the hyperplane will be n-1 dimensional. To illustrate this concept in this Section, we will consider two-dimensional examples where the hyperplane is a one-dimensional line. However, our dataset will contain higher-dimensional data, which cannot be visualized in this manner.

The separating hyperplane is defined by support vectors, which are the points closest to the decision boundary. These points are the most challenging to classify. The location of the support vectors determines the location of the separating hyperplane. Only the position of the support vectors affects the position of the separating hyperplane. The location of the optimal hyperplane and identifying the support vectors are two equivalent problems that can be solved as a quadratic programming problem. In the following section, we will explain the mathematical foundations behind SVMs, how the kernel trick is leveraged to apply SVMs to non-linearly separable datasets and discuss the concept of soft-margin SVMs.

Mathematical Formulation

The following definitions are based on the work of Boser et al. (1992), which introduced SVMs, and the work of Cervantes et al. (2020); Wang and Lin (2014).

The input for the SVM consists of labeled training samples x with a dimensionality of n, which corresponds to the number of features, and a label y. Thus, an input with p samples and classes A and B has the following form.

$$(x_1, y_1), (x_2, y_2), ..., (x_p, y_p) \tag{1.7}$$

$$with \quad y_k = 1 \quad if \quad x_k \in A \quad and \quad y_k = -1 \quad if \quad x_k \in B \tag{1.8}$$

Based on this training set, the SVM learns the parameters of the decision function $D(x)$ which separates the classes A and B.

$$x \in A \quad if \quad D(x) > 0 \tag{1.9}$$

$$x \in B \quad otherwise. \tag{1.10}$$

The decision function $D(x)$ separates the classes A and B by predicting the class memberships of all samples. The parameters of the decision function consist of a set of weights w_i with one weight for each feature. The decision function D is a linear combination of these weights predicting the class label y.

$$w \cdot x_i + b \geq +1 \quad when \quad y_i = +1 \tag{1.11}$$

$$w \cdot x_i + b \leq -1 \quad when \quad y_i = -1 \tag{1.12}$$

$$\tag{1.13}$$

This decision function is based on the idea of the maximum margin hyperplane. Figure 1.2 shows a two-dimensional example of a separating hyperplane between classes A and B.

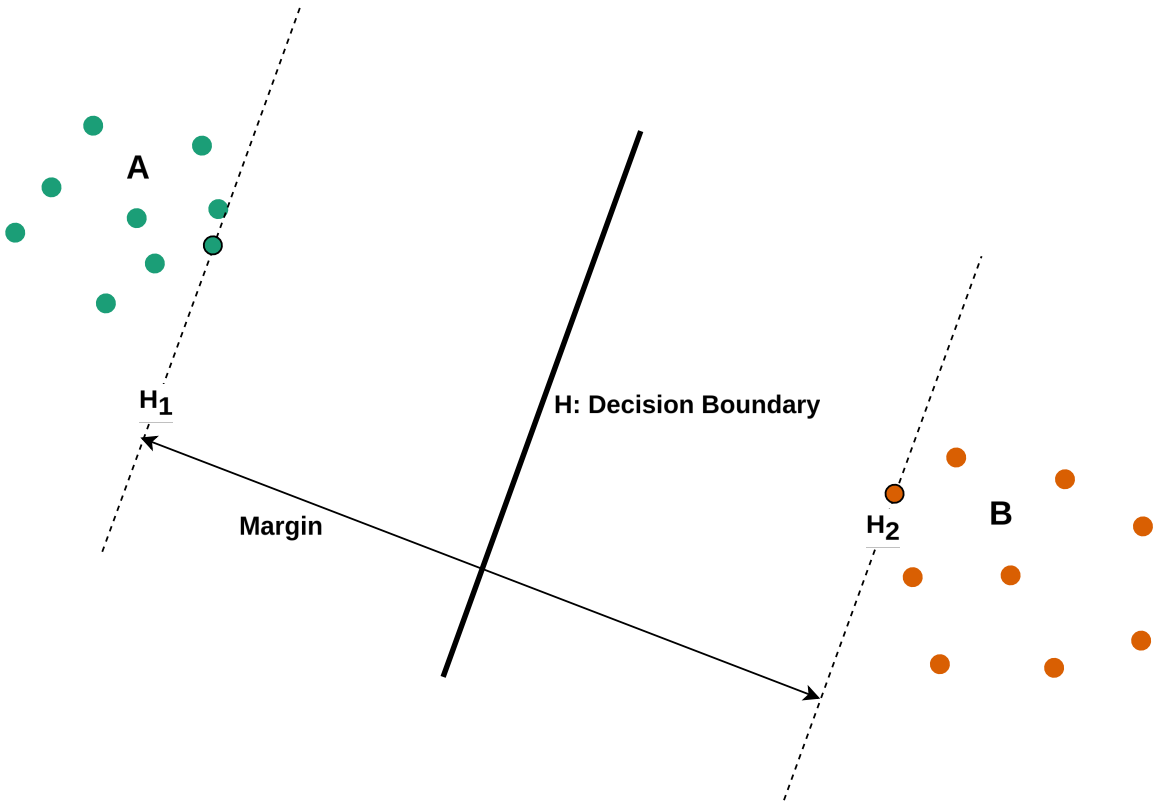


Figure 1.2: Visualization of Hyperplanes H, H_1, and H_2

A separating hyperplane is located within the feature space, or as seen later within a higher dimensional kernel space, and divides the space into two classes. Each class lies on its own side of the separating hyperplane. Thus, the decision to which class a new label belongs can be made by calculating on which side of the hyperplane the sample lies.

$H_0 : w \cdot x_i + b = 0$ is often called the separating hyperplane, the separating boundary or decision boundary and will hereafter be referred to simply as H. H is the hyperplane that divides the feature space into two separate classes. On one side of H lies the class A, and on the other side, the class B. $H_1 : w \cdot x_i + b = +1$ and $H_2 : w \cdot x_i + b = -1$ are located parallel to H_0 and lie on the support vectors of each class. The three hyperplanes are defined by weight vector $w = w_0, w_1, ..., w_n$ and the bias b. Instead of choosing an arbitrary hyperplane between both classes, the SVM searches for the optimal hyperplane. An optimal hyperplane maximizes the margin between H_1 and H_2. Thus, it lies in the middle of the classes A and B. This location is unique.

An important advantage of SVMs is that the set of non-zero weights is reduced to just a few when maximizing the margin. These few non-zero weights correspond to the support vectors, seen within Figure 1.2 as the outlined data samples directly on H_1 and H_2. Moving a sample that does not lie on either H_1 or H_2 does not change the weights and thus the position of H. Thus, only the support vectors are the samples that would change the position of the hyperplane if they are removed or their values change. Consequently, the problem of finding the optimal separating hyperplane is equivalent to finding the support vectors. This greatly reduces the complexity of the resulting classification model. The support vectors s fulfill the equation

$$w \cdot s + b = 1 \quad or \quad w \cdot s + b = -1. \tag{1.14}$$

The margin of separation M describes the distance between H and the closest data points, the support vectors, and thus the distance between H_1 or H_2.

The distance between the hyperplanes H and H_1 is $|W \cdot x + b|/||W|| = 1/||W||$. The total distance between H_1 and H_2 is thus $2/||W|| = M$. Thus, maximizing M is equivalent to minimizing $||W||$ with the extra condition that no data points exist between H_1 and H_2. The constraint that no data point exists between H_1 and H_2 is equivalent to the combination of Equation 1.12 and Equation 1.13. These equations describe that the hyperplanes H_1 and H_2 should correctly classify all training samples x_i, and no training samples lie within the margin. We also observe that Equation 1.12 and Equation 1.13 can be combined and result in the constraint for the SVM optimization problem, seen in 1.15.

$$y_i(x_i \cdot W + b) \geq 1 \iff y_i(W \cdot x_i + b) - 1 = 0 \tag{1.15}$$

Instead of minimizing $||W||$, we can also minimize $1/2||w||^2$, which is still equivalent to maximizing M. Thus, we obtain the SVM primal optimization problem of maximizing the margin and correctly classifying all training samples, which is now formulated as

$$\min_{w,b} f(x) : 1/2||w|| \tag{1.16}$$

$$s.t. \quad g(x) : y_i(w \cdot x_i + b) - 1 = 0 \tag{1.17}$$

To solve this optimization problem, the method of Lagrangian multipliers is used. As f is quadratic, its surface is a paraboloid with a single global minimum. The intuition behind this optimization problem is to find the intersection of f and g at a tangent point. The derivative at this point is 0, and thus, it will be a minimum or maximum with the constraint satisfied.

Lagrange multipliers are used to maximize or minimize a function f while maintaining another constraint g. To understand Lagrangian multipliers, we need to introduce some core concepts. The gradient of a function f, ∇f, is the extension of the concept of derivatives to vector calculus. The value at point p of ∇f gives the direction and the rate of the fastest increase. The gradient is a vector that collects all the partial first derivatives of a function.

A level set is defined as $L_c(f) = \{(x_1, x_2, ..., x_n)|f(x_1, x_2, ..., x_n) = c\}$. A level set is the set of all input values for f that results in output c.

We observe that the gradient of f is always perpendicular to the level curves of f. The intuition behind the Lagrangian multiplier method is that every solution for our problem will be a tangent point where our function g touches the level set where f is maximized. We can also see that the gradients of f and g are parallel but possibly differently scaled. This scaling parameter, which is called the Lagrangian multiplier, is denoted as α. We obtain the following equation.

$$\nabla f = \alpha \nabla(g) \iff \nabla f - \alpha \nabla(g) = 0 \tag{1.18}$$

The solution for Equation 1.18 is the solution of minimizing f while satisfying the constraint g. To find this solution, we define the Lagrangion as $L = f - \alpha(g)$ and require the derivate of L to be zero, $\nabla L = 0$. Now, the solution to our problem is always a saddle point of the Lagrangian function.

For the SVM optimization problem, we want to minimize $f(x) : \frac{1}{2}||w||^2$ with the constraint $g(x) : y_i(W \cdot x_i + b) - 1 = 0$. Following, in the SVM problem, the Lagrangian primal form for all i and $\alpha_i \geq 0$ where l is the number of training samples is as follows.

$$\min_{w,b} \quad L_P = \frac{1}{2}||w||^2 - \sum_{i=1}^{l} \alpha_i y_i (x_i \cdot w + b) + \sum_{i=1}^{l} \alpha_i \tag{1.19}$$

Additionally, we observe that the derivatives at this minimum are 0. The following equations are solved to find the solution of the optimization problem.

$$\frac{\nabla L_P}{\nabla w_i} = 0 \tag{1.20}$$

$$\frac{\nabla L_P}{\nabla \alpha} = 0 \tag{1.21}$$

$$\frac{\nabla L_P}{\nabla b} = 0 \tag{1.22}$$

For this solution to the SVM optimization problem, the Karush-Kuhn-Tucker conditions must be true. With f convex, the necessary and sufficient conditions for a normal point to be optimal are the existence of α and w such that

$$\frac{\nabla L_P}{\nabla w_i} = 0 \tag{1.23}$$

$$\frac{\nabla L_P}{\nabla b} = 0 \tag{1.24}$$

$$y_i(w \cdot x_i + b) - 1 \geq 0 \tag{1.25}$$

$$\alpha_i \geq 0 \tag{1.26}$$

$$\alpha(y_i(w \cdot x_i + b) - 1) = 0 \tag{1.27}$$

However, instead of solving the primal Lagrangian, we formulate the Lagrangian dual form of the SVM optimization problem. This has the advantage of removing the dependency on w and allowing us to employ the kernel trick as described later. From Equation 1.20 and Equation 1.21, we obtain that $w = \sum_{i=1}^{l} \alpha_i y_i x_i$ and $\sum_{i=1}^{l} \alpha_i y_i = 0$. We can obtain the dual form of the Lagrangian by substitution w.

$$\max_{\alpha} \quad L_D = max \sum_{i=1}^{l} \alpha_i - 1/2 \sum_{i=1}^{l} \alpha_i \alpha_j (x_i \cdot x_j) \tag{1.28}$$

$$s.t. \quad \sum_{i=1}^{l} \alpha_i y_i = 0 \quad and \quad \alpha_i \geq 0 \tag{1.29}$$

The dependency on w and b has been removed, and the dot product of x_i and x_j allows us to employ the kernel trick. Instead of minimizing over w and b, we can maximize over α After solving the problem in the Lagrangian dual form, we can determine the weights by

$$w = \sum_{i=1}^{l} \alpha_i y_i x_i \tag{1.30}$$

The SVM can now classify unknown samples x_j and decide which class they belong to according to our model. This is based on the training samples x_i. We remember that most

weights will be zero. Only the weights corresponding to our support vectors will be non-zero, and thus, our model only depends on them. This reduces the dimensionality of our solution. The decision rule is as follows.

$$D(x_j) = sign(w \cdot x_j + b) = sign((\sum_{i=1}^{l} \alpha_i y_i x_i \cdot x_j) + b) \tag{1.31}$$

With the current formulation, the SVM can solve classification tasks on two linearly separable classes. We can apply the kernel trick to extend the SVM to solve classification tasks on datasets that are in their original feature space and are not linearly separable. We use the dual form of the Lagrangian to insert the kernel function $k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$. This results in the following decision rule.

$$D(x_j) = sign(w \cdot \phi(x_j) + b) = sign((\sum_{i=1}^{l} \alpha_i y_i k(x_i \cdot x_j)) + b) \tag{1.32}$$

We introduced the original SVM optimization problem formulation and extended it with the addition of a kernel function to support classification for non-linearly separable datasets. This formulation is referred to as the hard margin SVM. The constraint 1.15 prohibits training samples from being located within the margin of the separating hyperplane H. In the case of a non-linear separable dataset, this constraint can not be satisfied, and no solution can be found. Furthermore, a constraint this harsh can lead to overfitting. The soft margin is an approach to deal with datasets that are not linearly separable. A soft margin allows the SVM to make a certain number of misclassifications. The soft margin introduces an error term for samples that lie on the wrong side of H or between H_1 and H_2 while keeping the margin as large as possible. Even in cases where the data might be linearly separable, this approach might lower the generalization error and make the SVM less prone to overfitting the training data. In our equations, we introduce a slack variable ξ as a measurement of misclassification for datapoint x_i. Now, our data points need to satisfy the new Constraint 1.33 for all i and $\xi_i >= 0$.

$$y_i(x_i \cdot w + b) \geq 1 - \xi_i \tag{1.33}$$

The left side of Equation 1.33 can be interpreted as the confidence of our classification. With a score greater than one, we have classified the point correctly. Every score lower than one should introduce a linear penalty, which can be parametrized by C. Thus, C controls how high the penalty for misclassified samples and samples between H_1 and H_2 is. Thus, we minimize a new loss function f.

$$f(x) : 1/2||w||^2 + C \sum_{i=1}^{l} \xi_i \tag{1.34}$$

The before-mentioned optimization using Lagrangian multipliers can be applied to this new loss function and constraint. Thus, we gain the ability to allow and parameterize misclassifications. This leads to better generalization and the ability to be applied to non-linearly separable datasets.

Parametrisation Our thesis uses the SVM implementation, SVC, provided by sklearn (Pedregosa et al., 2011). SVC itself is based on the implementation libsvm (Chang and Lin, 2011). The main parameters that control the optimization process are as follows:
The parameter C controls the error term introduced within the soft margin formulation of the SVM. It is a regularization parameter and controls the trade-off between the maximization of the margin and the minimization of the classification error on the training data. Thus, it influences the complexity of the decision boundary and balances it with the model's accuracy of the model when applied to the training set. The choice of C is important to avoid overfitting. A high C value emphasizes the correct classification of all training samples. This can lead to more complex decision boundaries and may overfit the model. A lower C value encourages simpler and smoother decision boundaries with a larger margin between the classes. While more robust to overfitting, it also might introduce misclassification as the model might underfit and fail to capture the patterns of the classes.

The parameter *kernel* specifies the kernel function we insert into the Lagrangian dual form. Within this thesis, we used the following kernel functions.

$$\textbf{linear} : \langle x, x' \rangle \tag{1.35}$$

$$\textbf{polynomial} : (\gamma \langle x, x' \rangle + r)^d \tag{1.36}$$

$$\textbf{rbf} : \exp(-\gamma \|x - x'\|^2) \tag{1.37}$$

The simplest kernel function is the linear kernel. It is fast and efficient in computing and provides interpretable results since the decision boundary lives within the original feature space and can be visualized. It is suitable for linearly separable datasets but fails to model more complex patterns within the dataset.

The Radial Basis Function (rbf), also referred to as the Gaussian kernel, implicitly maps the input features into an infinite-dimensional space. Non-linearly separable datasets can possibly be linearly separable within this kernel space. Thus, this kernel allows us to model more complex decision boundaries and thus capture more complex patterns within our dataset. These more complex boundaries are harder to interpret than the patterns identified by the linear kernel. This is further increased as the decision boundary lives within the kernel space and is thus not naively visualizable. Its computation is more intensive than the linear kernel. This kernel function introduces the parameter γ, as seen in Equation 1.37. Within the rbf kernel, γ is used to control the width of the Gaussian function. Its value determines the radius of the influence of a single training sample. A high γ value reduces the sphere of influence of a training sample to a small region. Thus, the decision boundary is more sensitive to individual points and gets more complex. Similar to a high C value,

it leads to a more uneven and overfitting-prone decision boundary. Low γ values increase the range of the influence of a single training sample. Thus, the decision boundary is more smooth and less complex. A small γ value corresponds to a Gaussian with a large variance and larger reach. While a high γ leads to low variance models with higher bias.

The polynomial kernel can be seen as a tradeoff between linearity and non-linearity. The decision boundary can adapt to more complex patterns with a polynomial shape. The degree of the polynomial controls the flexibility and complexity of the decision boundary. The choice of degree balances a more complex model, which might overfit, against a simpler one, which might underfit. Additionally, a high degree increases the computational complexity. This kernel uses the parameters *gamma* and *degree*, see Equation 1.36. Within this kernel, γ serves as a scaling factor for the dot product of the input vectors. The dot product of two vectors u and v is $u \cdot v = |u||v|cos\theta$. The dot product tells us the similarity in terms of the direction of the vectors. This works even in high dimensions. It is -1 when the vectors point in opposite directions. The dot product can be interpreted as projection u onto v (or vice-versa) and then taking the product of the projected length of u, which is $|u|$, with the length of v, which is $|v|$. If we look at the possible rotations of u while keeping v fixed, the dot product results in 0 when u is orthogonal to v as the projection of u onto v yields a vector of zero length. It results in the largest value of $|u||v|$ when u and v point in the same direction. When they point in opposite directions, the result will be the lowest possible value of —$|u||v|$. If we now divide $u \cdot v$ by the max value of $|u||v|$, we limit the range to [-1,1], and thus, it is scale invariant. Within the polynomial kernel, the γ parameter can make the kernel function more sensitive to the similarity of the input vectors. A higher γ increases the impact of the dot product and thus increases the influence of the similarity of the input vectors. This can lead to more complex decision boundaries, while a lower γ value results in smoother and more generalized decision boundaries.

1.4 Goals

In the following, we will explain the goals of our thesis and how we aim to achieve them.

G1: Are SVMs suitable for detecting instances in-between two known clusters? To gain insight into whether SVMs are suitable for detecting instances in between clusters, we will conduct a range of experiments on a synthetic dataset. As our motivation stems from the real-world scenario of isotope analysis for provenance research, this dataset will be based on an isotope dataset. We will train SVM models on this dataset and use the obtained structure, the separating hyperplane defined through the support vectors, to build an algorithm that aims to predict whether a given sample is the result of a linear mixing process between two known reference clusters.

G2: Is it feasible to estimate the progress? As we expect the in-between instance to be the product of a linear mixing process over time and with respect to the features, we

aim to approximate the progress of this mixing process based on the data point's position. Particular interest is given to instances with a far-advanced migration process. These points will be the hardest to detect and, thus, the most difficult to task.

G3: Given an isotope dataset of humans buried at a certain location but potentially originating from another known region, can we apply the results of G1 and G2 to this dataset? Within G1 and G2, we aim to prove that our concept works on a synthetic dataset. Afterward, we will apply it to a real-world dataset. This dataset will contain isotope ratios obtained from cremated human remains in a region around the Alps. The dataset is described in further detail in Section 3.8.1. For each individual, the location of burial is known, while their place of origin remains uncertain. Since the burial location may not coincide with the individual's provenance, the aim is to detect whether the individuals migrated to this place and identify their possible places of origin.

2 Related Work

Our thesis encompasses three separate research domains. The study of SVMs and the concept of the optimal hyperplane for classification, IBIs and their detection, and multi-isotope analysis for provenance research. Each of these areas is crucial to our work and merits equal consideration. To the best of our knowledge, no existing research has yet explored the integration of machine learning methods with the concept of IBIs to analyze the provenance of ancient human cultures.

2.1 SVM Applications

SVMs, as one of the most potent and robust classification and regression algorithms, are widely employed in multiple fields of applications. Active research on SVMs, their application fields, strengths, and ways to mitigate their weaknesses is continuously developing new ideas and improvements. These weaknesses include the challenges associated with large datasets and the quadratic runtime of SVMs, multi-class classification, and unbalanced datasets. Within many different real-world applications, SVMs proved their merit. In the following, we will highlight some prominent examples.

He et al. (2003), compared k-nearest neighbor, SVM, and adaptive resonance associative map for document categorization. SVM outperformed the other methods when applied to moderate-size datasets, and adaptive resonance associative maps only performed comparably on large and noisy data.

The research of Lee et al. (2012) is of special importance to us as they employed a two-step approach. The first step trains an SVM on text documents and obtains the support vectors for each text category. The second step predicts the category membership for new samples. The average distances between the sample and the support vectors are calculated, and the sample is assigned the category with the lowest average distance to its support vectors. Even if not pursued in this thesis, this idea is also applicable to our problem and, in-between instances, could be detected as samples with similar average distances to the support vectors of two classes.

A common application of SVMs is image classification (Verma et al., 2017; Afifi et al., 2020; Kaur et al., 2019; Hsieh and Liou, 2015). In medical imaging, SVMs were successfully used to identify kidney stones (Verma et al., 2017), mass classification of mammograms (Liu and Tang, 2013), and early detection of melanoma (Verma et al., 2017; Kassani and Kassani, 2019; Afifi et al., 2019). Expression recognition (Zhang et al., 2015) and the recognition of hand gestures (Hsieh and Liou, 2015; Raheja et al., 2016) also proved the ability of SVMs to classify image and video data.

Even problems with many classes, such as hand-written character recognition, were solvable through the use of SVMs (Bhowmik et al., 2009; Dinesh and Sabenian, 2019; Gao et al., 2011; Christlein et al., 2017). Hand-written character recognition is the process of automatically converting images of handwritten text into machine-encoded text. As each individual symbol of the used alphabet is its own class and the input data might be high dimensional, this is especially challenging.
Often, the strengths of SVMs are combined with other machine-learning approaches, such as distributed clustering (Kumar and Gopal, 2009), k-nearest neighbor(Verma et al., 2017; He et al., 2003), and Gaussian Mixture Models (Christlein et al., 2017).
We are currently not aware of research on finding IBI using SVMs. However, in the following, we will present SVM-related research that might be suitable for IBI detection.
The objective of the SVM, to fix an optimal hyperplane with the largest margin between two classes, should not be interpreted as a class membership probability. The SVM only optimizes in such a way that each class lies on a particular side of the decision boundary but does not fit a sigmoid function like logistic regression, which results in probability outputs. This is a significant drawback for applications that rely on understanding the uncertainty or confidence of predictions. Platt and others (1999) introduced an approach to convert the decision values of an SVM into probabilities. This made SVMs suitable for problems that relied on the postprocessing of class membership probabilities, such as some ensemble strategies. This approach is widely used to calibrate the output probability of binary classifiers and is known as Platt scaling. A logistic regression model is fitted to the output of the SVM to obtain a probability based on the SVM model. The logistic regression model, referred to as Platt Scaling, is defined as seen in Equation 2.1.

$$P(y = 1|x) = \frac{1}{1 + \exp(Af(x) + B)} \tag{2.1}$$

Where $P(y = 1|x)$ denotes the probability that sample x belongs to class 1. $f(x)$ is the output of the SVM decision function for x, and A and B are the parameters that are estimated by a maximum likelihood method. The logistic model is fit after training the SVM by using the decision function outputs as input and target probabilities based on the class labels.
As we aim to find in-between instances, calculating and analyzing the probability of class memberships might be a good idea. Within Section 1.3.3, we introduced multiple concepts of in-betweenness defined by predicates. Equation 1.5 describes a predicate based on a probabilistic model to identify IBIs. Based on the SVM decision function and the application of Platt scaling, we could obtain a probabilistic model and apply this predicate. We pursued a different approach that more directly leverages the SVM output. However, comparing the Platt scaling-based method with our approach might be interesting for further research.
When optimizing the dual Lagrangian, the SVM treats each class sample equally. However, in real-world applications, some training samples are often more important than the rest. Misclassifying noise should not have the same impact as misclassifying meaningful training

samples. Thus, Lin and Wang (2002) introduced the fuzzy SVM (FSVM). FSVMs are an extension of SVMs, incorporating the concept of fuzzy memberships, which reflect the degree to which a data point belongs to a particular class. The fuzzy membership is used to weight the slack variable within the SVM Lagrangian. Thus, the impact of samples with lower class membership within the optimization is lower, while mislabeling samples with higher class membership introduces a larger penalty. While this approach does not identify IBIs, it could reduce their impact on the SVM's training as IBIs would be expected to have lower class membership than samples that are not IBIs.
Instead of a maximum margin hyperplane, Cevikalp (2016) proposed two classifiers based on the best-fitting hyperplanes for classification. Rather than a single hyperplane with a maximum margin to both classes, multiple best-fitting hyperplanes are proposed. Each best-fitting hyperplane is close to the samples of one class while being as far as possible from the other class. Thus, they not only separate the classes but are also optimally positioned relative to the distribution of a class. The classifier proved to be particularly suited to handling large-scale problems, open set recognition, and object detection tasks. Native SVMs work with the assumption that the area between both distributions, the area within the margin, can be separated, and new labels within this area should be classified. This research places particular focus on the area between two distributions. Within this area, we would suspect IBIs to occur. Best-fitting hyperplanes could be applied to separate the distribution of two classes and the space in which IBIs may occur, while traditional SVMs would still classify IBIs to either class.

2.2 Multi-Isotope Analysis

Machine learning methods have been applied to multi-isotope fingerprints to overcome the shortcomings of single-isotope fingerprints.The study *Data Fusion Between Land and Sea: Multi-Isotope Fingerprints of Viking Animals and Modern Plants* (Gohring et al., 2024) uses a Gaussian Mixture Model (GMM) to conduct multi-isotope analysis, integrating the concept of IBIs to enhance the detection of sea spray-affected isotopes beyond the capabilities of conventional machine learning methods.

The sea spray effect occurs when sea salts are taken up by plants and subsequently consumed by animals, including humans. Then, terrestrial plants and animals show a seemingly marine isotope signal. The sea spray effect impacts multiple isotopes, making it challenging to differentiate between purely terrestrial and marine origins based solely on traditional single isotope analysis. The research aims to understand if the existence of in-between characteristics can be combined with domain knowledge and which insights can be gained from this observation compared to methods such as clustering (Gohring et al., 2024). The proposed research question of interest for us is whether archaeological individuals affected by sea spray are identifiable as IBIs and to evaluate if the IBI concept may provide unique insights into bioarchaeological research.

The study focuses on a region near the Baltic Sea in northern Germany, historically significant as a Viking Age and medieval age trading center (Gohring et al., 2024). Isotope signatures of a wider area around the Baltic Sea are used to complement this dataset. The datasets contain isotope data measured in bone samples from archaeological terrestrial herbivorous mammals, marine mammals as well as modern plant leaves.

A three-step approach was employed to identify IBIs and individuals influenced by sea spray. The first step involved analyzing the one-dimensional isotope systems of terrestrial herbivorous and marine mammals. These were expected to cluster into two distinct groups, with the marine mammals' lowest measured isotope value defining the lower marine limit. If a terrestrial mammal's isotope signal is higher than this limit, especially within multiple isotope dimensions, it is considered affected by sea spray.

The second step is applied to the multi-isotope fingerprint. A GMM model is trained to separate the group into two clusters. Afterward, a predicate to identify IBIs is applied. The predicate defines a sample as an IBI if the distance between the sample and the mean of the clusters separately is smaller than the distance between the means of both clusters. Furthermore, it is assumed that identifying an IBI in the full-dimensional feature also implies that this IBI can be identified in a lower-dimensional feature space, leading to the criterion that a sample is an IBI if the predicate holds for at least one feature dimension.

The third step uses a GMM model to create three clusters. A cluster of terrestrial herbivorous mammals and marine mammals is expected. The third cluster should contain the samples affected by sea spray.

The researchers examine the results of these three steps and evaluate whether the inclusion of the IBI concept leads to the identification of more sea spray-affected individuals. The GMM clustering produced a third cluster containing individuals affected by sea spray. However, additional individuals were identified as IBIs and showed sea spray signals within at least one dimension. These individuals could not be identified as too marine based on the lower marine limit or the GMM-based approach.

Thus, the results of this research show that the IBI approach enabled the detection of additional sea spray candidates compared to the individuals found within the third cluster. These individuals could not have been identified using traditional machine learning methods, such as GMM models or outlier detection. Thus, the idea of IBIs was successfully implemented.

Even though this research uses a different semantic meaning behind IBIs, describing sea spray-affected mammals as in-between marine and unaffected terrestrial mammals, it applies the IBI concept to an isotope dataset and thus is of particular interest to us. Furthermore, it supports the need for the concept of IBIs. It validates that defining a class of samples as in-between and applying an approach to identify this class can identify samples that traditional machine learning approaches could not.

Grupe et al. (2020) studied the multi-isotope fingerprints of archaeological human remains from the Alpine region during the late Hallstatt period and early Fritzens-Sanzeno culture. The researchers applied multi-isotope analysis, focusing on radiogenic strontium and lead

isotope ratios, to differentiate between local and non-local individuals. This approach was applied to a dataset of skeletal remains from 92 burials. During this time period, cremations were the common burial practice. Thus, using tooth enamel to identify migrated individuals was unfeasible; compact (mostly cremated) bone was sampled instead.

The Inn-Eisack-Adige-Brenner passage was one of the most frequented Alpine transects around which the Fritzens-Sanzeno culture emerged. This research aims to give insight into whether the spread of this culture resulted from migration or acculturation processes.

Grupe et al. (2020) applied GMM clustering to analyze the multi-isotope fingerprints. This allowed for grouping individuals into distinct clusters based on their isotopic similarities. The property of GMM clustering that assigns probabilities to cluster assignments is particularly useful for this use case. These probabilities were used to identify potential migrants, as a lower probability indicates an insecure cluster assignment, which occurs in the case of a mixed isotope signal. This method effectively identified spatial resolution within the dataset, reflecting the geographical origins of the skeletal remains. The GMM clustering resulted in three clusters, differentiating between inner alpine, south Tyrolean, and a unique cluster with three archaeologically conspicuous graves (Grupe et al., 2020). Identifying this unique cluster suggested the non-local origins of the associated individuals due to their distinct isotopic signatures compared to the other two identified clusters. These individuals were archaeologically notable for their grave goods, deviating from other graves in the region.

This study is particularly relevant to our research as it demonstrates the effectiveness of multi-isotope analysis and machine learning methods, such as GMM clustering, to identify migrated individuals within the dataset. This research is central to our thesis, as we will apply our approach for detecting migrant candidates on the same dataset from the Inn-Eisack-Adige-Brenner passage. Its findings are central to the discussion of our results, and we will evaluate whether our results align with the results of Grupe et al. (2020).

Another application of multi-isotope analysis for provenance analysis was conducted by Tschetsch et al. (2020). This research investigated the use of multi-isotope fingerprints sampled from archaeological animal bones to determine an isoscape for a geologically complex Alpine region. This Alpine region was also examined within the research of Grupe et al. (2020). Motivated by the limitations of single isotope analysis, which struggles with geographical redundancies due to similar $^{87}Sr/^{86}Sr$ isoscapes and cannot alone provide precise geographical origins but rather only indicate a non-local status of an individual. The researchers used a combination of strontium, lead, and oxygen isotope ratios to overcome this. A Gaussian Mixture Model was applied to distinguish between different regions.

Their study used archaeological animal bones from the Inn-Eisack-Adige-Brenner passage. The same research area is used within Grupe et al. (2020). By mapping this area and its isotopic ratios, they aimed to establish a baseline for bioavailable historical isotope ratios, which could be applied to human archaeological finds to evaluate transalpine mobility and cultural transfer. Their research confirmed that a combination of strontium and lead isotopes could differentiate between the northern and southern Alpine regions, which could not be achieved with strontium alone.

This is especially interesting for us, as our dataset stems from the same Alpine region. Their research motivates the need for a multi-isotope approach and the inclusion of lead isotopes alongside strontium to differentiate the geographical regions of the Inn-Eisack-Adige-Brenner passage.

2.3 Cluster In-betweenness

To the best of our knowledge, the concept of IBIs is novel and has only recently been proposed by Kazempour et al. (2024). Consequently, there is no existing research on this topic. However, we found the following research on soft labels for image classification. The concept of soft labels within this research bears a resemblance to the concept of IBIs and can serve as additional motivation for the introduction of the concept of IBIs.

In a recent study, Schmarje et al. (2022) investigated the importance of soft labels in image classification. Acquisition of labeled data is difficult due to noisy and ambiguous human annotations (Grossmann et al., 2022). Humans are prone to subjective interpretations and mistakes during the annotation task. When comparing human annotations against known ground truth, they found that even a majority vote can not overcome this situation as, in some situations, the majority of human interpretations differ from the ground truth (Schmarje et al., 2022).

This leads to poor data quality, especially in image data. Schmarje et al. (2022) propose a data-centric image classification benchmark with multiple annotations per image that allows researchers to investigate and quantify the impact of data quality issues due to mislabeling (Schmarje et al., 2022). It measures the relation between the input labeling effort and the performance of the applied machine learning method. This aims to understand better how labels should be created to obtain the best training results.

The benchmark uses a labeling and an evaluation phase. The labeling phase is divided into two steps. At the beginning, the labels are estimated, and the second step tries to improve the estimated labels. The evaluation phase measures the impact of the provided label on different machine-learning models. The authors found that multiple annotations can increase the performance of the machine learning method. They claim that one annotation is insufficient to capture the underlying ground truth. Thus, soft labels are needed to deal with the challenges of ambiguous or noisy data due to mislabeled samples (Schmarje et al., 2022).

This is relevant to our use case because IBIs, which are mixtures of multiple classes, can be interpreted as soft labels. Within this research, some pictures of cats would be labeled as dogs, frogs, and deer. Thus, it would not be too far-fetched that this image could be interpreted as an in-between instance of cats and other animals. Following this idea, the results of this research could also motivate the usage of IBI labels to increase the performance of machine learning algorithms. In essence, it shows that a single class membership can not capture the real world, and thus, multiple annotations, soft labels, or the realization

that some instances are in-between instances of multiple classes are needed to capture the underlying ground truth.

Even when creating a synthetic dataset, and thus a perfect ground truth exists, a soft label might increase the dataset's quality (Grossmann et al., 2022). Grossmann et al. (2022) compared the embedding space of a synthetic dataset with a hard and soft label. The embedding perfectly captured the data generation process when using a soft label. The hard label created more clusters than anticipated. Thus, they conclude that soft labels give a more exact representation of the expected feature space. In contrast, the hard label forces the algorithm to create an approximation of the feature space. Thus, even in the absence of noise, such as in a synthetic dataset, soft labels, and possibly the concept of IBIs, might be valuable for training machine learning algorithms (Grossmann et al., 2022).

3 Approach

In the following chapter, we introduce our notion of IBIs based on the isotopic fingerprint remodeling use case in Section 3.1. We then present our SVM-based approach to detecting whether a point matches our definition of IBIs within Section 3.2.

3.1 In-between Instances based on Isotopic Remodelling

In Section 1.3.3, we identify multiple possibilities to define an IBI. The choice of the IBI-defining predicate should be based on the individual use case. This choice plays an important role when approaching the question of how to discern between the three groups of outliers, samples corresponding to the classes, and IBIs. Each predicate introduced follows a similar pattern. Each predicate uses outliers as candidates for IBIs and defines a notion of similarity to multiple clusters. Thus, the IBI is defined as an instance that is different enough not to be assigned to either class but still shares similarities with both classes.
As we aim to find migrants within an isotope dataset from multiple locations, we will define a predicate to capture this use case. We have seen in Section 1.3.1 that the isotopic fingerprint of a human who has migrated to a new location initiates a linear remodeling process.

A simplified example of this mixing process of signatures from two locations can be seen in Figure 3.1. The blue clusters correspond to the isotopic signals of two locations based on the 208Pb/204Pb and 87Sr/86Sr ratios. Let us assume an individual migrates from the location corresponding to the cluster in the bottom right to the location corresponding to the cluster in the upper left. The individual's bone remodeling process will begin as soon as they arrive at the new location and adjust their diet or drinking water sources, resulting in a stepwise adjustment of the individual's isotopic fingerprint. The sequentially colored data points correspond to different years after the migration. The mixing process begins within the cluster corresponding to the individual's origin. Visually speaking, the individual's signature will follow a linear path between both clusters. After 20 years, the process is completed, and the isotopic fingerprint corresponds to that of the new location. While this might not capture every detail of the real-world scenario, we try to balance the realism and simplicity of our model. As described in Section 1.3.1, many factors influence the isotopic fingerprint of a human. Thus, we will make the assumption that the linear remodeling process of bone, and thus the linear remodeling process of the isotopic fingerprint, follows this simplified scenario. We try to capture the instances during the mixing process with the following predicate definition.

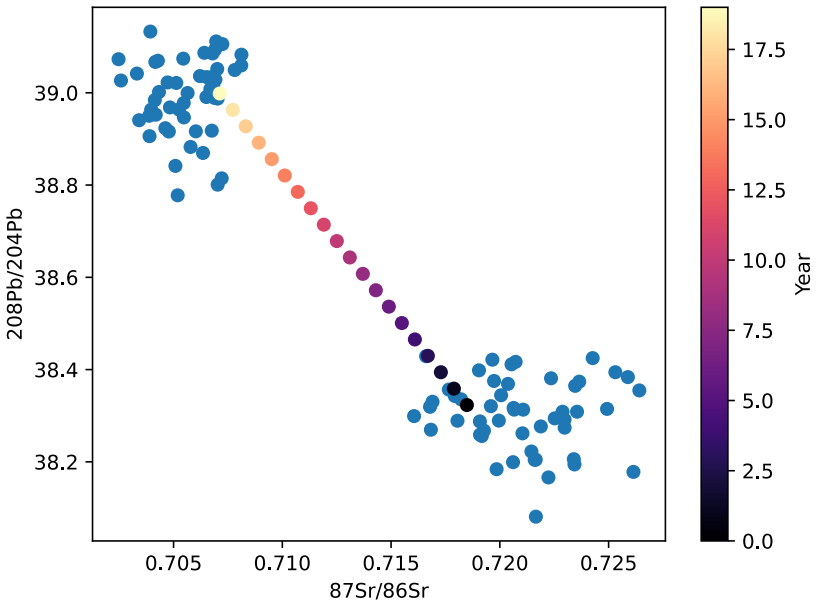


Figure 3.1: Bivariate $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$ Plot of the Linear Mixing Process after Migration with the Years since Migration color coded

Given Gaussian distributions $A = N(\mu_a, \sigma_a^2)$ and $B = N(\mu_b, \sigma_b^2)$ for given reference clusters C_A and C_B. The predicate is defined for an object $o \in O$ as follows.

$$\theta(o, A, B) = \exists a, b : a \in A \wedge b \in B \wedge a \neq b \wedge \tag{3.1}$$

$$\exists f(x) : \alpha x + \beta \wedge a, b, o \in f \wedge \tag{3.2}$$

$$dist(a, o) < dist(a, b) \wedge dist(b, o) < dist(a, b) \tag{3.3}$$

where $a \in A$, and analogously $b \in B$, denotes that object A is the result of the underlying mechanism of the Gaussian model A. We introduce the limitation $a \neq b$ that excludes cases where the start and endpoint of the transformation are the same, for example, possible when the clusters overlap. If o is an IBI, then a function f exists, which is of a linear nature, and the start, endpoints, and candidate o lie on the line defined by this function. Furthermore, $dist(a, o) < dist(a, b) \wedge dist(b, o) < dist(a, b)$ ensures that o is between the start and endpoint if it is an IBI. The start and end points, a and b, of an IBI must be elements from the underlying distributions of the reference clusters to ensure that o is a mixture of C_A and C_B. The concept of defining an IBI through an unknown start and endpoint is visualized within Figure 3.2. We chose to restrict ourselves to Gaussian distributions as isotopic signatures for locations are assumed to follow an underlying Gaussian distribution, as described in Section 1.3.1. It is important to note that the start and endpoints are unknown for a possible IBI o.

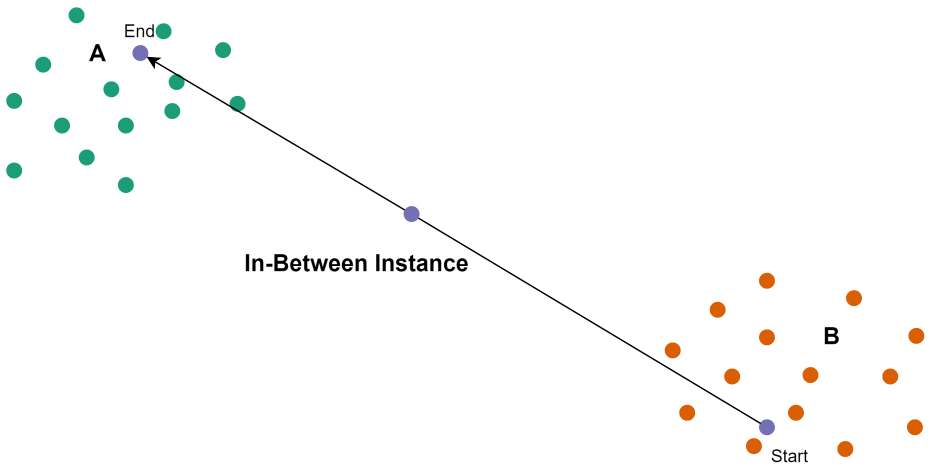


Figure 3.2: Visualization of IBI and Start and Endpoint(in purple) between Clusters A and B

Thus, we can't apply this predicate to identify possible IBIs or, in our use case, migrants. In the following, we will introduce our approach to overcoming this situation.

3.2 The Concept for an Algorithm to Detect in-between Instances

We aim to provide an algorithm to detect IBIs adhering to our definition according to Section 3.1. As this definition captures the use case of finding migrants in an isotopic dataset, our algorithm will attempt to be a possible solution to this goal. Based on this definition, we can define the space in which IBIs occur. For the sake of argument, we will assume that we have perfect knowledge of the structure of both clusters and their underlying Gaussian distributions. This does hold true for the synthetic dataset as we generated it. The space in which IBIs may occur is oriented along the axis on which both clusters lie to one another. As they are Gaussian distributed, this would be the axis that can be found when connecting the known means of the distribution. Along this axis, the space is constricted by the border of each cluster. Points that lie between both clusters are considered to be in-between both clusters according to our definition introduced in Section 3.1. It is important to note that some IBIs may be outside of this space and within the clusters, as the start and end points of the transformation do not necessarily lie on the border of the clusters, as seen in Figure 3.1. Therefore, a trade-off between finding nearly all IBIs and not falsely labeling samples belonging to one of both clusters is unavoidable. The objective of our algorithm is to confine this space as much as possible while ensuring that it does not include the space in which non-intermediate instances occur, such as the data points of a cluster. We will explain our concepts using 2-dimensional examples. These concepts can also be applied to higher-dimensional feature spaces.

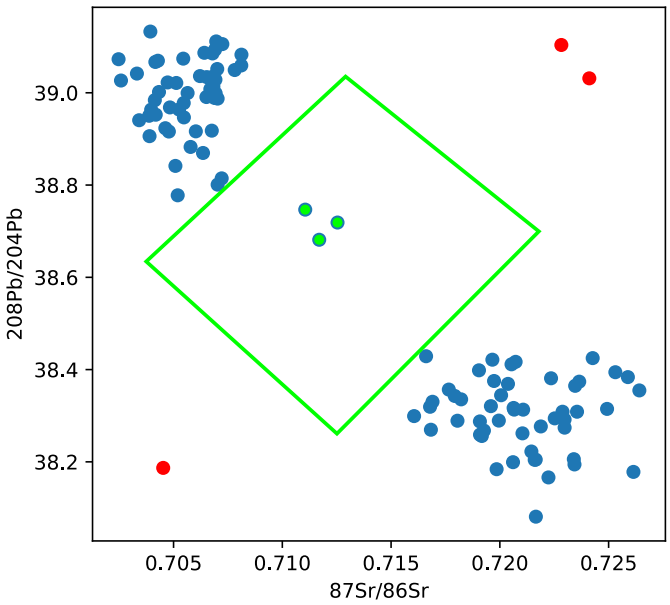


Figure 3.3: Plot of the Space in which IBIs may occur, bounded within the green Borders, the reference Clusters in blue, and Outliers which are not IBIs in red

Figure 3.3 illustrates the space in which IBIs may occur according to our predicate. This space does not necessarily contain all IBIs within a real-world scenario, as some IBIs may be well within the distributions of the clusters, as mentioned in Section 3.1. The green data points are examples of outliers, which are IBIs. They lie between both clusters and can be seen as a mixture of the characteristics of both clusters, yet they are not similar enough to a single cluster to be counted as an inlier. The red data points are outliers as well; however, they do not adhere to the established definition of an IBI, as defined in Equation 3.1 and thus do not represent migrants between the two clusters. These data points do not contain a mixture of the characteristics of both clusters. They might instead be migrants from an unknown location.

To identify points within this area, we build an algorithm based on the structure that SVMs already provide us. In Figure 3.4, we can see a possible result of applying an SVM with a linear kernel to the dataset containing the blue reference clusters. Each cluster would correspond to the local isotope signature of a location. The solid line is the hyperplane that separates our data space and serves as the decision boundary for both classes.

The dotted lines correspond to the parallel hyperplanes defined by the support vectors, which are illustrated by the black data points. The space between both dotted lines already provides a good approximation of the migrant space. We will use this space as a starting point for defining an IBI space.

However, instead of using the hyperplanes defined by the support vectors as boundaries, we

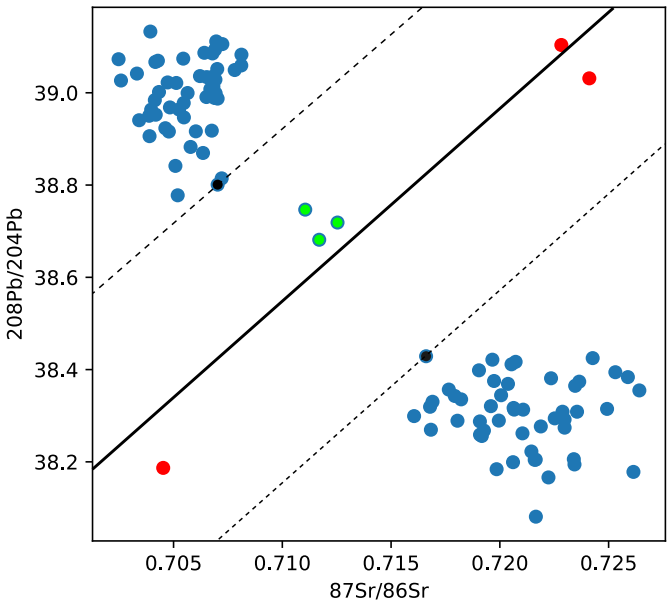


Figure 3.4: Result of applying an linear SVM with the resulting Hyperplane as solid Line and the parallel Hyperplanes of the support Vectors as dotted Lines

will use the distance to the decision boundary to limit the IBI space. This will give us greater flexibility and the ability to introduce a new parameter, the threshold for the distance to the decision boundary, hereafter referred to as $T1$, that controls how similar an IBI may be to one of the clusters. This structure, provided by the SVM, can differentiate between points within the clusters and possible migrants. However, we can see that we cannot differentiate between the outliers, shown in red, and the IBIs, shown in green. Thus, we need to confine our bounding space further. We have to introduce an orthogonal boundary to the SVM's decision boundary to do this.

In Figure 3.5, we can see a possible solution for the bounds, visualized as the pink lines, orthogonal to the SVM hyperplane. We search for the orientation in which the clusters lie to one another and confine the migrant space further, as seen in purple. If a data point lies within the bounds based on the hyperplane obtained by applying the SVM and our cluster-oriented bounds, we declare it as a possible IBI. The following will explain how we applied these ideas and the resulting pipeline to find IBIs.

3.3 Realisation of the Concept

We make some core assumptions to realize the ideas introduced in Section 3.2. We assume that our dataset is Gaussian distributed. As seen in Section 1.3.1 this corresponds to our use case of finding migrants in an isotope dataset. As isotopic signatures rely on underlying nat-

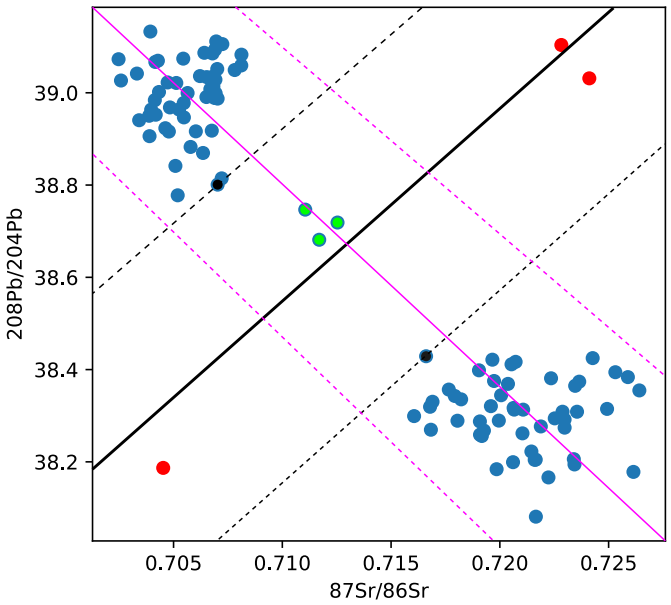


Figure 3.5: Possible Solution for Cluster oriented Bounds as shown in Pink

ural processes and are thus assumed to be Gaussian distributed (Wilhelmson and Ahlström, 2015), it is possible to exploit this later to choose appropriate thresholds for our distances. Furthermore, we assume that our dataset is labeled and, for the time being, does not include statistical outliers. Consequently, as the set of outliers is a superset of the set of IBIs, the dataset to train the SVM does not include IBIs. SVMs belong to the class of supervised machine learning. Therefore, they require a labeled dataset to train the model. Excluding outliers and IBIs allows us to assume an SVM model that separates both classes, which, in the case of the isotope dataset, corresponds to both locations. Thus, this separation is not affected by outliers and IBIs, which lie between the clusters. IBIs could be chosen as support vectors, and thus, the position of the decision boundary might be based on the position of the IBI. Consequently, the hyperplane should be constructed in a manner that maximizes the margin between the two classes without exhibiting a bias towards one class due to a set of IBIs that have been erroneously labeled as instances of the other class. This assumption is primarily for the purpose of testing our approach with synthetic data. When we apply our approach to real-world isotope data, it is important to recognize that the labels may not be free of mislabeled IBIs, outliers, and noise. This may influence our ability to recognize migrants, as the decision boundary may be skewed towards one location. However, we assume that the majority of the data consists of non-migrants. This assumption and the use of a soft margin may result in the decision boundary being evenly placed between both clusters.

Based on these assumptions, our concept can be divided into two steps. The first step

confines the IBI space using the hyperplane provided by the SVM. The second step confines the IBI space using the cluster connecting vector, hereafter referred to as CCV.
The full implementation of our approach is publically available at `https://github.com/janb15/There-may-be-migrants-in-between-Support-Vector-Machines-and-cluster-in-betweenness`.

Confinement based on the SVM Model

As motivated in Section 3.2, in this step, we will use the model provided by the SVM to delimit the IBI space. This model provides us with the support vectors, which are specific data points from our training set that define the position of the decision boundary. Additionally, it provides the weights, which are the coefficients of the vector that determine the coordinates of the decision boundary in the kernel space. Finally, the model outputs the decision function, which is used to classify new data points based on the established decision boundary. Even though the SVM optimizes the margin and thus maximizes the distance of the support vectors to the decision plane, we do not immediately obtain a function that calculates the distances of a point to the decision plane. The SVM does not explicitly compute the distances to the decision boundary but rather translates this optimization problem into its Lagrangian dual form. Rather, the Lagrangian dual form computes a dot product within the potentially infinite-dimensional kernel space. The output of the decision function y has the following form.

$$y(x) = \sum_{i \in SV} \alpha_i k(x, x_i) + b = \langle w, \phi(x) \rangle_H + b \tag{3.4}$$

Where the weights w live in the kernel space. In the case of a linear kernel, we can obtain the distance of point x to the decision boundary by dividing $y(x)$ through the norm of w. In the case of a non-linear kernel, such as the radial basis kernel, we cannot obtain the exact distances to the decision boundary.

The Pre-Image Problem In kernel SVMs, a non-linear mapping from the input space to the feature space deals with non-linear separable classes. The reverse mapping from the kernel feature space to the input space is of special interest in calculating the distance to the decision boundary, which lives within the kernel feature space. Thus, a representation within the original feature space could allow us to calculate the exact distances of the possible IBI to the decision boundary. Finding this mapping would also allow us to visualize the decision boundary. Unfortunately, this reverse mapping generally does not exist (Mika et al., 1998) as for data points living in the kernel space, a corresponding value might not exist within the original feature space (Honeine and Richard, 2011), and even if it exists, it might not be unique. The pre-image problem deals with this shortcoming and consists of finding approximate solutions for data in the input space based on their features in the kernel space (Honeine and Richard, 2011). There exist multiple approaches to solve this problem (Honeine and Richard, 2011; Kwok and Tsang, 2004; Mika et al., 1998). In some cases, an exact

pre-image can be computed. One case where an exact solution exists is for invertible kernel functions, such as the polynomial kernel with an odd degree (Honeine and Richard, 2011). In general, however, only an approximate solution for this problem exists.
We could apply some of the known techniques for the pre-image problem for an approximate solution. For our use case, a different approach might be more suitable as the output of the decision function $y(x)$ is proportional to the distance of x from the decision boundary. As our approach relies on the output of the decision function rather than the exact distances, we can still apply our method using the decision function's output instead of the distance to the decision boundary
For a more intuitive understanding and better interpretability, we will still use the distance to the decision boundary for our explanation. The linear kernel poses a special case where we can calculate the exact distance from the decision boundary. We will remember that we substitute the distance to the decision boundary with the output of the decision function.

Defining a Bound for the Distance to the Separating Boundary The threshold for the distance to the separating boundary influences how sharply we separate the space in which IBIs may occur from the clusters. A larger threshold may result in the labeling of some samples of the clusters as IBIs. Conversely, a lower threshold may lead to the omission of some IBIs and an increase in false negatives. In the following, we will refer to this threshold as $T1$.
A potential threshold for the distance to the separating boundary could be the distance between the decision boundary and the support vectors. The support vectors represent data points from the clusters that are situated close to the decision boundary within the kernel space. Consequently, orthogonal to the decision boundary, they are situated far from the cluster distribution. Points with a higher distance from the decision boundary than the support vectors will be closer to the cluster distribution in this orientation. Therefore, we assume that they are indeed points belonging to the cluster and not IBIs. However, this method has some drawbacks. The assumption that support vectors are the points furthest from the cluster center in the orientation of the other cluster is only valid in certain cases for linear kernels. For instance, a Gaussian kernel may result in a circular separating boundary around one of the clusters. Consequently, the orientation of the support vector may be arbitrary. Additionally, soft margins may also influence our interpretation of support vectors.

To address these limitations, we developed an alternative approach for setting $T1$. This methodology is based on analyzing the distribution of distances to the separating boundary of the samples belonging to a cluster. For a cluster C and separating hyperplane H, we calculate the distance of each point $x \in C$ to H. Thus, let $D = \{distance(x, H) | \forall x \in C\}$. It is assumed that since the cluster is Gaussian-distributed, the set of distances, D, is also Gaussian-distributed. When executing the experiments on the threshold calculation, we

examine if the set of distances is Gaussian distributed by evaluating Q-Q Plots and the kurtosis and skewness of the distribution.

Once the set of distances, D, has been obtained, the mean and standard deviation of D are calculated. The aim is to set a threshold to exclude data samples from within the cluster. It is hypothesized that the mean of the distances minus three times the standard deviation might be a suitable threshold. This applies the three-sigma rule of thumb as described in Section 1.3.2. However, we prioritize identifying potential IBIs over the risk of mistakenly labeling cluster samples as IBIs. We also note that the sample might still be close to the origin locations cluster at the beginning of the mixing process of isotopic signatures. Given these considerations, we will also test the performance of our approach with thresholds based on a lower multiplier of the standard deviation of D. If the objective is to predict a sample, it is possible to obtain its position relative to the separating boundary by applying the decision function of the SVM to it. Subsequently, the threshold based on the cluster on this side of the separating boundary can be applied.

Confinement based on the Cluster connecting Vector

As previously stated in Section 3.2, it is necessary to further restrict the space in which data points are labeled as IBIs. Based on the model provided by the SVM, we confined the space in which IBIs occur orthogonal to the separating boundary. This will differentiate samples belonging to one of the two clusters and IBIs. Additionally, we will label outliers located behind the clusters in the orientation of the clusters in relation to each other as non-IBIs. However, we might still mislabel outliers that are not IBIs, as seen in Figure 3.4. To identify these points, we realize the ideas presented in Section 3.2 based on a cluster connecting vector to identify the orientation in which the clusters lie to one another.
A representative for each cluster, designated as R_1 and R_2, is identified. As the assumption is made that the clusters are Gaussian distributed, the mean of each cluster is selected. In the subsequent step, a vector is calculated between the representatives and the distance of the data point x to this vector. This represents the orientation of the clusters in relation to each other. Consequently, we calculate the vector between the two cluster representatives as $v = R_1 - R_2$. Subsequently, we determine the vector $w = x - R_1$ between the data point x and one of the representatives. The following equations give the distance d to the cluster connecting vector where $\cdot$ denotes the dot product between two vectors. This distance measures how far a point deviates from both clusters in this particular orientation.

$$t = \frac{(w \cdot v)}{(v \cdot v)} \tag{3.5}$$

$$d = \|w - t * v\| \tag{3.6}$$

In terms of visual representation, we define a straight tube in which both clusters are situated. According to our IBI definition, a data sample that is not within this tube should not be the result of a linear mixing process of both clusters. Therefore, they are outliers and do

not exhibit the characteristics of an IBI as defined by our predicate. The threshold for this distance defines the girth of this tube, visualized as the dotted green lines in Figure 3.6. The larger the threshold, the more confident we are that all possible IBIs are included. Nevertheless, an increase in the threshold may also result in an increased number of false positives, as outliers may be incorrectly identified as IBIs.

We can estimate a threshold for this distance, in the following referred to as $T2$, by examining the distances D of the points of each cluster to the CCV. For a cluster C and the CCV V, we define $D = \{distance(x, V) | \forall x \in C\}$. Similar to the threshold calculation for $T1$, we assume that D is Gaussian distributed as C is expected to be Gaussian distributed. Analogous to the threshold for the distance to the separating boundary, we calculate the mean and standard deviation and apply the three-sigma rule, introduced in Section 1.3.2, for a possible threshold. This time, we use the outer boundary of the mean plus three times the standard deviation of D. We will also test different multipliers within our preparatory experiments. Applying this idea to both clusters, we get two possible thresholds. For example, $T2_A$ for the cluster A and $T2_B$ for the cluster B. We could choose the maximum of both thresholds. This approach could be seen as defining a subspace and analyzing the variance within this subspace. However, when the spreads of both clusters are highly

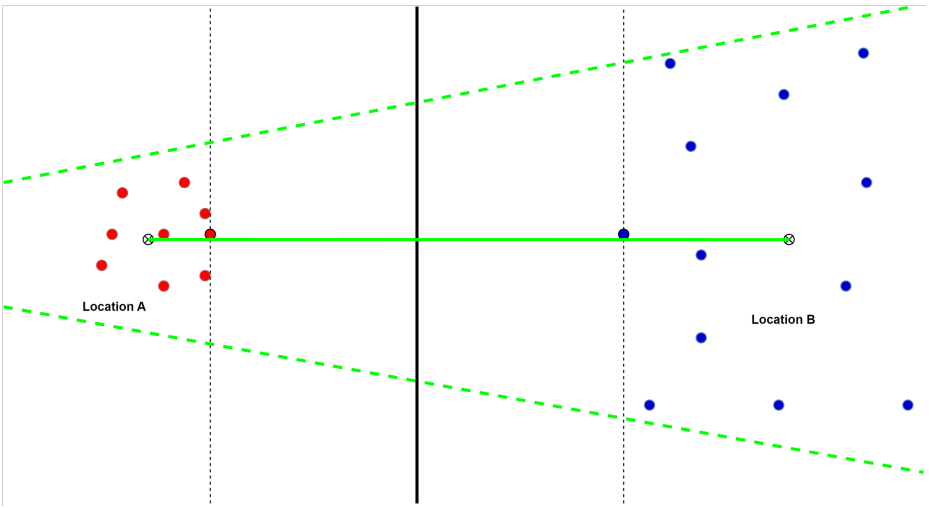


Figure 3.6: Scenario where the Spread of the Clusters differs with the adaptive Threshold as dotted green Lines

different, a more sophisticated method is required. Figure 3.6 illustrates a scenario where the spread of the clusters differs. The spread of the blue cluster is larger than the red cluster. Consequently, the threshold for the blue cluster is larger than the threshold for the red cluster. Selecting the threshold based on the blue cluster would risk including outliers as IBIs. Selecting the threshold based on the red cluster would result in the misclassification of some IBIs as outliers. Our objective is to identify every IBI, which necessitates the threshold to be based on the blue cluster. However, according to our definition, the space between the green and black dotted lines where IBIs occur adapts to the spread of the clusters and can

not be captured using a static threshold. To overcome this problem, we introduce a dynamic threshold based on the position of the sample relative to the CCV. Let us take the scenario from Figure 3.6 and a sample X for which we want to predict if it is an IBI. The threshold calculation is shown below.

$$X_A = x - center(A) \tag{3.7}$$

$$proj_C X_A = \frac{(X_A \cdot CCV)}{(C \cdot CCV)} CCV \tag{3.8}$$

$$frac = \frac{\|proj_{CCV} X_A\|}{\|CCV\|} \tag{3.9}$$

$$threshold = T_A + ((T_B - T_A) * frac) \tag{3.10}$$

We chose one cluster and calculate the vector X_A between its center and X. As long as an appropriate threshold is selected, the choice of cluster center does not affect the calculation. Afterward, we calculate the projection, $proj_C X_A$ of the vector X_A onto the CCV. We measure the distance traveled along the CCV compared to the full length of the CCV, $frac$, as the fraction of the length of the projected vector and the CCV. Lastly, we use the thresholds for clusters A and B, T_A and T_B, and calculate the new dynamic threshold. Within this threshold calculation, we also check if $frac$ is below one. If it is below one, the sample X lies between the cluster centers and might be an IBI. Thus, we set the threshold according to our calculations. If $fraq$ is larger than one, the sample X lies behind the center of the closest cluster and thus should not be labeled as IBI, and we set the threshold to minus infinity, a value that can not be satisfied within the threshold check.

Migration Time Prediction The migration of an individual from one location to another initiates a linear mixing process of the isotope signatures of both locations. Initially, the individual's isotope fingerprint closely resembles the origin location's isotope signature. Over a period of approximately 20 years, it adapts to the new location until it is no longer distinguishable from the isotope signal of the new location. The objective is to estimate the number of years ago the migration occurred. To this end, estimating the progress of the linear mixing process of the isotope signals is necessary. In order to determine the progress, we would examine the start and endpoint of this mixing process and calculate how far advanced the current position of the sample is along the vector between the start and endpoint. Although the exact start and endpoints are not obtainable, we assume the mixing process begins at the isotopic signature of the origin location and ends at the new location. Therefore, cluster representatives can be used as proxies for these points.
A similar concept was previously employed when calculating the adaptive threshold for the distance to the CCV. We defined the CCV as the vector between the cluster centers, as we assume Gaussian distributions we chose the means of both locations. Afterward, we calculated the projection of the sample to be predicted onto the CCV and the fraction of this projection and the CCV. Within this use case, the CCV serves as an example of a mixing process within a perfect model. It defines the subspace and, thus, the direction in which the

mixing process progresses.
Let us assume we identified a migrant candidate x. The origin location cluster's mean is m_o, and the mean of the location cluster in which x was buried is denoted as m_b. We calculate the vector CCV as $CCV = m_o - m_b$. We calculate the estimated years since migration as following:

$$X_{origin} = x - m_o \tag{3.11}$$

$$proj_{CVV} X_{origin} = \frac{X_{origin} \cdot CCV}{C \cdot CCV} CCV \tag{3.12}$$

$$frac = \frac{\|proj_{CVV} X_{origin}\|}{\|CCV\|} \tag{3.13}$$

$$years \quad since \quad migration = 20 * frac \tag{3.14}$$

X_{origin} is the vector between the location of the sample x and the center of its origin cluster m_o. The fraction of the projection of this vector onto the CCV, $proj_{CVV} X_{origin}$, and the full CCV describes the fraction how far the mixing process has progressed within the subspace defining the direction of an ideal mixing process with the mean of the cluster of the origin as start and the mean of the cluster in which the individual was found as the endpoint. Now we only need to rescale this fraction by 20 years, which we assume a perfect mixing process lasts until the fingerprint fully adapted to the new location, to obtain the approximation of the years that have passed since migration.

3.4 Setup for Experiments on Synthetic Data

We evaluate our approach on a synthetic dataset with multiple experiments. We aim to cover the most important parameters and use cases. In the following, we will explain the setup for the experiments on the synthetic dataset.

Dataset We generate a synthetic dataset based on the assumptions discussed in Section 3.3. We modeled our 6-dimensional dataset after the isotope signature dataset used in Gisela Grupe's research on the genesis and spread of the early Fritzens-Sanzeno culture (Grupe et al., 2020). We calculated the means of the original isotope signature dataset and approximated the standard deviation as the range of the data divided by four, assuming that nearly all samples in a Gaussian distribution fall within four standard deviations of the mean. Therefore, we assume that the range of the distribution divided by 4 is a rough estimate of the standard deviation. At this point, we do not need to know the exact value of the standard deviation, it is just a starting point. For our first experiments, we aimed for a linearly separable dataset without overlapping clusters. Thus, we reduced the standard deviation of both clusters within some dimensions and adjusted their means. We then checked for overlap and concluded that we had created a non-overlapping dataset in each dimension individually. The original and the non-overlapping adjusted datasets can be seen

Table 3.1: Mean and approximated Standard Deviation of Gaussian Distributions for the two reference Clusters of the Synthetic Dataset

Isotope Ratio	μ	σ	Adjusted μ	Adjusted σ
87Sr/86Sr	0.71537/0.71273	0.0033975/0.001375	0.7207/0.70573	0.0023975/0.001375
208Pb/204Pb	38.4853/38.8076	0.073625/0.07545	38.2853/39.0076	0.073625/0.07545
207Pb/204Pb	15.6478/15.6909	0.015925/0.023525	15.6458/15.8019	0.015925/0.023525
206Pb/204Pb	18.5347/18.6493	0.13005/0.081725	18.2447/18.9493	0.13005/0.081725
206Pb/207Pb	2.4595/2.4733	0.003775/0.0023	2.4545/2.4783	0.002775/0.0023
208Pb/207Pb	1.1845/1.1885	0.007925/0.005625	1.1645//1.2085	0.007925/0.005625

in Table 3.1. The first entry contains the mean or standard deviation for the first cluster, and the second entry contains the corresponding value for the second cluster.

We generated random samples from the Gaussian distributions with the shown means and standard deviation. We used the scikit-learn (Pedregosa et al., 2011) *make_blobs* function to generate these samples. We set the *random_state* parameter to 0 to allow reproducibility of our results. Unless stated otherwise, we generated 200 samples per cluster.
We split the dataset into a training and a test split with 75% of the samples as the training set and 25% of the samples as the test set.

Additionally, we generate a set of IBIs. We follow the predicate definition for an IBI within our use case, as defined in Section 3.1. Thus, each IBI is associated with a start and end point within the clusters' distributions. To generate the start and end points, we follow the procedure for generating our dataset, which involves generating new samples of the distributions of the reference clusters. Each point on the vector between the two aforementioned points is in accordance with the IBI definition. Twenty evenly spaced points are generated on the interval between the two points. Additionally, these data points correspond to the years since the migration occurred within the isotope analysis use case. When ordered along the axis between the start and end point, the first point corresponds to a migration that has occurred one year before, and so on. An example of the generated IBIs can be observed in Figure 3.7. The blue samples represent the reference clusters, and the samples following the line between them represent the generated IBIs. We can visually identify the color-coded linear mixing process after the migration. We also realize that the points close to the start and end of the mixing process lie firmly within the distributions of the clusters. We chose to generate the IBIs this way as it is closer to the real-world use case of isotope analysis for provenance research. We relax the condition that each IBI has to be an outlier to obtain a more realistic test set. We must understand how advanced the linear mixing process can be while our algorithm still correctly detects the IBIs. The IBIs close to the clusters make up the hard-to-predict instances and are of special interest to us. The number of IBIs we generate for each experiment depends on the amount of cluster samples in the test set. We aim to obtain a 50/50 split of positive and negative samples within our

test set.

The choice of the start and end points is disproportionately important as we generate 20 IBIs per pair of start and end points. To avoid cases where our algorithm performs particularly well or poorly due to the position of the start and end points, we execute each experiment multiple times with different seeds for generating the migrants. For most experiments, we use seeds from 1 to 100 and run each experiment 100 times. We then calculate the mean of the metrics we use to evaluate the performance of our approach.

The test set, which contains generated IBIs and cluster samples, has one significant limitation. Ideally, we would like to test our algorithm against three distinct types of samples. These three types are 1) non-outliers from both clusters underlying Gaussian distributions, 2) outliers that are IBIs, and 3) outliers that are not IBIs. Unfortunately, we are unable to generate outliers that are not IBIs and are still of interest to us. One possible approach would be to generate points and then check whether they are in the vicinity of one of the two clusters. However, it is also necessary to determine whether these points are IBIs. Currently, there is no other algorithm that can detect IBIs that meet our our definition. Consequently, our algorithm can only be used to determine whether these points are IBIs. However, evaluating our algorithm on a test set labeled with our algorithm would be useless. Another possibility would be to define a larger space where IBIs cannot occur. However, this would result in trivial instances and thus would only embellish our results. One could also argue that our test set does not strictly adhere to our definition of IBIs. When generating IBIs, we sample points on a straight line between a start and an endpoint. The first and last of these points may still be near our clusters and might not be identified as outliers. However, our IBI definition has the condition that a candidate for an IBI is an outlier. We chose this tradeoff to better model the real-world use case of isotopic signature analysis for provenance research. In this use case, the isotopic signature of individuals who migrated 15 or more years ago has aligned mostly to the new location. However, these individuals are especially interesting for us as they are the hardest to predict. Thus, we include IBIs that are close to our clusters to model these instances that are hard to predict.

Normalization Data preprocessing is a crucial step before applying a machine learning algorithm. Many of these algorithms are based on distances between data points. Having multiple dimensions with highly varied scales can lead to the problem that some dimensions have a higher impact on the total distance than others. This often does not represent the semantics behind the dataset, as a higher value does not imply higher importance and thus should not be valued stronger for the distance metric result.
Various normalization techniques exist, and we will take a closer look at some of the more widely used techniques. Our choice is based on a comparison study for multiple normalization methods for supervised classification (Raju et al., 2020).

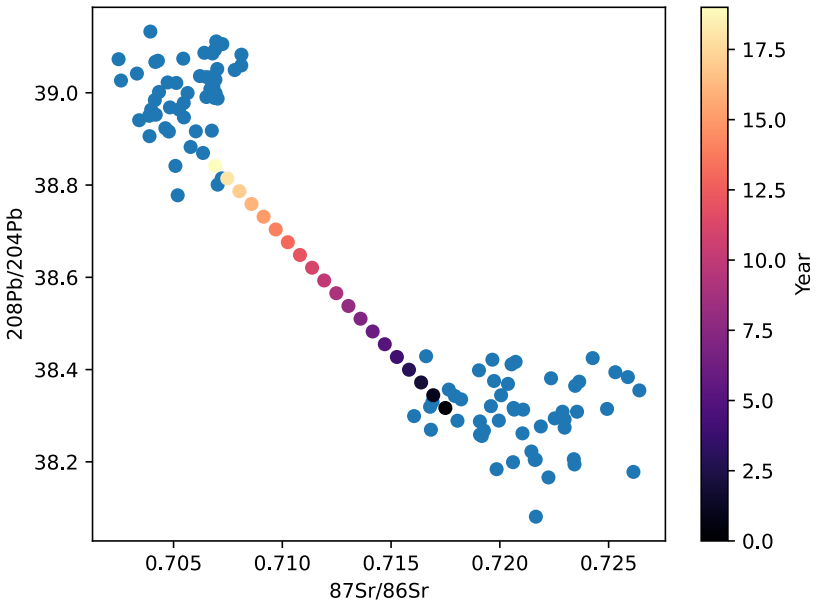


Figure 3.7: In-Between Instance Generation

The *MinMaxScaler* transforms the dataset to fit between the range of *min* and *max*. 0 and 1 are often used as *min* and *max* values. This transformation will preserve the shape of the dataset as it does not introduce distortion. Its transformation is defined as follows.

$$X_{std} = (X - X_{min})/(X_{max} - X_{min}) \tag{3.15}$$

$$X_{scaled} = X_{std} * (X_{max} - X_{min}) + X_{min} \tag{3.16}$$

$$\tag{3.17}$$

With X_{min} = minimum value in feature X and X_{max} = maximum value in feature X

The *StandardScaler* rescales the data such that the new mean is 0 and the standard deviation is 1. The transformation of the *StandardScaler* is defined as follows.

$$X_{scaled} = (X - \mu)/\sigma \tag{3.18}$$

$$\tag{3.19}$$

With μ as the mean of the feature and σ as the standard deviation of the feature.

Lastly, the *RobustScaler* applies a technique similar to the *MinMaxScaler* but tries to decrease the influence of outliers. Its transformation is defined as follows.

$$X_{scaled} = (X - X_{median}/IQR_{1,3}(X) \tag{3.20}$$

$$\tag{3.21}$$

$IQR_{1,3}$ is the interquartile range between the first (Q_1) and third (Q_3) quartile. The $IQR_{1,3}$ measures statistical dispersion. This range contains 50% of the dataset. Q_1 is defined as the median of the data's lower half, the 25th percentile, and Q_3 is the median of the upper half, the 75th percentile of the data.
The *MinMaxScaler* may compress the transformed data into a narrow range when outliers are present. The *StandardScaler* is similarly affected by outliers. Additionally, the spread of the transformed dataset on each feature might become very different. The *RobustScaler* is an adaption to deal with the presence of outliers within the dataset. Its transformation discards a small set of very large marginal outliers as it is based on the $IQR_{1,3}$. The *MinMaxScaler* does not reduce the importance of outliers. As IBIs are outliers, we try to keep the information contained in them. Thus, a normalization method that does not reduce the impact of outliers might be more suitable for our use case. When using a Gaussian kernel within the SVM, the algorithm assumes all features are centered around 0 and have variance in the same order. Additionally, SVMs initialize the weights to 0 or a small random value close to 0. Applying the *StandardScaler* and thus centering the features at mean 0 with standard deviation 1 makes it easier for the SVM to learn the weights (Raschka, 2015). However, the *StandardScaler* does distort the relative distances between feature values. Comparisons between these transformations have concluded that regardless of the choice of transformation, normalizing the dataset results in much more accurate results when applying SVMs (Raju et al., 2020). However, the *MinMaxScaler* performed the best in the scaling method comparison of Raju et al. (2020). We chose the *MinMaxScaler* for our experiments with a min value of 0 and a max value of 1. We train and apply the **MinMaxScaler** on the train set and apply the trained normalization model on the test set, including the migrants. We created a contour plot of the decision function to inspect the difference between applying an SVM model on a non-normalized and normalized dataset. Our dataset is 6-dimensional, so we can not simply plot the decision boundary within a two-dimensional plot. To overcome this problem, we created a 6-dimensional grid on our feature space and calculated the value of the decision function evaluated on each grid point. For each pair of dimensions, we collapse the other four dimensions by calculating the mean of the decision function values. Thus, we can plot 2-dimensional approximations of the structure of the decision boundary. For the following visualizations, we used a grid with a granularity of 10, resulting in 10^6 samples. The SVM has been trained with a linear kernel, and the parameter C is set to 10E10.

While this visualization cannot exactly reproduce the decision boundary position and orientation but rather approximates it, this can still help us understand the benefits of normalization. We can immediately see that the decision boundary of the SVM, trained on a normalized dataset, is located firmly between both clusters within every pair of dimensions.

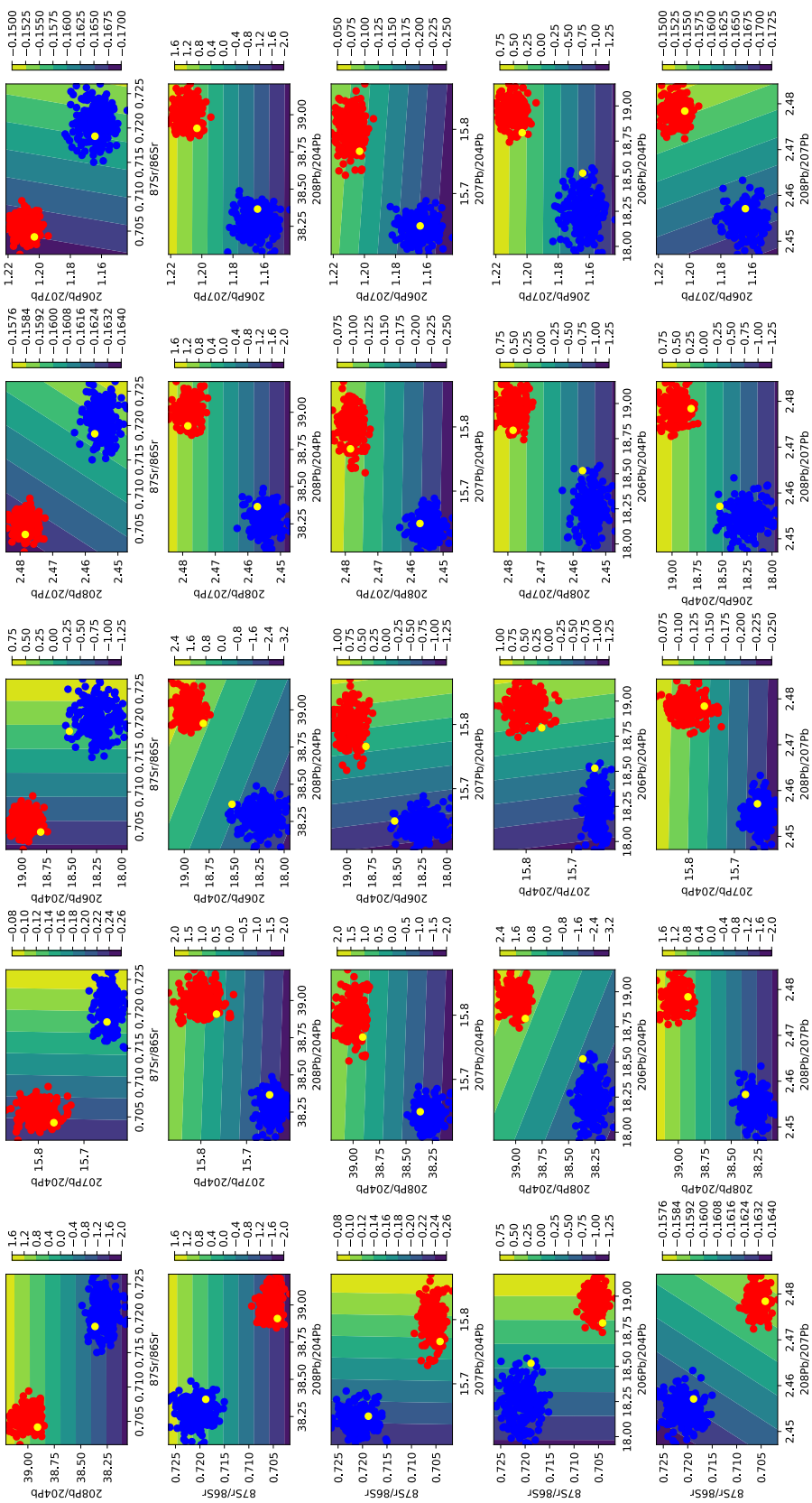


Figure 3.8: Contourplot of the Decision Function Values without Normalization

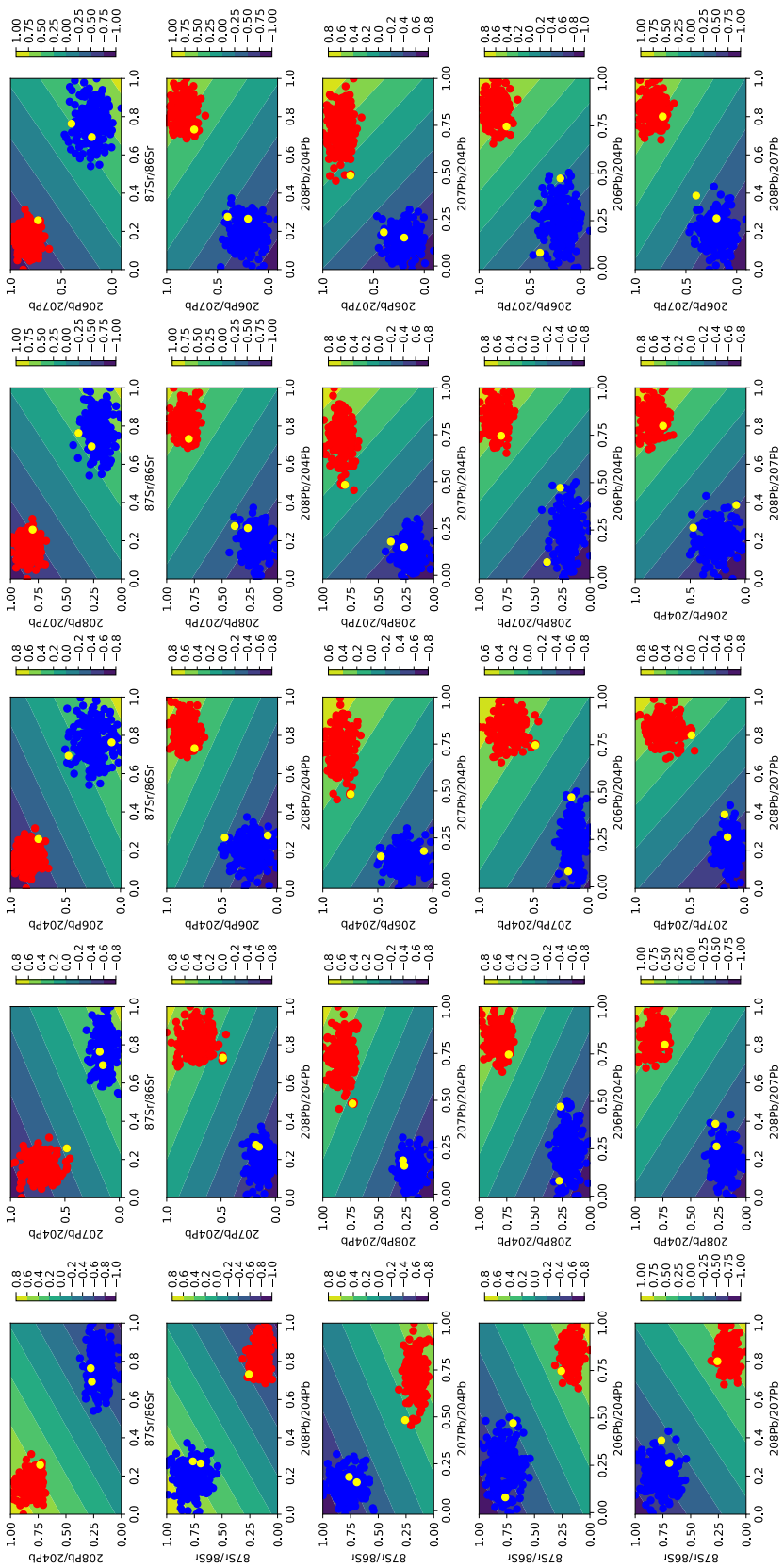


Figure 3.9: Contourplot of the Decision Function Values with Normalization

We can also see that the orientation of the decision boundary adapts to the orientation in which the clusters lie to one another. This fact is especially important for us as we use the distance to the decision boundary as the boundary for IBIs within this orientation. Although not exactly the same, the scales of the result of the decision function applied to the grid also range from -1 to 1 for every dimension pair.
Looking at the contour plot for the un-normalized dataset, Figure 3.8, we can see that the orientation of the decision boundary does not match the orientation of the clusters within each dimension pair. Examining the features with the highest absolute values, such as the combination of $^{208}Pb/^{204}Pb$ and $^{206}Pb/^{204}Pb$, reveals that the position of the decision boundary and its orientation matches with the scaled version for this dimension combination. However, in dimensional pairs such as the first shown subplot in Figure 3.8 with dimensions $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$, the decision boundary seems to be oriented horizontally without normalization and as such only takes the feature on the vertical axis into account, the $^{208}Pb/^{204}Pb$ which has higher absolute values and thus a larger difference between both clusters. Additionally, the scale of the values produced by the decision function varies highly between the dimensional pairs. In the previously mentioned dimension pair, it ranges from -3.2 to 2.4, while others may, for example, range from -0.1725 to -0.1500 for dimensions $^{208}Pb/^{207}Pb$ and $^{206}Pb/^{207}Pb$. We conclude that, as expected, the features with higher absolute values dominate the calculation of the objective function and, as such, also impact the position of the decision boundary more than features with lower absolute values. Additionally, we see that normalization solves this problem, and each feature has a similar weight after normalization.

Traning the SVM and applying the Migrant Detection After applying the normalization, we proceed to train the SVM on the train set. For this, we utilize the SVM implementation, SVC, provided by the Python library scikit-learn (Pedregosa et al., 2011), which is based on libsvm (Chang and Lin, 2011). Given the simplicity and separability of the synthetic dataset, a grid search with cross-validation is not employed, in contrast to the approach taken with the isotope dataset. In the experiments, a limited number of parameters are utilized. After obtaining the SVM model, the thresholds are calculated, the migrant detection is applied, and the test set is evaluated to determine whether each sample lies within the IBI space. The specific parameters and threshold calculation approaches are stated for each experiment.

Evaluation We evaluate the results of our experiment using four metrics as described below. We apply these metrics to the binary classification problem of deciding whether a sample is an IBI. In the following, TP is the number of true positives, which is the number of IBIs our algorithm correctly identifies as such. TN is the number of true negatives. True negatives are non-IBI, which our algorithm correctly identifies as such. FP is the number of false positives. These are the samples our algorithm labeled falsely as IBI. Lastly, FN is the

number of false negatives, the IBIs our algorithm could not detect and labeled as non-IBI. Accuracy is defined as follows.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.22}$$

The accuracy describes the fraction of correct predictions of all predictions. A higher accuracy can tell us that our predictions are truthful. Meanwhile, a lower accuracy implies more errors in our predictions.

The formula for calculating precision is:

$$Precision = \frac{TP}{TP + FP} \tag{3.23}$$

The precision, as the fraction of true positives against all positive labeled samples, is a metric to gain insight into how many positive labeled samples were IBIs. Recall is determined by:

$$Recall = \frac{TP}{TP + FN} \tag{3.24}$$

The recall provides insight into how well our model can identify positive labels. We measure the proportion of IBIs we detect out of all IBIs in the test set. A lower recall indicates that we miss more IBIs, while a higher recall indicates that our algorithm finds most IBIs.The F1-Score combines recall and precision into a single metric and is defined as its harmonic mean. Its best score is 1, and its worst score is 0.

$$F1 - Score = \frac{2 * TP}{2 * TP + FP + FN} \tag{3.25}$$

3.5 Experimental Analysis of Threshold Calculations

As described in Section 3.3, we presented multiple approaches to set the thresholds for the distances to the CCV and the separating boundary. In this experiment, we want to evaluate which approach best suits our experiments on the synthetic dataset and results in the best approximation of the migrant space. We use the dataset described in Section 3.4. We use an SVM model trained with a linear kernel and setting the parameter $C = 10E10$. This is equivalent to using a hard-margin SVM for the most realistic use cases. We use a 75-25 train-test split and obtain a model that predicts the test set perfectly. Thus, no further parameter tuning is needed. We use 400 samples in total, 200 per cluster. Thus, our test set contains 100 samples. We generate 100 migrants to obtain a test set containing 100 positive and 100 negative samples. The migrants are generated such that the progress of migration is evenly distributed. This is done by choosing five start and end points and generating 20 migrants between each start and end point combination. Since we randomly generate these five starting/ending points from the underlying Gaussian distribution of both clusters, the random seeds that influence which points are chosen affect the resulting migrants generated

disproportionately. Thus, we execute each experiment 100 times with different random seeds. To evaluate the performance and applicability of the different thresholding approaches, we conducted three sub-experiments.

Sub-experiment 1 The first sub-experiment uses static thresholds for the distance to the separating boundary and the CCV. The threshold for the distance to the separating boundary is set as the maximum of the distances from the separating boundary to the support vectors. The threshold for the distance to the CCV is set based on the distribution of distances to the CCV of each cluster. As described in Section 3.3, we calculate two thresholds, one for each cluster, as the mean of the distances plus x times the standard deviation of the distances. The multiplier x will hereafter be referred to as $T2_m$. As the threshold $T2$ is calculated for each cluster separately, the maximum of both values is selected. The values [0.1, 1, 2, 3, 4, 5] are chosen as the $T2_m$ values.

Sub-experiment 2 The second sub-experiment employs the threshold based on the distribution of distances to the separating boundary, hereafter referred to as $T1$, and applies the same calculation for $T2$ as in the first experiment. The aforementioned calculation is conducted in accordance with the methodology outlined in Section 3.3. It should be noted that the dynamic threshold adaptation is not applied to the threshold for the distance to the separating boundary. For each data point, we check which side of the separating boundary it lies, and the threshold is selected based on the cluster on that side. The value of 3 was chosen for $T2_m$, and the values [0.1, 1, 2, 3, 4, 5] were selected for the multiplier for the standard deviation of the distances to the separating boundary, hereafter referred to as $T1_m$.

Sub-experiment 3 The third and last sub-experiment calculates the thresholds using the same approach as the second sub-experiment. We also employ this experiment using the same multipliers as in the second sub-experiment. However, in this sub-experiment, we use the approach for a dynamic threshold for the distance to the CCV to calculate a new threshold based on the position of the data point to be predicted.
We additionally executed this experiment with 20 multipliers evenly spaced between 1 and 3.

Within the evaluation of these experiments, we will compare the spread of both clusters. As our dataset is multi-dimensional, the variance and standard deviation can not be applied. Instead, we will use the following multidimensional measures of dispersion. The trace of a nxn Matrix A is defined as $\sum_{i=1}^{n} a_{ii}$. The elements of the trace of the covariance matrix denote the variance of each dimension. Intuitively, a measure of spread would be given by the sum of these variances, which is the trace. The total variation of a random vector X is defined as the trace of the variance-covariance matrix. This is equivalent to the sum of the population variances. This measure of spread does not take the correlation between variables into account. Thus, if the correlation increases, the dispersion of samples might decrease while the total variation stays the same.

The determinant of the variance-covariance matrix of a d-dimensional random vector variable X is defined as its generalized variance (Wilks, 1932). The generalized variance can be

interpreted as a measure of overall multidimensional scatter (Sengupta, 2004). This measure of spread takes the correlation of the variables into account. Decreasing the correlation increases the generalized variance and vice versa.

3.6 Evaluation of the Migration Time Prediction

We conducted the following experiment to evaluate the effectiveness of our approach for estimating the number of years since migration. We created two clusters, each containing 200 samples, using the same routine as within the experiments on the synthetic dataset. Two million migrants were generated to simulate individuals who have migrated between these locations, represented by the clusters. For each generated migrant, we predicted the years that have passed since migration. We assumed that all migrants were found within cluster 0. The ground truth labels are directly derived from the generation process of the migrants. When generating the migrants, we evenly spaced 20 samples between the start and endpoints. Thus, the years that have passed since migration increase by one for each sample.

To evaluate the performance of our predictions, we calculated the Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) metrics. The MSE measures the average squared difference between the predicted years since migration and the actual years. The RMSE is defined as the root of the MSE and provides an error measure within the same units as the predicted value, in this case, years. This makes the RMSE particularly useful as it provides a more interpretable measure of prediction errors in the context of the dataset. Additionally, we grouped the migrants into bins based on the years that have passed since migrations. We evaluated whether we can identify groups such as very early (1-5 years), early (6-10 years), late (11-15 years), and very late (16-20 years) migrants.

3.7 Experimental Analysis of Dataset Changes

Following the initial experiments conducted to evaluate the optimal threshold calculation for our approach, a series of experiments were conducted to examine the performance of our approach in response to varying changes in the dataset, such as changes in density, sample size, and spread.

The following section will present the rationale behind the experiments conducted. It will also describe the experiments' objectives and the insights sought to be gained from their execution. Given the numerous parameters available for selecting experiments, we aim to choose a set of experiments that encompasses our algorithm's most common use cases. This selection is intended to give us a better understanding of how effectively our approach can identify IBIs. While it is not feasible to evaluate every possible parameter and use case within the scope of this thesis, our goal is to achieve a solid grasp of the strengths and limitations of our approach.

Unless otherwise noted, we will set up the experiments as described in Section 3.4. We will use the dynamic threshold adaption for the threshold of the distance to the CCV with a multiplier of 3. For the confinement based on the SVM model, we will use the threshold based on the distance distribution of the clusters to the separating boundary with a multiplier of 2. The choice of the thresholds and the multipliers for the standard deviation is motivated and founded by the results of the experimental analysis of threshold calculations, as described in Section 3.5. Unless otherwise stated, we use a linear kernel and set the parameter C to 10E10 for the training of the SVM.

3.7.1 First Experiment: The Impact of the Cluster Sample Size

This experiment tests the impact of the sample size per cluster on the performance of our approach. This will allow us to understand whether the clusters' density impacts the performance. Our approach, e.g., when calculating the thresholds, is based on the assumption that the reference clusters are Gaussian, so the number of samples might impact how well the distributions can be estimated and captured by our algorithm and the SVM. As previously stated in Section 1.3.2, the sample size markedly influences the capacity to estimate distributions.

Machine learning algorithms based on probability measures or the Law of Large Numbers (Révész, 2014) tend to need a large sample size to obtain desirable results (Meng and Zhao, 2015). The Law of Large Numbers describes the relation between sample size and its mean to the true value. It states that the average of the results obtained from a large number of independent and identical random samples converges to the true value if it exists (Dekking, 2005). However, this only holds true for a sufficiently large sample size. However, the model of the SVM is only based on the position of a subset of samples, the support vector, and thus inherently simple and not dependent on a large number of samples for training. Thus, we expect this aspect of our approach to be independent of the sample size. This experiment will evaluate whether this also holds for our threshold calculation based on an approximated distribution where the Law of Large Numbers could be applied. We executed this experiment using our standard configuration without further adjustments. We generated datasets containing the sample sizes 50, 100, 200, 400, 1000, 2000, 20000, 50000, 100000, 150000, 200000, 300000, and 500000. Each reference cluster contained half the samples before splitting into 75% train and 25% test sets. As we generate migrants as sets of 20 samples per start and end point pair, we set the number of migrants as the multiple of 20, which is closest to the number of cluster samples in the test set. Thus, we obtain a test set with close to 50% positive and negative labels.

3.7.2 Second Experiment: The Impact of Unbalanced Cluster Sample Sizes

Within this experiment, we will evaluate the impact of unbalanced cluster sample sizes on the performance of our approach. Many real-life use cases of classification, such as spam mail detection (Wang et al., 2021), face the problem of unbalanced classes. Unbalanced datasets

can cause the problem of low accuracy for the minority class when applying classification algorithms (Wang et al., 2021). In our case, a higher sample size for a cluster also generated a more dense cluster. Thus, this experiment also compares differences in the densities of the reference clusters. We executed this experiment using our standard configuration without further adjustments. We set the number of samples in the first cluster as 200 while keeping the number in the second cluster flexible. We executed this experiment with sample sizes 25, 50, 100, 200, 500, 1000, 10000, 25000, 50000, 75000, 100000, 150000, and 250000 for the second cluster.

3.7.3 Third Experiment: The Impact of the Migration Progress

Since we model our definition of an IBI after the use case of detecting migrants in an isotope dataset, we are interested in the effect of the progress of the isotope signature remodeling process that occurs after migration. Directly after the migration occurred, the signal has not yet been completely remodeled and is similar to the origin location. Afterward, due to bone remodeling, the signal is mixed with the signal of the new location. After approximately 20 years, the rebuilding has concluded, and the isotope signal of the migrant has fully aligned to the new location. The signals of migrants who migrated 15 years or longer before burial are especially of particular interest to us, as they are the hardest to find. We generate the IBIs by generating 20 points between a start and an endpoint. These 20 points can be interpreted as years that have passed since migration. As we do not use a label, such as the burial location, the idea of years that have passed since the migration, when applied to the synthetic IBIs, is symmetrical when applying our migrant detection. Thus, the first and last samples generated between the start and endpoint represent the difficult-to-predict migrants who migrated 15 or more years before burial, visualized in Figure 3.1. This does not apply to the later use of our algorithm on the real isotope dataset. Within this dataset, a burial label is provided, allowing us to adjust the thresholds and detect early migrants more confidently. This experiment will label the generated IBIs with years according to their position between the start and endpoint. We will evaluate the performance of our approach on migrants from groups of years and each year individually. The groups of years we used are 1 to 4, 5 to 8, 9 to 12, 13 to 16, and 17 to 20. As the distance threshold to the separating boundary is essential for detecting instances close to the cluster, corresponding to IBIs from the first and last years, we calculate this threshold using a multiplier of 1.5 and 2 in this experiment.

As we evaluate the performance of our algorithm on samples that should be located close to the reference clusters, which are the first and last years, we additionally calculate the Mahalanobis distance of each sample to its nearest cluster. The Mahalanobis distance is a distance measure that provides insight into the similarity between multivariate samples and distributions. This distance measure is particularly useful because it accounts for the variance and covariance of data distributions. It provides a way to determine the distance between a point and a distribution, not just another point. It calculates the distance of a test point from the mean of the distribution, normalized by the distribution's covariance. This normalization is vital as it ensures that the scale of the measurements and the correlations

between variables are appropriately considered, making this distance measure scale-invariant and rotation-invariant. Scale invariance refers to the property that the measurement scale of variables does not affect the distance computation. Rotation invariance means the computed distance between points remains constant even if the coordinate system is rotated.

The Mahalanobis distance D from a point x to a distribution with mean μ and covariance matrix Σ is defined as:

$$D(x) = \sqrt{(x-\mu)^\top \Sigma^{-1}(x-\mu)} \tag{3.26}$$

3.7.4 Fourth Experiment: The Impact of Spread Differences between Clusters

Differences in the spread of both clusters have multiple effects on our approach. On the one hand, the migrant space can no longer be approximated using a static threshold for the distance to the CCV. This threshold needs to be tapered and adjusted to the position of the predicted sample in the orientation in which the clusters lie to one another. On the other hand, the clusters may begin to overlap in one or multiple dimensions. Depending on the degree of the overlap, the space in which IBIs may occur can collapse, and the concept of IBIs, as defined by our predicate, may no longer be applicable. In this experiment, we will evaluate how well our approach can adapt to clusters with different spreads that may be overlapping. We will use the synthetic dataset and increase the standard deviation to generate the Gaussian distributed samples, using an equal multiplier for all dimensions. We used multipliers between 1 and 3 with a stepsize of 0.1. We executed these experiments with a linear and a Gaussian kernel. For the parameter C, we used 1 and 10E10 as values.

3.8 Identifying Migrant Candidates within Archaeological Isotope Datasets

3.8.1 Dataset Description

We applied our algorithm to identify migrant candidates on two multi-isotope datasets as described below.

Fritzens-Sanzeno Culture The first dataset contains the isotope fingerprints of 92 human skeletons from the Fritzens-Sanzeno culture, dating back to the late Hallstatt period. Most of these individuals were cremated. The dataset was published within the work of Grupe et al. (2020). Each multi-isotope fingerprint consists of $^{87}Sr/^{86}Sr$, $^{208}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, $^{206}Pb/^{204}Pb$, $^{208}Pb/^{207}Pb$, and $^{206}Pb/^{207}Pb$. The individuals stem from burial sites along the Inn-Eisack-Adige passage and can be grouped into two areas. The inner Alpine sites contain the locations Wörgl (n=11) and Kundl (n=48). The southern Tyrolean sites contain the locations Moritzing (n=11), Pfatten (n=4), Latsch (n=16), and Brixen (n=2). Their geographical location can be seen in Figure 3.10. We apply the SVM to separate these two

geographical regions based on the full 6-dimensional multi-isotope fingerprints. The full Fritzens-Sanzeno dataset is contained within the appendix in Table 1.



Figure 3.10: Map of Fritzens-Sanzeno Archaeological Sites (Grupe et al., 2020)

Extended Isotope Dataset In addition to the Fritzens-Sanzeno culture dataset, we also used a second, larger dataset. This dataset contained the isotope ratios of 301 individuals from areas in southern Germany and along the Inn-Eisack-Adige passage. The Fritzens-Sanzeno culture dataset is contained as a subset within this extended isotope dataset. The full extended dataset is contained within the appendix in Table 2. As seen in Table 3.2, we

Table 3.2: Burial Site Groups of the extended Dataset

Region	Samples	Sites
Southern Germany (1)	146	22
Inneralpine (2)	122	17
Southern Tyrolean (3)	33	4

separated this dataset into three distinct groups. The southern German burial sites contain 146 samples, the Inneralpine burial sites contain 122 samples, and the southern Tyrolean burial sites contain 33 samples. Parts of this dataset have been published within Grupe et al. (1997), Toncala et al. (2017), and Grupe et al. (2020).

3.8.2 Applying the Migrant Detection on the Archaeological Isotope Dataset

When applying our approach to the isotope dataset, we face new challenges. This dataset consists of many different burial sites, and each individual burial site may be very sparse due to small sample sizes. Furthermore, the dataset is not as easily separable due to considerable overlap. We expect an unknown number of migrants within this dataset. There is no labeling for migrants known, and the proportion of migrants is unknown beforehand.

Thus, the choice of the SVM parameters is non-trivial. On the one hand, we might operate based on our knowledge about the parameters' influence. We might, for example, seek to obtain a smooth decision surface to reduce the impact of the migrant samples and set the parameters accordingly. On the other hand, we can operate purely data-driven and search for the parameters that best separate the dataset based on the burial location. Another possible idea would be to generate synthetic migrants between the locations. We could select a model that best predicts these migrants.

To overcome these challenges, we propose a three-step approach. We chose a purely data-driven approach using multiple models to predict migrants with threshold parameters to create a ranking of samples we expect to be migrants.

Model Selection The choice of parameters for the training of the SVM is non-trivial. To decide which models to use to predict the migrant labels, we conducted parameter tuning using an exhaustive grid search. We then trained the SVM model with different parameters to predict the burial site group.

Cross-validation is used to evaluate the generalizability and prevent the overfitting of machine learning models. It is a data resampling method and belongs to the family of Monte Carlo methods (Berrar and others, 2019). Even with complex patterns, training an SVM model that captures these patterns is often relatively easy as arbitrary complex decision boundaries can be introduced. However, when applying this model to unseen samples, this complex model might not generalize well and predict these unseen samples. Thus, the model is overfitted. To avoid this, the model should be evaluated on a dataset different from the set we used to train the model. Thus, the dataset is split into train and test sets. Cross-validation goes one step further and uses multiple train and test sets. In k-fold cross-validation, k disjoint train sets are generated from the original full dataset. The different combinations of k-1 train sets and 1 test set are called folds.
In each step, we use a single fold to train the SVM on k-1 train sets and validate the model on the remaining set. The validation step calculates a score, in our case, the F1-Score, based on the model's predictions.
Afterward, this step is repeated for every fold, and an average validation performance is calculated across all k folds. We apply a stratified random sampling to select the training sets. A stratified random sampling creates test sets in which the proportions within each individual subset reflect the proportions within the full dataset (Berrar and others, 2019).

Table 3.3: Parameter Grid for Cross-Validation

Parameters	Kernel	Min	Max	Type	Steps	Scale
C	linear	1^{-2}	1^{10}	Real	100 (20)	logarithmic
gamma	rbf, polynomial	1^{-9}	1^{4}	Real	100 (20)	logarithmic
degree	polynomial	2	4	Integer	3	linear

Syarif et al. (2016) conducted research on the parameter optimization of SVMs. A grid search with logarithmic scales for the parameters C and gamma with ten steps between 0.001 and 10000 found almost near optimal parameter combinations. The drawback of this approach is the high number of parameter combinations and, thus, models to be trained. This leads to high training times, which we also observed. We aimed to use a parameter grid with the finest resolution while completing the cross-validation within a reasonable time. The chosen grid can be seen within Table 3.3.

We use a logarithmic scale for the parameter C ranging from 1^{-2} to 1^{10} with 100 steps. Similarly, a logarithmic scale is applied for the gamma parameter, ranging from 1^{-9} to 1^{4}. For the polynomial kernel, we used degrees 2,3 and 4. Parameter C is applied across all kernels, and gamma is used only for the rbf and polynomial kernels. The degree parameter is only relevant to the polynomial kernel. Additionally, we must reduce the number of steps for the C and gamma parameters to 20 for the polynomial kernel due to excessively long runtimes. Moreover, larger gamma values result in increasingly high feature values within the kernel space, which can lead to overflowing errors. The cross-validation is executed with 5 stratified folds, and the F1-Score is used to evaluate the performance of each model. Before fitting the model, we apply a MinMaxScaler to rescale the dataset to range from 0 to 1.

Migrant Detection After selecting the ten best models, we apply the migrant detection to the full dataset. We separately retrain the model with the parameters of the ten best models from the parameter search separately. We set the multiplier for the distance threshold from the CCV to 3. We use values from 0 to 3 on a linear scale with 30 steps, resulting in a step size of 0.1 for the multiplier for the threshold of the distance to the separating boundary. Thus, we apply the algorithm 30 times for different parameters on the full dataset for ten models separately. In total, we obtain 300 labels for each sample. This allows us to measure the certainty of our prediction. We expect samples that already get labeled as migrant candidates at a low $T1_m$ to be more confident. As we increase the multiplier, the width of the IBI space increases, and the IBI space grows. Thus, a sample labeled as a migrant candidate for $T1_m = x$ will also be labeled a migrant candidate for every $T1_m = y$ with $y > x$. Samples that get labeled early, with a small $T1_m$, are placed more firmly in the middle of the IBI space. They are the samples closest to the separating boundary. Under the assumption that the separating boundary separates the underlying distributions of the isotopic fingerprints of the locations, they are the most dissimilar from both locations while still in-between.

In this use case we have the additional information of a burial label. This label marks the burial site of the human but does not guarantee that this human did not migrate to the place of death beforehand. We can use this label to increase the migrant space, particularly the threshold for the distance to the separating boundary. Let us assume that the sample in question originates from the burial site designated as x, which is associated with cluster 0. If the sample is located on the side of the separating boundary where cluster 0 lies, we proceed as usual. The threshold should be used to determine whether the isotopic fingerprint of this sample belongs to the reference cluster 0. However, if the sample lies on the side of cluster 1, we apply a different threshold since, based on the burial site label, it does not belong to this cluster. The semantic behind the threshold changes from discerning between samples from this reference cluster and IBIs to discerning from outliers and IBIs. If the sample is situated within the cluster, it can be assumed to be an IBI that has migrated to cluster 0 in recent years. Conversely, if the sample is situated behind the cluster, it can be presumed to be an outlier or to have migrated from an unknown location. Consequently, the aforementioned approach is employed, whereby a multiple of the standard deviation is added to the threshold calculation for $T1$ instead of subtracting it. This results in a boundary shift either towards, within, or beyond the cluster.
It is crucial to highlight that this approach is not applicable to our visualization of the IBI space. When calculating predictions for a grid of points within our feature space, applying the concept of a burial label is not feasible. Consequently, this aspect is not represented within the current visualization of the IBI space.

Prediction Ranking Applying the migrant detection on all ten models, as described in Section 3.8.2, yields 300 labels per sample. For each of the optimal ten parameter settings identified during parameter tuning, we obtain 30 labels for the distinct settings of $T2_m$. We commence by examining the outcomes of each model separately and calculating for each sample at which $T2_m$ value the initial positive labeling occurred, which we refer to as $T2_m^{min}$. For samples with a small $T2_m^{min}$, we are most confident that this sample is the isotopic fingerprint of a mixture of both locations' isotopic signatures, as explained in Section 3.8.2. Thus, they are the migrant candidates we are most confident in. Samples are then ranked per model based on $T2_m^{min}$. Lower $T2_m^{min}$ achieve a higher rank. To fuse the predictions of all models, we calculate the mean of the ranks across all ten models per sample. This yields a ranking across all models, with samples with a higher rank representing those in which we are most confident in identifying a genuine migrant.

4 Results and Discussion

4.1 Results and Discussion of the Experimental Analysis of Threshold Calculations

The following section presents the results of the experimental analysis of the different threshold calculation approaches. The SVM parameters and dataset we used are described within Sections 3.5 and 3.4.

Results of the Experiments on the threshold calculation The results of the first sub-experiment can be seen in Table 4.1. This experiment employs the static threshold approaches for both the distance to the CCV and the distance to the separating boundary, as described in Section 3.5. The threshold to the separating boundary is set based on the distance to the support vectors. We group the results by $T2_m$ and calculate the mean for all seeds. The first column shows the multiplier for the threshold calculation of $T2$. Columns 2-5 consist of the metrics we obtain when predicting the test set. A multiplier of 0.1 exhibits the lowest scores. Increasing the multiplier yields increasingly better results until a multiplier of 3.0 is used. Further increases yield exactly the same labelings and scores. The best F1-Score of 0.857626 is obtained using the multipliers 3.0, 4.0, and 5.0. Excluding the F1-Score, precision is the highest metric for all experiments, followed by accuracy and recall.

In Table 4.2, we see the results of the second sub-experiment. This experiment uses the static threshold approaches for both the distance to the CCV and the distance to the separating boundary, as described in Section 3.5. The threshold to the separating boundary is set based on the distance distribution to the separating boundary. Again, we group the results by the multiplier, in this case $T1_m$, and calculate the mean for all seeds. The first column shows the multiplier for the threshold calculation of $T1$. Columns 2-5 consist of the metrics we obtain when predicting the test set. We obtain the lowest F1-Score, of 0.551207,

Table 4.1: Results for Sub-Experiment 1

$T2_m$	Accuracy	Recall	Precision	F1-Score
0.1	0.8309	0.6718	0.985163	0.796847
1.0	0.8675	0.7450	0.986731	0.848625
2.0	0.8734	0.7568	0.986945	0.856458
3.0	0.8743	0.7586	0.986976	0.857626
4.0	0.8743	0.7586	0.986976	0.857626
5.0	0.8743	0.7586	0.986976	0.857626

Table 4.2: Results for Sub-Experiment 2

$T1_m$	Accuracy	Recall	Precision	F1-Score
0.1	0.80725	0.9545	0.737292	0.831907
1.0	0.89840	0.8768	0.916334	0.895986
2.0	0.87500	0.7600	0.986999	0.858518
3.0	0.81735	0.6347	1.000000	0.776327
4.0	0.75490	0.5098	1.000000	0.675139
5.0	0.69030	0.3806	1.000000	0.551207

Table 4.3: Results for Sub-Experiment 3

$T1_m$	Accuracy	Recall	Precision	F1-Score
0.1	0.75445	0.9789	0.675585	0.799420
1.0	0.86935	0.9287	0.830091	0.876543
2.0	0.90930	0.8386	0.976683	0.902163
3.0	0.85970	0.7294	0.986463	0.838479
4.0	0.81005	0.6201	1.000000	0.765321
5.0	0.75265	0.5053	1.000000	0.671164

with a multiplier of 5.0, and the highest F1-Score, of 0.895986, with a multiplier of 1.0. We can see that the F1-Score decreases as we increase the multiplier beyond 1.0. The same holds true for accuracy and recall. Recall already decreases when the multiplier is increased to 1, while precision has a contrary trend and increases steadily as the multiplier increases until we reach the perfect precision score of 1.

In Table 4.3, we present the results of the third sub-experiment. The adaptive threshold calculation is applied within this experiment, as described in Section 3.5. The table's structure is analogous to the table depicting the results of the second sub-experiment. In this sub-experiment, we obtained the best F1-Score with a multiplier of 2.0 and the worst F1-Score with a multiplier of 5.0. The general trends are similar to those in the second sub-experiment. Again, the recall decreases steadily as we increase the multiplier. The accuracy reaches its peak at a multiplier of 2.0 and decreases thereafter. The precision increases with a higher multiplier and reaches the perfect score of 1.0 at a multiplier of 4.0. In the second sub-experiment, it reached this point earlier at a multiplier of 3.0. We also observe that this experiment yields the highest F1-Score out of the three sub-experiments, 0.902163. The second sub-experiment reaches an F1-Score of 0.895986, and the first sub-experiment reaches an F1-Score of 0.857626.

Figure 4.1 illustrates the results of the third sub-experiment with 20 multiplier values between 1.0 and 3.0. With a higher multiplier, the precision increases, and the recall decreases. The accuracy and F1-Score increase early on before reaching their highest value at a multiplier of 1.842105, with values of 0.91660 and 0.910759, respectively. Afterwards, they begin to decrease slowly. At a Multiplier of 1.5, all four metrics intersect.

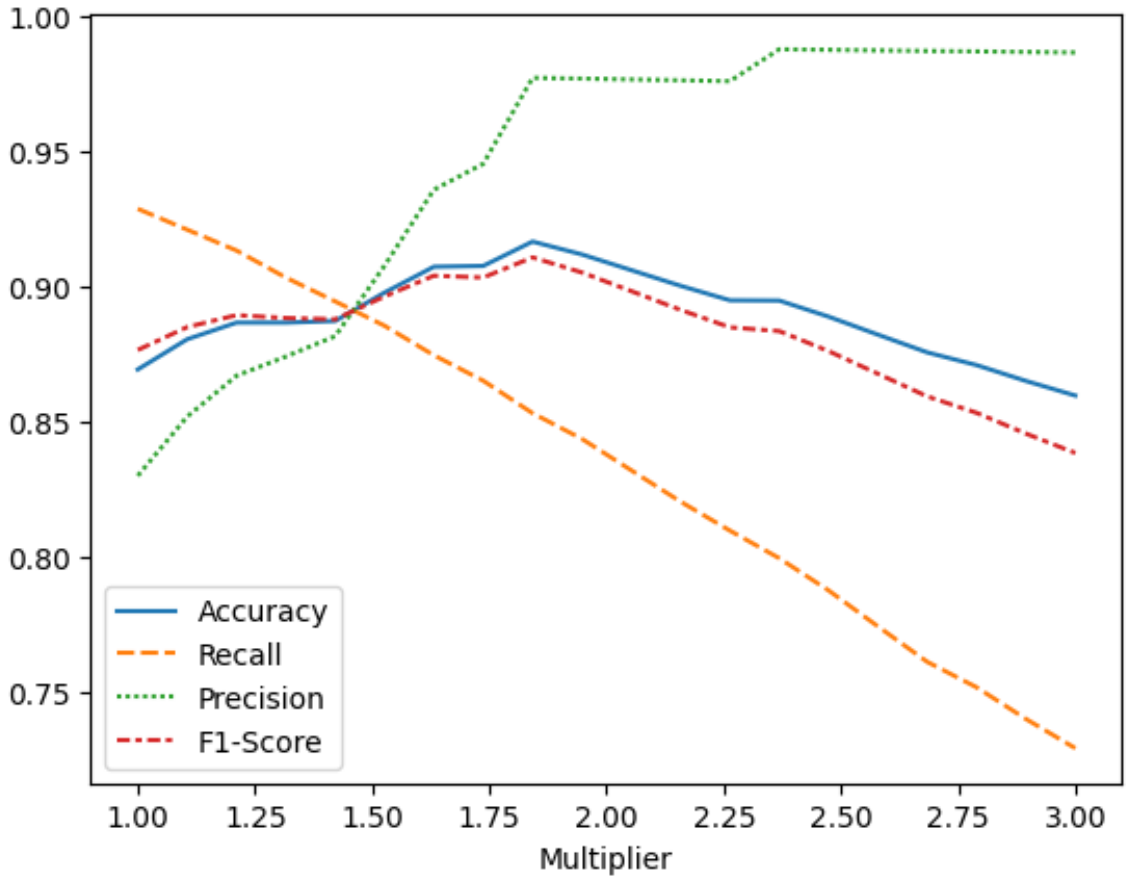


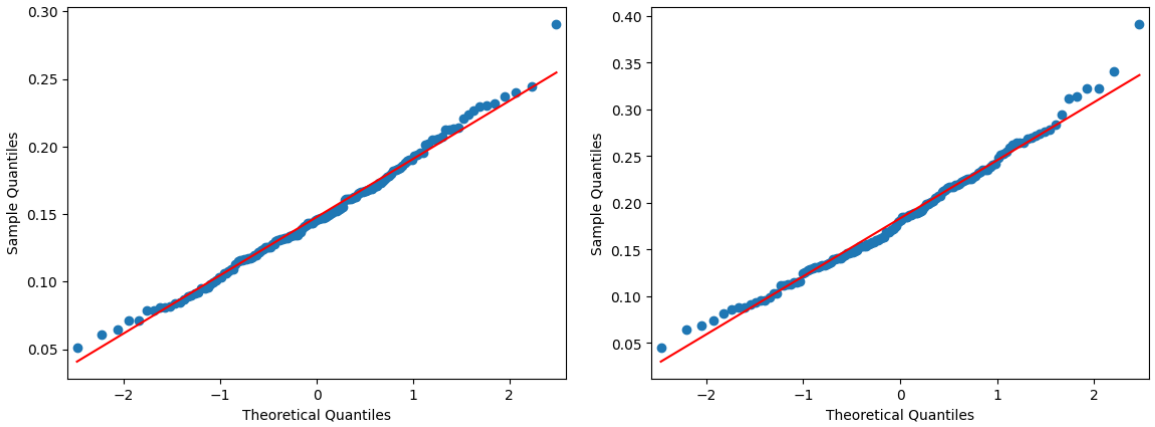
Figure 4.1: Results of the third Sub-Experiment Fine Tuning

Discussion of the Experimental Analysis of Threshold Calculations In the following, we will discuss the results of the experimental analysis of threshold calculations.

Overall Performance We achieved desirable results with all three configurations. The best F1-Score results for the three sub-experiments are 0.857626 (at $T2_m$ values of 3.0, 4.0, and 5.0), 0.895986 (at a $T1_m$ value of 1.0), and 0.902163 (at a $T1_m$ value of 2.9) for the first, second, and third sub-experiments. While all show that our model performs well on the synthetic dataset, we observe that the adaptations to the threshold calculations further increase our test results. The modification of the calculation of $T1$ from being based on the distance to the support vectors to being based on the distribution of distances to the separating boundary resulted in the greatest performance enhancement, with an increase of approximately 0.05 on the F1-Score. While a larger multiplier for the decision boundary distance threshold decreases the recall, it also increases the precision. A larger multiplier decreases the area where we label data points as IBIs as it changes the boundary at the cluster edges and moves it closer to the decision boundary. Thus, a larger multiplier leads to fewer cluster samples being labeled as IBIs, while a smaller multiplier leads to more IBIs correctly identifying at the edge of the clusters. The trend of the recall and precision of the second and third sub-experiments support this theory. A higher number of true positives, for example, through finding more IBIs at the cluster edges, increases the recall, and a lower number of false positives in relation to the true positives would increase the precision, for example, when labeling fewer cluster samples as IBIs. Overall, our approach has a high precision of around 0.9-1.0. This indicates that the positive instances, which we have designated as IBI, are, in the majority of cases, correctly identified as IBIs and not mislabeled cluster samples. This increases our confidence in the positively labeled instances. Thus, the boundary of our migrant space in the orientation in which the clusters lie to one another, which is based on the distance to the separating boundary, successfully differentiates cluster samples from IBIs. The recall of our model, albeit lower, seems satisfactory. Values between 0.8 and 0.9 seem obtainable using a multiplier between 1 and 2 without compromising too much on precision. Our model accuracy for all parameters ranges between 0.69 and 0.90. Only very high (4 or 5) or low (0.1) multipliers produce accuracies lower than 0.8. Our model's accuracy seems robust against most realistic threshold changes, while recall and precision seem more susceptible to smaller changes in the threshold calculation.

Gaussian Distance Distributions Except for the threshold to the separating boundary in the first sub-experiment, all thresholds are based on distributions of distances. As our dataset consists of Gaussian distributions, we assumed that the resulting projections, when calculating the distances to the CCV or the separating boundary, are of Gaussian nature, too. In the following, we will examine the distances we calculated and evaluate if this assumption holds true.

Several methods exist to investigate whether a distribution is Gaussian. Graphical methods, such as histograms, stem-and-leaf plots, box-and-whisker plots, P-P plots, and Q-Q plots, are often used to identify Gaussian distributions visually (Das and Imon, 2016). They either graphically compare the observed distribution with a theoretical normal distribution or highlight characteristics of the observed distribution, such as the bell shape. We will take a closer look at quantile-quantile plots, also called Q-Q plots. The quantiles of the observed distribution are compared against the quantiles of a standardized theoretical distribution of the specified family of distributions (Das and Imon, 2016). In our case, we compare the quantiles of the distance distributions against those of a Gaussian distribution. Quantiles are used to partition the range of a probability distribution into continuous intervals with equal probabilities. For example, the special case of quartiles partition the range of the probability distribution into four groups of equal probability (25%). If the quartiles of the observed distribution follow the line that describes the quartiles of the theoretical distribution, it is assumed that the specified family reasonably describes the data distribution (Das and Imon, 2016).



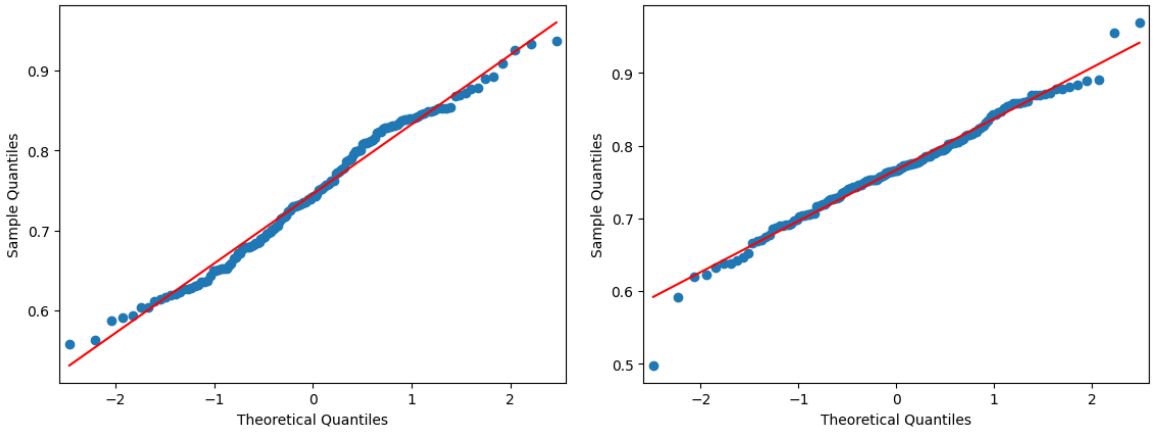
(a) Cluster 0

(b) Cluster 1

Figure 4.2: Q-Q Plot of Distances to CCV from the Threshold Calculation Experiments for Cluster 0 and Cluster 1 separately

The Q-Q plots of the distances to the CCV for cluster 0 and cluster 1 separately can be seen in Figure 4.2. The dots mark the quantiles of the distribution of distances to the CCV compared against a Gaussian distribution, visualized as the red line. We can observe that the distribution follows the red line. The middle part is very close to the Gaussian distribution, while smaller outliers are in the outer parts. Smaller outliers within the outer parts are not unexpected for Gaussian distributions. Overall, the distributions appear to be Gaussian based on the visual inspection of the Q-Q plots.
Another method to identify whether a distribution is Gaussian is based on descriptive measures. Skewness and kurtosis are, in practice, widely used to identify Gaussian distributions even though there is no general rule defining the values that indicate normality (Hatem

et al., 2022). Skewness describes how asymmetrical a distribution is (Hatem et al., 2022). A distribution is skewed if, visually speaking, it stretches towards the left or right. A perfectly symmetrical distribution has a skewness of 0, and a Gaussian distribution should possess a skewness of close to 0. On the other hand, Kurtosis compares the observed distribution to the theoretical normal distribution (Hatem et al., 2022). Distributions with a high kurtosis tend to have heavy tails or outliers, while a low or negative kurtosis indicates lighter tails. The kurtosis of a Gaussian distribution should be close to 3, or in case of excess kurtosis, where three is subtracted, close to 0. Additionally, we compute the skewness and kurtosis of both distributions. The distances to the CCV for cluster 0 have a kurtosis of 3.0095 and a skewness of 0.3209. For cluster 1, a kurtosis of 3.1128 and a skewness of 0.4186 were computed. These values fall well within the range that a Gaussian distribution would produce.
This reaffirms our assumption that the distances to the CCV are Gaussian.



(a) Cluster 0

(b) Cluster 1

Figure 4.3: Q-Q Plot of Distances to separating Boundary

For the distances to the separating boundary, the Q-Q plots are shown within Figure 4.3. The distribution of both clusters follows the line representing the theoretical Gaussian distribution. We can, however, observe that the distributions deviate more than the distributions of the distance to the CCV, see Figure 4.2. Again, we observe higher deviations at the ends and, in the case of cluster 0, also some deviations along the entire line. We would still interpret this as sufficiently close for both distributions to appear Gaussian. The skewness of the distribution of the distances to the separating boundary for cluster 0 is -0.0096, and its kurtosis is 2.143. For cluster 0, the skewness is -0.2414, and the kurtosis is 3.973. These results are harder to interpret because no general rules define the values that indicate normality (Hatem et al., 2022). While the kurtosis and skewness of the distances to the CCV are very close to the kurtosis and skewness of a theoretical Gaussian distribution, the values for the distances to the separating boundaries deviate more. However, both

deviate less than 1 from the optimal values. Thus, we expect these values to fall within the range of a Gaussian distribution with a small random sample size.

Dynamic Threshold Calculation The third sub-experiment utilizes our method of a dynamically adapting threshold for $T1$. We develop this method to deal with clusters with different spreads. In this scenario, a static threshold fails to capture the linear mixture process after the migrant, which is the basis of our IBI definition. We can see a slight performance increase within the results of this experiment compared to the second sub-experiment, which uses the same setup with a static threshold. We see a slight shift in which multiplier produces the best result. For the second sub-experiment, the multiplier 1.0 produces an F1-Score of 0.896, while the multiplier 2.0 produces the best F1-Score of 0.902 using a dynamic threshold. However, the differences between the best scores are on such a small scale that we cannot infer that the dynamic threshold performs better in other use cases. This might be due to the nature of our dataset, as the spread of both clusters within our dataset is similar. When looking at measures of spread for multi-dimensional data, we observe that both clusters have quite similar spread. The trace of the first cluster is 0.0224, while that of the second cluster is 0.01257. The determinant of the first cluster is 4.74E-23, while that of the second cluster is 4.69E-24. These low values indicate that both clusters are quite compact. While a difference is measurable, the total difference is still low. It can be reasonably assumed that the lack of improvement in performance observed with the adaptive threshold approach is a consequence of this experiment not fully exploiting its advantage.
In addition, the statistical measures from which the thresholds are calculated are as follows. The mean distance to the CCV for cluster 0 is 0.183419, while for cluster 1 it is 0.147887. The standard deviation of these differences is 0.062181 and 0.0429952, respectively. Resulting in $T2 = 0.369962$ for cluster 0 and $T2 = 0.276872$. The threshold for cluster 0 is 33.622% larger than the threshold for cluster 1. The adaptive threshold is designed to address this discrepancy more effectively than the static threshold, which is based on the maximum value of both. However, the primary disadvantage of simply selecting the maximum is that it may include outliers that are not IBIs. As previously discussed within Section 3.4, the primary limitation of our test set is that it lacks this type of sample. Therefore, we conclude that our experiment fails to accurately reflect the advantages of the dynamic threshold. Nevertheless, the results were not inferior, and as we anticipate that the dynamic threshold will yield superior outcomes when applied to a dataset comprising clusters with disparate spreads, we will continue to employ it.

4.2 Results and Discussion of the Evaluation of the Migrant Time Prediction

In the following, we will present and discuss the results of the evaluation of the migrant time prediction, as described in Section 3.6. Our experiment's MSE is 0.996, and the RMSE is 0.998. These results indicate a high level of accuracy in our predictions. The low values of the

Table 4.4: Results of the Migration Date Prediction Experiment with Bins of Years

Years	F1-Score
1-5	0.941354
6-10	0.944733
11-15	0.952256
16-20	0.936873

MSE and RMSE suggest that our approach effectively captures the isotopic mixing process. With an RMSE of close to one, we predict the years since migration with an estimated error of one year.

The results of our experiment with the migrants grouped within bins of years are shown in Table 4.4. These results demonstrate that our algorithm has performed consistently well across different time intervals since migration. The F1-Scores are all relatively high, indicating strong precision and recall in identifying migrants within each time bin. While slightly lower, even the bins of very early and very late migrants produce satisfying F1-Scores. Thus, the prediction does not seem to be strongly affected by the extremes of migration progress.

The experiment demonstrates the effectiveness of our approach in estimating the time since migration based on isotopic signatures. It is important to note that this experiment utilized two reference clusters, simplifying cluster centers' approximation compared to real-world scenarios.

4.3 Results and Discussion of Experimental Analysis of Dataset Changes

In the following, we will present the results of our experiments to evaluate the impact of dataset changes.

4.3.1 First Experiment: The Impact of the Sample Size per Cluster

Figure 4.4 shows the first experiment's results. Within this experiment, we evaluated the impact of different sample sizes on the performance of our approach, as described in Section 3.7.1. We observe satisfactory performance across all configurations. The highest F1-Score of 0.91 was obtained at 100 samples per cluster, and the lowest F1-Score of 0.85 at 25. The highest F1-Score coincides with the highest precision of 1.0. The difference in performance with sample sizes larger than 100 is very low, with a difference of 0.017 between the highest and lowest F1-Score. Overall, the precision scores are the highest out of all metrics, followed by the accuracy and F1-Score, which are close to each other, while the recall is substantially lower. Very low sample sizes, 25 and 50, produce worse results than larger ones. Generally, we see higher fluctuation at lower sample sizes and consistent performance with a sample

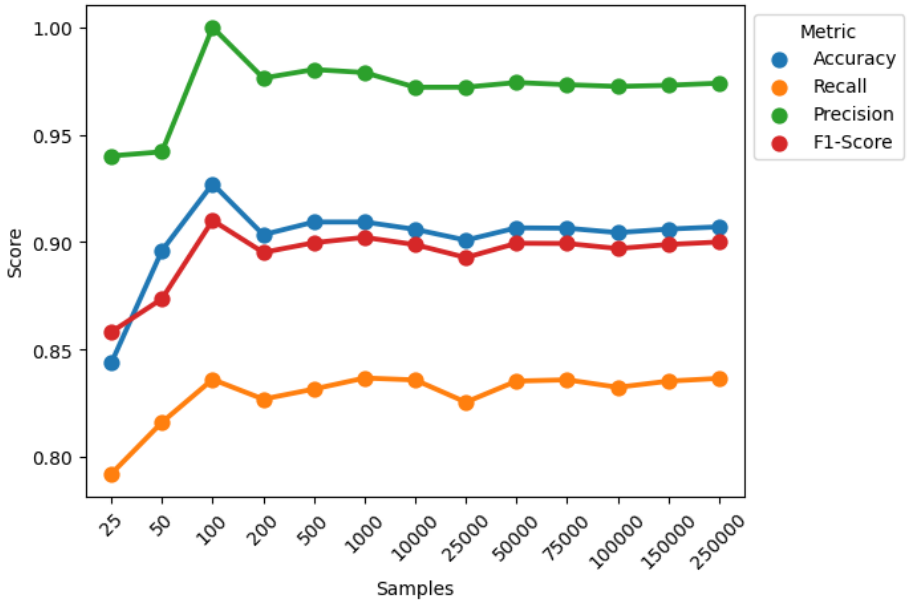


Figure 4.4: Results of the Experiment Sample Size per Cluster with the x-axis showing the Sample Size for both Clusters

size of at least 200. It is noteworthy that the x-axis of this plot is non-linear. Thus, the fluctuation of all four metrics at the beginning happens within the span of sample sizes 25-200, while the larger range of sample sizes between 500 and 250000 produces stable results.

Discussion As presented within this section, our algorithm performs satisfactorily across all sample sizes. We observe that the performance of our model is increasingly stable with higher sample sizes. It is important to note that we did not use multiple seeds for the data generation, contrary to the generation of the IBIs. Thus, the outlying performance of our method at 100 samples for cluster 0 might be an outlier where the dataset generation produced two clusters that are especially well suited for our algorithm. One potential explanation for the phenomenon of unstable results with low sample sizes is that the approximation of Gaussian distributions may not be adequately reliable with such small sample sizes, impacting the efficacy of our threshold calculation. However, the application of GMM models to isotope datasets comprising approximately 25 to 50 samples has demonstrated that even modest sample sizes can yield satisfactory train-test errors (Göhring et al., 2016, 2018; Zacherl et al., 2023). Given that we trained the Support Vector Machine (SVM) using C=10E10, a setting expected to produce a low number of support vectors, it is unlikely that the SVM training itself was adversely affected by the smaller sample sizes. Instead, the diminished performance at smaller sample sizes may be attributable to how the model provided by the SVM is utilized within our approach. The approximation of the distribution of distances to the separating boundary for threshold $T1$ may be more susceptible to instability, potentially leading to the emergence of a wider range of thresholds for smaller sample sizes. This could

result in fluctuating metrics.
In summary, while very small sample sizes can degrade the performance of our method, sizes of approximately 100, or possibly 200, if the result from the 100 sample setting is considered an outlier, yield better and more consistent outcomes. This setting might produce two reference clusters whose shape allows for a better prediction of in-between instances. Even though we execute the experiments multiple times with different seeds for the IBI generation, the reference cluster generation uses a static seed, which could be the reason for this outlier in performance. The performance of our algorithm appears to be largely independent of the sample size, with the exception of very small sample sizes. Furthermore, all metrics show good or very good results and our algorithm seems to be able to identify most IBIs while producing a very low number of false positives. Larger sample sizes, beginning at a sample size of 10000, lead to more stable results.

The contour plot in Figure 4.5 visualizes an approximation of the detected IBI space for a sample size of 200, calculated as described in Section 3.4. Our algorithm successfully identifies the space between the two reference clusters, where IBIs are intuitively expected to occur. This observation holds true across all dimensions, not just individual pairs of dimensions. Although this is only an approximation, it indicates that our approach effectively captures the IBI space within the full-dimensional dataset.

4.3.2 Second Experiment: The Impact of Unbalanced Cluster Sample Sizes

The results of this experiment are shown in Figure 4.6. This experiment evaluates the impact of unbalanced classes and density differences, as described in Section 3.7.2. Cluster 0 contains 200 samples for every configuration, and the sample size of cluster 1 is adjusted accordingly. The accuracy for all configurations is between 0.90 and 0.91. Similarly, albeit a bit lower, the F1-Scores range between 0.89 and 0.91. As already observed in the first experiment, recall is generally lower than the other metrics at 0.83 to 0.85. Thus indicating a higher false negative number. The highest scoring metric is precision at 0.94 to 0.98. Precision also shows the highest differences for different configurations, with a maximum difference of 0.04 between the configuration with 25 samples for cluster 1 and the configuration with 1000 samples for cluster 1.

Discussion From these results, it is evident that our approach remains robust against unbalanced datasets. Notably, there is no observed degradation in algorithm performance with increasingly unbalanced sample sizes per class. In fact, the model demonstrates enhanced stability, particularly with larger sample sizes in cluster 1. Precision and accuracy exhibit minimal fluctuations when sample sizes exceed 10,000 for cluster 0 and stay close to constant at a high value for the rest of the experiment. This performance is comparable to that observed with similar sample sizes in a balanced setting.

Furthermore, smaller sample sizes consistently result in more variable outcomes, even when the classes are increasingly unbalanced as the sample size grows, indicating that our model's performance is more stable with larger datasets. This experiment also provides

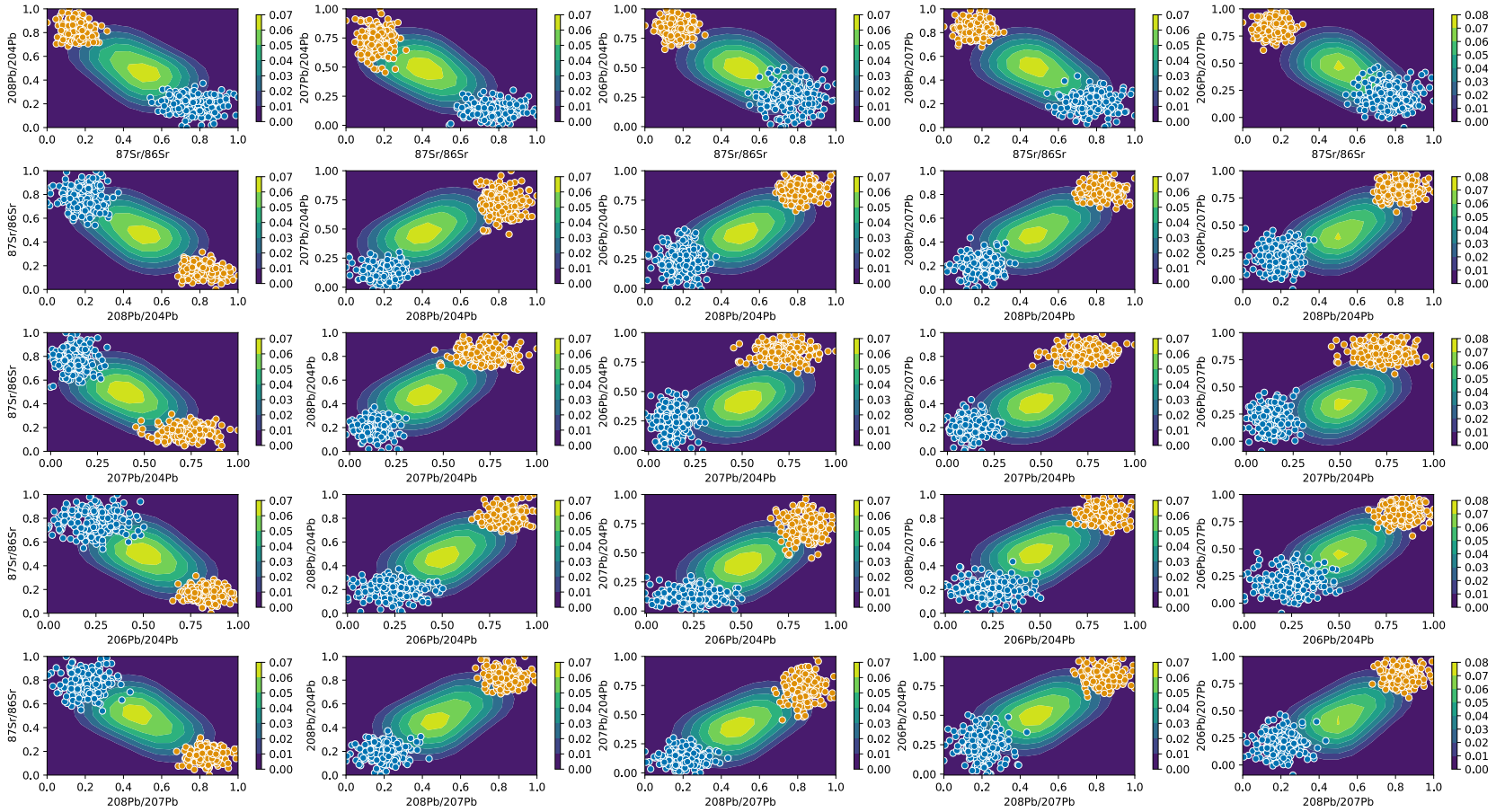


Figure 4.5: Contourplot visualizing the detected IBI Space on the Synthetic Dataset for a Sample Size of 200 with a Linear Kernel and the SVM Parameter C=10E10

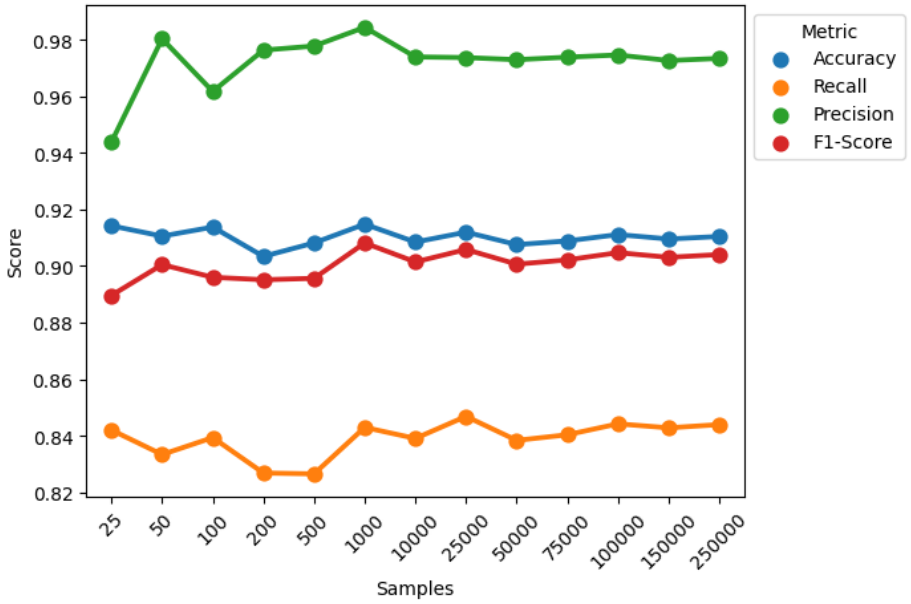


Figure 4.6: Results of the Experiment Unbalanced Clusters with the x-axis showing the Sample Size for Cluster 1 and Cluster 0 containing 200 Samples

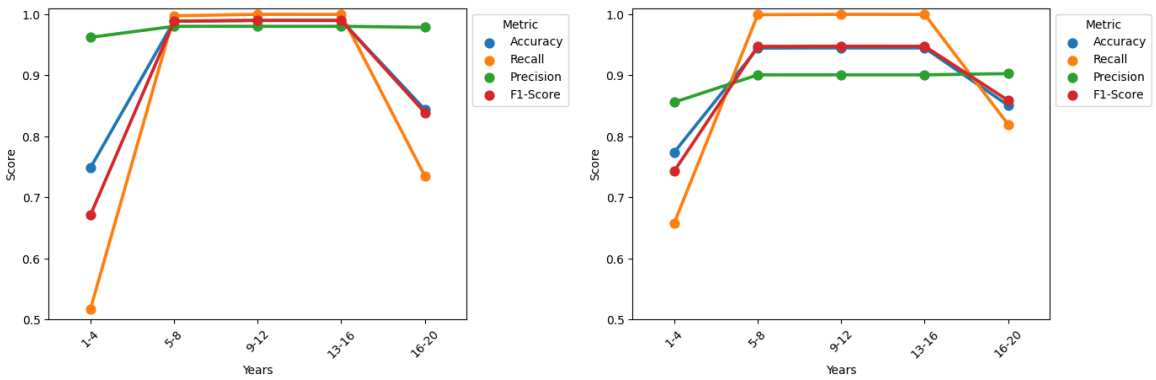
valuable insights into our model's capability to handle variations in the density of reference clusters. Since there is no observed performance degradation trend with increasing class imbalance, we conclude that our approach is well-suited for datasets with imbalanced classes. We expect this to be the reason, as we calculate the thresholds $T1$ and $T2$ for both clusters separately. Thus, our threshold calculation is unaware of possible unbalanced classes and not influenced by them.

4.3.3 Third Experiment: The Impact of the Migration Progress

Groups of Years The graphs shown within Figure 4.7 present the results of our experiment on the impact of the migration process. Within this experiment, we evaluated if our algorithm can identify the IBI instances that are close to the reference clusters, as described in Section 3.7.3.

For the configuration *T1=2*, we observe a distinct increase of all metrics except precision, which is already very high for all years, at the 5-8 years bracket. The performance then reaches a peak until it declines at the 16-20 years bracket, albeit slower than the increase at 5-8 years. The lowest score was obtained at the 1-4 years bracket. Detached from this trend is the precision score, which starts and remains high over the full course of the experiment. It is noteworthy that the recall reaches a perfect score of 1.0 at the 9-12 bracket and 0.99 at the bordering brackets, while the precision remains stable at around 0.98 for all brackets, except for a slightly lower precision of 0.96 in the 1-4 years bracket.

The configuration *T1=1.5* yields similar results as seen on the right subplot within Figure 4.7. Again, we see a distinct increase in the performance, followed by stable good results



(a) T1 = 2 (b) T1 = 1.5

Figure 4.7: Results of the Experiment on the Impact of the Migration Progress by Groups of four Years

which end in a decrease at the 16-20 year bracket. We observe less variability in the scores. The recall ranged from 0.65 to 1.0, while the recall for the configuration *T1=2* ranged between 0.51 and 1.0. The precision increases slightly when moving from the 1-4 years to the 5-8 years bracket but overall stays stable at around 0.98.

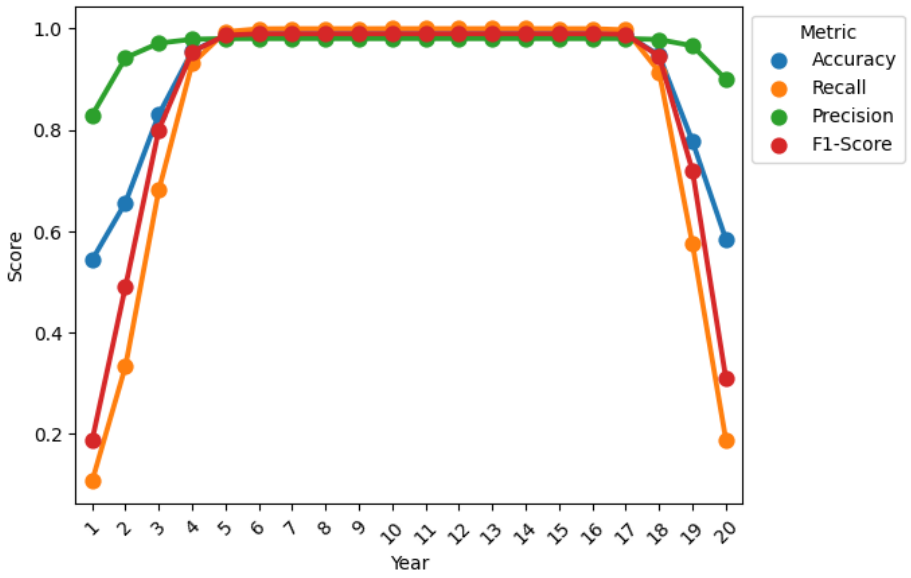


Figure 4.8: Results of the Experiment on the Impact of the Migration Progress for every Year individually

Per Year Figure 4.8 depicts the experiment results with every year predicted separately, giving us a more detailed look at the performance of our approach when predicting migrants who migrated different years ago. We observe a distinct increase in every metric for the

first 5 years, followed by stable and high scores until the 17th year is reached, where the trend reverses, and the scores begin to decrease sharply, which is comparable with the results shown in Figure 4.7. The recall metric exhibits the greatest degree of variability across the evaluated period. It ranges from 0.1 for year one to 1.0 for years 10 to 14. The F1-Score ranges between 0.18 (year 1) and 0.99 (years 5-17), and the accuracy ranges between 0.54 (year 1) and 0.99 (years 5-17). The lowest variability can be observed for the precision, which ranges only between 0.83 (year 1) and 0.98 (years 4-18). Thus also having the lowest score ceiling.

Furthermore, an additional experiment was conducted with the parameter C=1. The objective was to assess the impact of a smoother decision boundary on the performance of our approach within this experiment. The results of this execution were found to be highly consistent with those of the previous experiment, as seen in Figure 4.9.

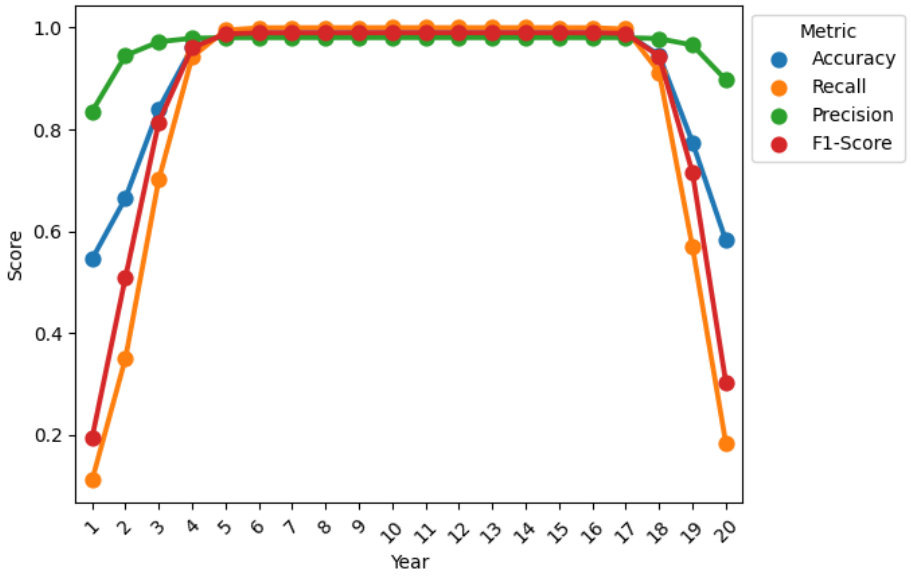


Figure 4.9: Results of the Experiment on the Impact of the Migration Progress for every Year individually with the SVM Parameter C=1

Discussion The results of this experiment confirm our assumption that migrants whose isotopic fingerprints have already largely adapted to the new location are the hardest to identify. As the synthetic dataset is symmetric and does not use a label, such as the burial site, to identify the location where the individual is found, this also holds true for the first years after migration. This is shown through low recall and accuracy at the earliest and latest years. As the negative samples of our test set consist of samples from both clusters, the ability to predict them is not impacted, and the precision is consistently high. The slight decrease in precision in the first and last few years can be attributed to a lower total number of positive predicted instances due to a decreased ability to detect IBIs in these

years. Resulting in a higher impact of the constant number of false positives within the precision calculation, which can be seen in Equation 3.23.
Increasing the IBI space by decreasing the multiplier for $T1$ allows us to correctly identify more IBIs at an earlier stage by increasing the IBI space towards the reference clusters. However, this comes with the drawback of reduced precision. The mean precision for the configuration with the multiplier for $T1$ set to 1.5 is 0.892, while it is at 0.976 for the configuration where the multiplier for T2 is set to 2. When observing the performance of our model to predict samples for each year individually, we recognize the inability to detect IBIs in the first, second, and last years, as the recall scores of these years are 0.106, 0.333, and 0.188. Years 3 and 19 form the transition where the model begins to correctly identify IBIs. The model's ability to identify IBIs from the years 6-17 seems to be close to perfect, with scores over 0.99.

We suspect that the samples from the first and last years are too similar to the closest reference cluster and thus cannot reliably be separated without falsely labeling cluster samples as IBIs. To test this hypothesis, we compare the Mahalanobis distance of the generated IBIs to the distribution of the reference clusters and the Mahalanobis distance of the cluster samples to the distribution of the reference clusters.

We calculate the Mahalanobis distance of each generated migrant per year to both reference clusters and select the lower distance as seen in Figure 4.10. Additionally, we calculate the Mahalanobis distance for each cluster sample to its own cluster. The mean Mahalanobis distances of the cluster samples 0 and 1 are 6.658 and 6.639, respectively, and the standard deviations are 4.0780 and 3.818 for the first and second clusters, respectively. The mean distances plus three times the standard deviation would result in 18.891821 and 18.09295.

In Figure 4.10, we note that during the initial three years and the final two years, the Mahalanobis distances do not exceed the mean plus three standard deviations of the distances of the cluster samples. Particularly, the data points from the first two years and the last year show Mahalanobis distances below 10, indicating a high degree of similarity to the distribution of the reference location. This suggests that these points might be too similar to the reference distribution to be effectively distinguished as IBI without also labeling points from the reference clusters as IBI.

Although we can observe the symmetric trend that the first and last years are the hardest to predict, we can see that this constraint is harder for the first years. We measure a recall below 0.9 for the first years and only for the last 2 years. Examining the Mahalanobis distances, as shown in Figure 4.10, we can see that the largest distances are found for year 12 instead of 10, as would be the case for a symmetric dataset. In addition, the distances for the first years are smaller than those for the last years. The greater the distance to the nearest reference cluster, the easier it is for our algorithm to recognize the sample as an IBI. We expect this asymmetry in the distance to the reference cluster to be the reason for the asymmetry in our predictions. In Section 4.1, we computed the multidimensional measures of dispersion for the reference clusters 0 and 1. We saw that the trace of the clusters is

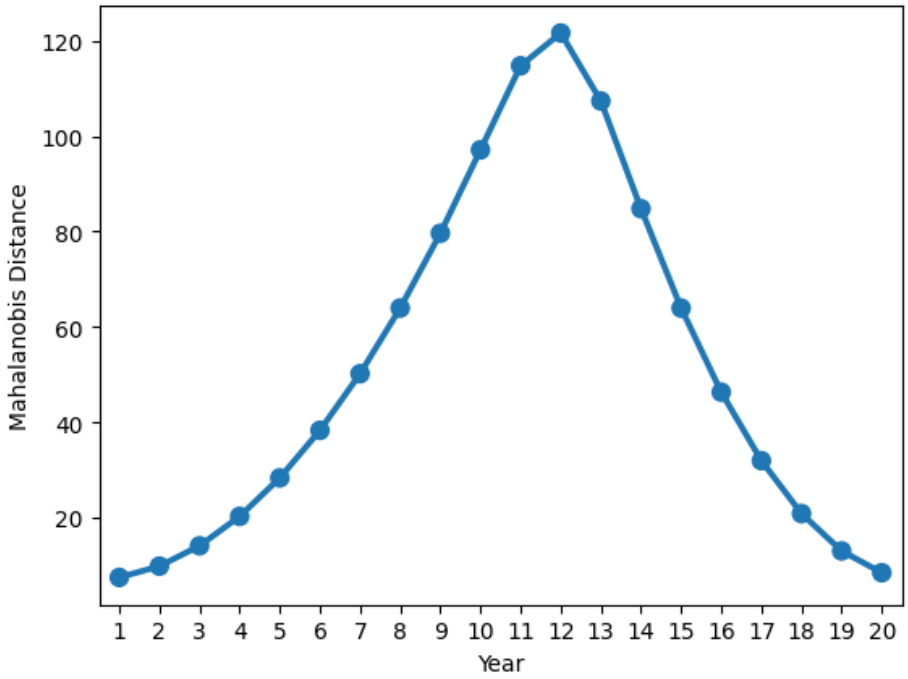


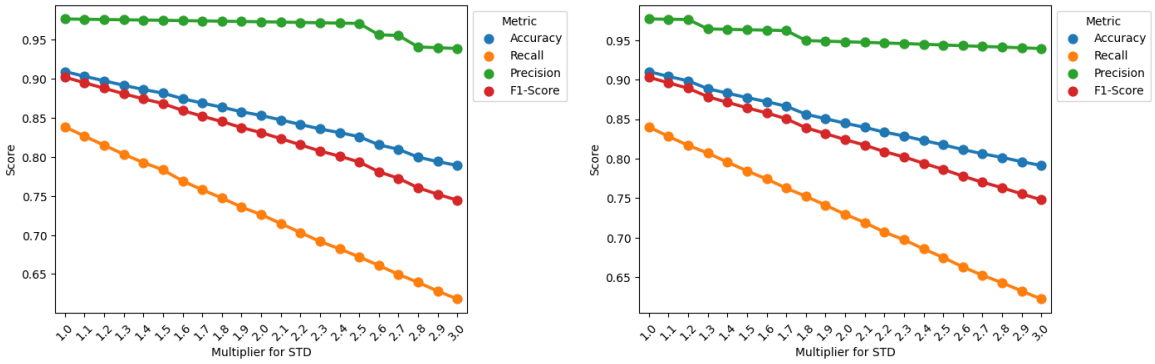
Figure 4.10: Mahalanobis Distance of generated IBIs to closest Cluster

0.0224 and 0.01257 for clusters 0 and 1, respectively. The determinant of the first cluster is 4.74E-23, while that of the second cluster is 4.69E-24. Thus, cluster 0, the cluster closest to the generated IBIs of the early years, has a larger spread than cluster 1 due to different standard deviations in the cluster generation process. The larger spread of cluster 0 could lead to smaller distances between the IBIs from the early years and cluster 0, making these cases more difficult to detect. In the following discussion of the results of the experiment on the effect of spread differences between the reference clusters, see Section 4.3.4, we will observe how this phenomenon affects our ability to detect IBIs.

The results of this experiment confirm that the multiplier of the standard deviation for the threshold to the separating hyperplane influences our ability to detect late and early migrants. It serves as a tradeoff between possibly mislabelling samples from the reference cluster as IBIs and detecting IBIs close to the underlying distribution of the reference cluster. Thus, decreasing this parameter increases the sensitivity of our model but decreases the precision. Notably, within a real-world scenario, we would have additional information about the individual's burial place. Thus, only individuals that migrated close to 20 years ago would be hard to predict. The results of this experiment demonstrate that our approach is capable of reliably capturing IBIs resulting from a migration that occurred up to 18 years ago.

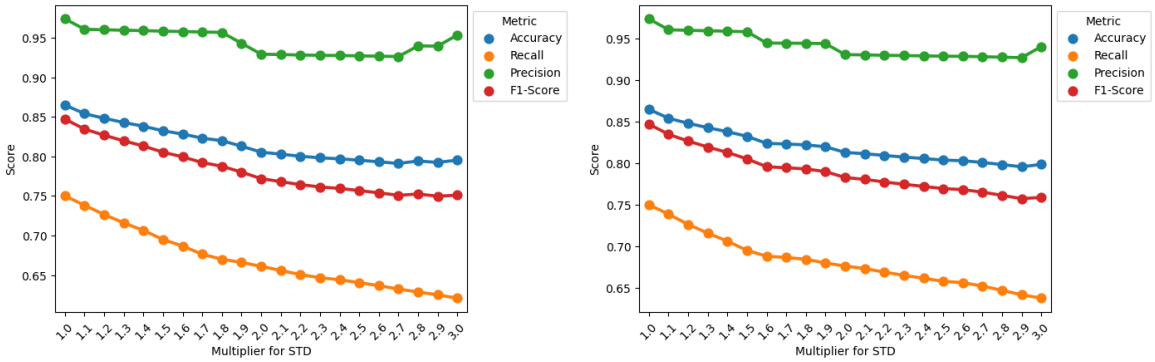
4.3.4 Fourth Experiment: The Impact of Spread Differences between Clusters

In the following section, we will present the results and discussion of our algorithm when applied to a dataset with differences in spread between both clusters. The x-axis in Figures 4.11 and 4.12 shows the multiplier we use to increase the standard deviation (STD) and, thus, the spread of the first cluster. The y-axis contains the scores of our metrics.



(a) C=10E10 (b) C=1

Figure 4.11: Results of the Experiment Spread Differences and Overlap with a Linear Kernel and the Parameter C=10E10 in Subplot (a) and Parameter C=1 in Subplot (b)



(a) C=10E10 (b) C = 1

Figure 4.12: Results of the Experiment Spread Differences and Overlap with an rbf Kernel and the Parameter C=10E10 in Subplot (a) and Parameter C=1 in Subplot (b)

In Figure 4.11, we can observe how our algorithm performs if applied with a linear kernel. For most metrics, the performance is not dependent on the choice of C, and the general trends of all metrics are independent of the choice of C. The accuracy starts just below 0.95 and steadily decreases as the multiplier for STD increases. It shows a fairly consistent downward trend throughout the experiment's range. The recall also decreases with the increase in the multiplier for STD, starting just above 0.85 and continuing to decrease to just above 0.65. It has the steepest decline among all the metrics. The precision begins

around 0.95, and like the other metrics, it decreases as the multiplier increases. However, it declines slightly less steeply than recall and accuracy, maintaining a relatively higher score even at higher multiplier values. The precision decline begins later for the harder margin, C=10E10, but both settings produce the same general trend. Consequently, the F1-Score begins around 0.90 and follows a similar downward trajectory, ending slightly above 0.75 at the highest multiplier values.

Figure 4.12 shows how our algorithm performs if applied with an rbf kernel. Precision remains relatively stable in both configurations compared to other metrics, with configuration C=1 maintaining a slightly higher and more stable score throughout the multiplier range. There seems to be a slight increase in the precision at the highest multiplier values. Accuracy and F1-Score exhibit a downward trend in each configuration as the multiplier increases. However, the starting values and rates of decrease vary, with C=10E10 starting slightly higher but decreasing at a steeper rate. The recall shows a considerable and steady decline in both settings. The recall decreases at the steepest rate of all metrics.

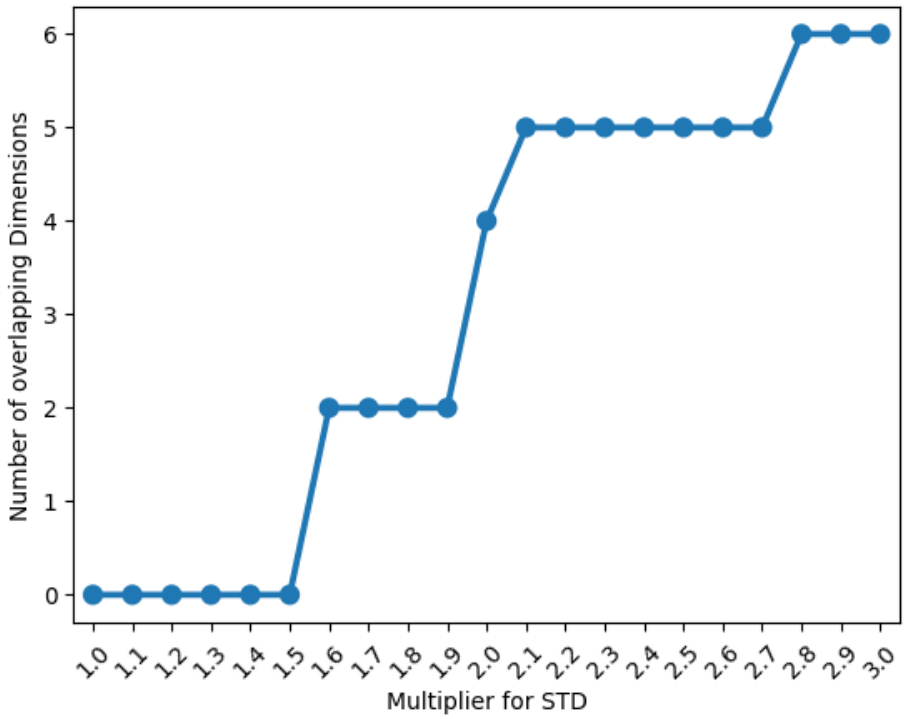


Figure 4.13: Bivariate Plot of the Multiplier for the Standard Deviation and the Number of Overlapping Dimensions

We calculated the number of overlapping dimensions for the different multipliers of the standard deviation, as presented in Figure 4.13. This was done by calculating the range of each cluster per dimension. Afterward, we checked if the range of the clusters overlapped within this specific dimension. For each multiplier, we calculated how many dimensions the clusters overlap. We see that at a multiplier of 1.6, the clusters begin to overlap in at least

two dimensions. The clusters overlap in every dimension at the highest multiplier values of 2.8, 2.9, and 3.0.

Even though the dataset overlaps within multiple dimensions, the SVM separated the classes without misclassification on the test set for all configurations.

Discussion All four metrics in all four configurations show a negative correlation with an increased difference in spread. This suggests that increasing the spread within one cluster leads to a consistent degradation in the algorithm's performance across all measured metrics, as seen in Figures 4.11 and 4.12. The metrics start from relatively high values, indicating good performance at lower multiplier values, but performance deteriorates as the multiplier increases. The steepest decrease is observed for the recall, which might indicate that the model becomes progressively less capable of detecting IBIs as the spread difference increases.

In Figure 4.14, we visualize the clusters for a Multiplier for the STD of three. This bivariate

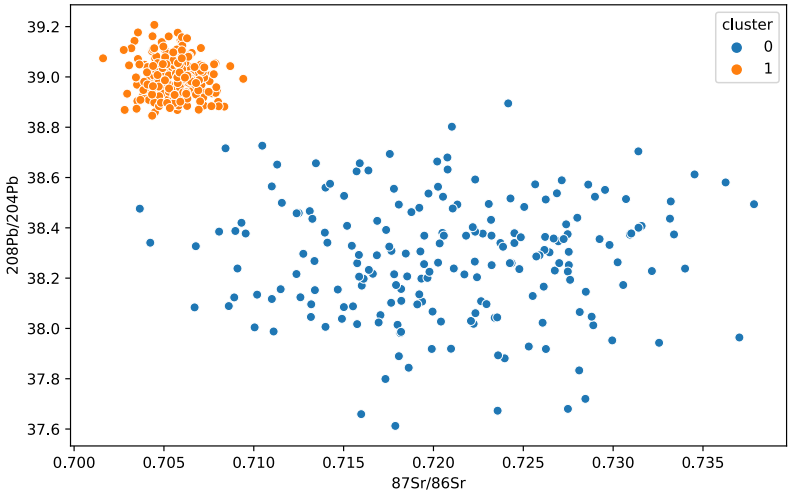


Figure 4.14: Bivariate Scatterplot of $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$ at a Multiplier for STD of 3

plot contains the $^{87}Sr/^{86}Sr$ on the x-axis and the $^{208}Pb/^{204}Pb$ on the y-axis. The orange cluster is the cluster with constant spread, and the blue cluster was generated by multiplying the standard deviation of the Gaussians by three. As the spread of one cluster increases, the area between both clusters where no samples of either cluster are located decreases. The decision boundary will be located closer to the orange cluster's center with higher density in Figure 4.14. Thus, the theoretical IBI space also decreases. When calculating the distance threshold to the separating boundary, we calculate the set of distances for each cluster to the separating boundary. When the separating boundary is located closer to the orange cluster, the distances will be smaller. Thus decreasing the threshold and the IBI space. Additionally, we create a new IBI instance by generating a start and endpoint from the distribution of

each cluster and selecting a point on the vector between them. As the spread of the blue cluster is larger, the space for the stand or endpoint in the blue cluster increases. Thus, in relation to the boundary thresholds calculated by our algorithm, the space in which we generate IBIs increases.

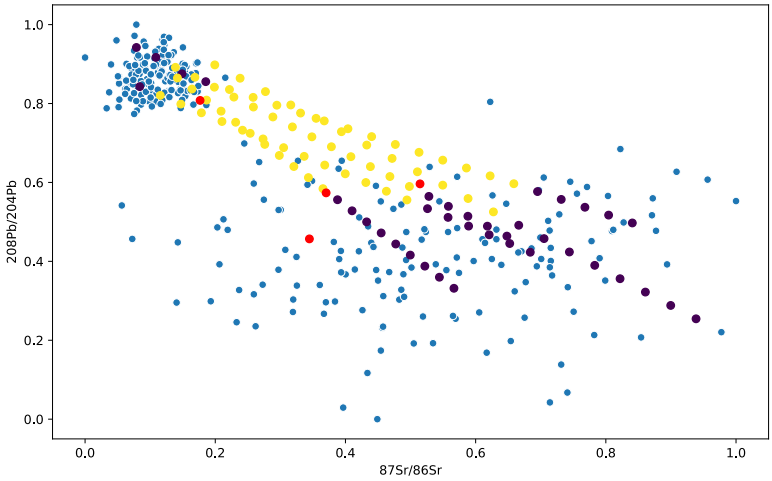


Figure 4.15: Bivariate Scatterplot of $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$ including IBIs which are color-coded by their Label

Within Figure 4.15, we can see the results of the labeling at a multiplier value of 3 with a linear kernel and the parameter C=10E10. It stands to note that this was executed for 100 different IBI sets within the experiment, and this plot only contains the IBI set for the seed zero. The blue samples correspond to the clusters, and the purple and yellow samples correspond to the IBIs. Purple IBIs were mislabeled as non-IBIs, and yellow IBIs were correctly classified. For each start and endpoint, we generated 20 IBIs between them. This creates the linear relationship of the IBIs. The red samples are the support vectors. As suspected, the IBIs within the clusters are mislabeled as non-IBIs. We can also observe the phenomena that we expected. As the lower cluster has a far higher spread, the start points of the IBI generation lie well within this cluster. Additionally, the first few IBIs on the vector between the start and the endpoint remain within the large spread cluster. We also see that the lower cluster spreads over a large portion of the feature space, decreasing the space between both clusters. Thus, a larger portion of the vector between the start and the endpoint lies within this cluster and is hard to identify.

As for differences between the four configurations, we observe that the recall starts at a substantially better value of just below 0.85 when using a linear kernel. When using an rbf kernel, the recall starts lower at around 0.75. However, independent of the choice of the kernel, the recall declines to around 0.6 at a multiplier of 3.0. Thus, the rbf kernel overall performs worse but has a slower decline in performance than the linear kernel.

We suspect that the rbf declines slower due to its better ability to capture the underlying patterns of the dataset at a higher spread. Even though both kernels can separate both classes, the rbf kernel might place the decision boundary more consistently as the spread changes. It may create a circular decision boundary around the cluster with higher density. This circular boundary shape may be more robust to changes within the spread of one cluster even though the linear boundary performs better, it may be more volatile and susceptible to spread changes.

However, the linear kernel shows a more stable performance regarding precision. The harder margin especially leads to consistent precision, which only begins to decline at the 2.6 mark for the standard deviation multiplier. This suggests that the configuration of a linear kernel and C=10E10 might be more robust to changes in the STD multiplier, maintaining more consistent performance. Thus, the performance difference might be due to a larger IBI space when using the rbf kernel instead of a more volatile placement of the linear decision boundary. A larger IBI space can mislabel more cluster labels as IBI and thus reduce the precision while correctly identifying more IBIs in relation to the increased spread, and thus, a slower decline in recall.

A lower C value also leads to a smoother decision surface, while a higher C value creates a more complex decision surface. The smoother decision surface might underfit on the training set, which could explain why the precision begins to drop earlier for lower C values.

While both configurations show declining trends as the multiplier for STD increases, the C=10E10 configuration generally maintains higher stability and performance, particularly in precision. The C=1 configuration, while starting with higher or similar scores, tends to degrade more noticeably across the metrics as the multiplier increases.

We observe that the rbf kernel exhibits increased precision at the highest multiplier values, which leads us to test even higher multiplier values up to 100. However, we did not observe a continuation of this trend. Precision remained between 0.9 and 1, while all other metrics steadily decline. The uptick of the precision for the rbf kernel at a multiplier of 3.0 might happen due to fewer false positives. Increasing the spread even further while keeping the sample size equal leads to decreased density within the cluster. Meanwhile, the IBI space might not increase in similar size. Thus, even if the cluster with a higher spread overlaps with the IBI space, fewer samples from this cluster are located within the IBI space due to the lower density. This leads to fewer false positives and, thus, an increase in precision.

When examining Figure 4.13, we observe that at the highest multiplier values, the clusters overlap in all dimensions separately. However, an overlap in multiple dimensions does not necessarily imply an overlap within the full-dimensional feature space. This still indicates that our approach deteriorates when the clusters overlap. We argue that this does not invalidate our approach, as the concept of in-betweenness is based on a region between the clusters. If the clusters overlap is too drastic, a region will not exist between the clusters, and thus, there will be no IBIs. However, with our procedure to generate IBIs, we can still generate samples that we label IBIs. These samples will not be discernible from the

reference clusters as they will be located within the overlap. We propose future research on this problem and an evaluation on a set of IBIs that only includes outliers.

In summary, our approach seems to deteriorate with higher spread differences. However, this could be due to the nature of the synthetic dataset. As seen in Figure 4.14, the cluster with the larger spread seems to encompass most of the feature space, and the space in which IBIs may occur seems to decrease. To evaluate if this is the reason, future work could include an experiment where the distance between both clusters is increased according to the increase in spread. This would ensure that for every configuration, a space between both clusters exists where IBIs may occur.

4.3.5 Overview of the Results of the Experimental Analysis of Dataset Changes

Table 4.5: Ability of our Algorithm to handle different Scenarios

Varying Sample Sizes	**Differences in Density**	**Late Migrant Detection**	**Differences in Spread**
Yes	Yes	Up to 18 Years	Only limited

Table 4.5 presents the results of our experimental evaluation. Our approach demonstrates resistance to varying sample sizes. Although very low sample sizes cause some instability in the results, the performance remains high across all tested sample sizes. Furthermore, our method effectively handles differences in density between reference clusters. Similar to the experiment on sample sizes, we observed more fluctuations in results for low sample sizes and, consequently, low density within a single cluster. Nevertheless, the performance remains robust across all configurations. Given our particular interest in migrants who migrated approximately 20 years ago, we assessed whether our approach could detect them. Unfortunately, our approach struggles to detect migrants located very close to or within the cluster. Therefore, we lack confidence in detecting migrants who migrated 19 or more years ago, which is still a satisfying result. These migrants have largely adapted to their new locations and do not exhibit isotopic signals that are distinguishable from the reference cluster, as shown in Figure 4.10. When applying our algorithm, the domain experts can use the parameter $T1_m$ to decide a tradeoff between more secure labeling, with fewer false positives, and higher sensitivity, with fewer false negatives. Additionally, our evaluation revealed that our approach's performance deteriorates when the spread of one reference cluster increases. This degradation may be due to the increasing overlap of reference clusters and the subsequent collapse of the IBI space rather than an inability to handle differences in spread. Overall, our experimental evaluation confirms that our approach is capable of identifying migrants within an isotope dataset.

4.4 Results and Discussion of Applying our Approach on the Fritzens-Sanzeno Dataset

In the following, we will present the results of our experiments on the Fritzens-Sanzeno dataset.

4.4.1 Results and Discussion of the Parameter Tuning on the Fritzens-Sanzeno Culture Dataset

Table 4.6: Best Models of the Parameter Tuning on the Fritzens-Sanzeno Culture Dataset

Kernel	C	gamma	Average F1-Score
rbf	2.656E8	4E-6	0.973818
rbf	3.511E8	4E-6	0.973818
rbf	6.136E8	4E-6	0.973818
rbf	1.0723E9	4E-6	0.973818
rbf	1.874E9	4E-6	0.973818
rbf	2.477E9	4E-6	0.973818
rbf	3.275E9	4E-6	0.973818
rbf	4.329E9	4E-6	0.973818
rbf	5.722E9	4E-6	0.973818
rbf	1E10	4E-6	0.973818

Table 4.6 includes the ten best-performing models resulting from the parameter tuning process on the Fritzens-Sanzeno Culture Dataset. It is evident that all models utilize a Gaussian (rbf) kernel, which appears to be the most effective at capturing the underlying patterns of this dataset. This is not unexpected, given that the underlying processes are expected to be Gaussian. Furthermore, the gamma value is constant among all models at 4E-6. A low gamma value increases the radius of influence for a single training sample. This reduces the complexity of the separating boundary and decreases the tendency to overfit.

The models differ only with respect to the C parameter. All models use a comparably high C value, which ranges from 2.656E8 to 1E10. Consequently, a substantial penalty for misclassification is necessary to capture the structure of this dataset. This results in more complex decision boundaries, which may potentially lead to overfitting. To mitigate this, we employed 5-fold cross-validation, and the F1-Score is computed as the average of all five folds.

The F1-Score of all models is consistently high at 0.973818. When examining the F1-Scores for each test set of the cross-validation, all models perform the same for all test sets, with scores of 0.96, 1.0, 1.0, 1.0, and 0.91 for the five different test sets, respectively.

This indicates that the model's performance is robust with respect to variations in C, provided that the value of C remains sufficiently high. The F1-Score of 0.973818 is sufficiently

Table 4.7: Best Models per Kernel of the Parameter Tuning on the Fritzens-Sanzeno Culture Dataset

Kernel	C	gamma	degree	Average F1-Score
rbf	1.0723E9	4E-6	None	0.973818
rbf	5.722E9	4E-6	None	0.973818
poly	1.624E6	6.952E-3	2	0.957913
poly	7.848E-1	3.793E0	3	0.957913
linear	7.055E2	None	None	0.923984
linear	9.326E2	None	None	0.923984

high to justify the conclusion that these models accurately capture the structure of the Fritzens-Sanzeno Culture Dataset.

However, we observe minimal variation with respect to the kernel, C parameter, and gamma parameter within the ten best models. All models utilize the Gaussian kernel, a large C value, and a gamma value of 4E-6. We anticipate that the separating boundary of all models will be similar. This is further supported by the fact that all models produce the exact same average F1-Score. To obtain more varied models and decision boundary structures, we also selected the two best models per kernel for the following migrant prediction as shown within Table 4.7.

The models with the rbf kernel represent a subset of previously discussed models. We observe satisfactory performance among all selected models. As we know, the rbf kernel excels at capturing the patterns of this dataset with a F1-Score of 0.973818. Nevertheless, also the polynomial kernel with a score of 0.957913 and the linear kernel with a score of 0.923984 are also capable of separating the reference clusters of the Inner Alpine and Southern Tyrolean burial sites. Even the linear kernel, although to a lesser extent, appears to be capable of dealing with the complexity of this dataset.

Upon examination of the parameters of the polynomial kernel, it becomes evident that there is a greater variability in the parameters. The first of the two configurations using a polynomial kernel employs a relatively high C value of 1.624E6 and a low gamma value of 6.952E-3 with a degree of 2. While the high C value results in a more complex model, the low degree of 2 offsets this. The second polynomial model employs a low C value of 7.848E-1, resulting in a smoother decision boundary, offset by a higher degree of 3 and a higher gamma value of 3.793E0. The configurations for the linear kernel exhibit comparable C parameter values, namely 7.055E2 and 9.326E2. As these values are not situated at either end of the spectrum, they are more challenging to interpret, and no decisive conclusion about their influence on the decision boundary can be made. We observe that SVMs can separate the Inneralpine and southern Tyrolean regions based on the multi-isotope signals within this dataset.

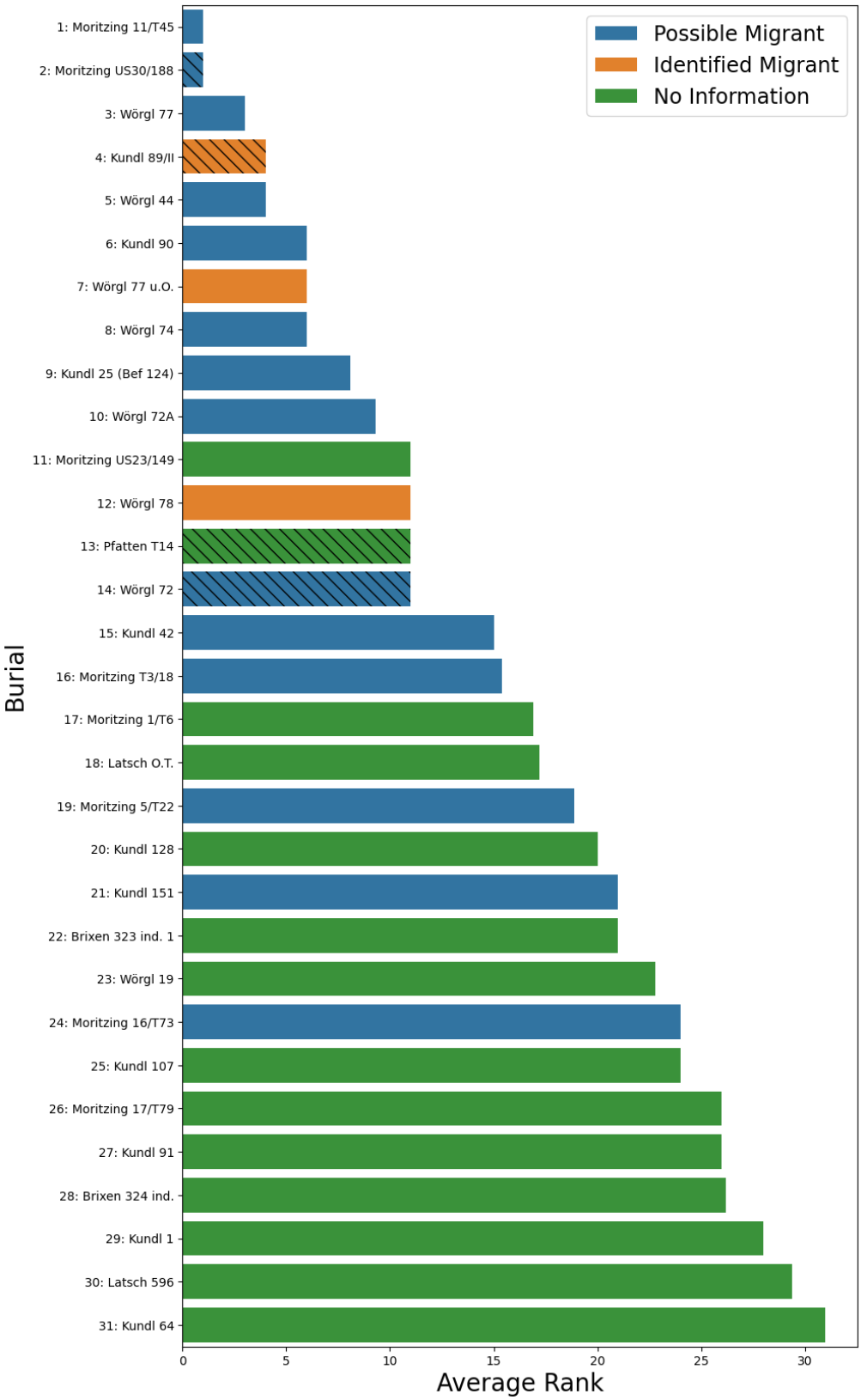


Figure 4.16: Migrant Ranking of the Fritzens-Sanzeno Culture Dataset using the Models shown within Table 4.6

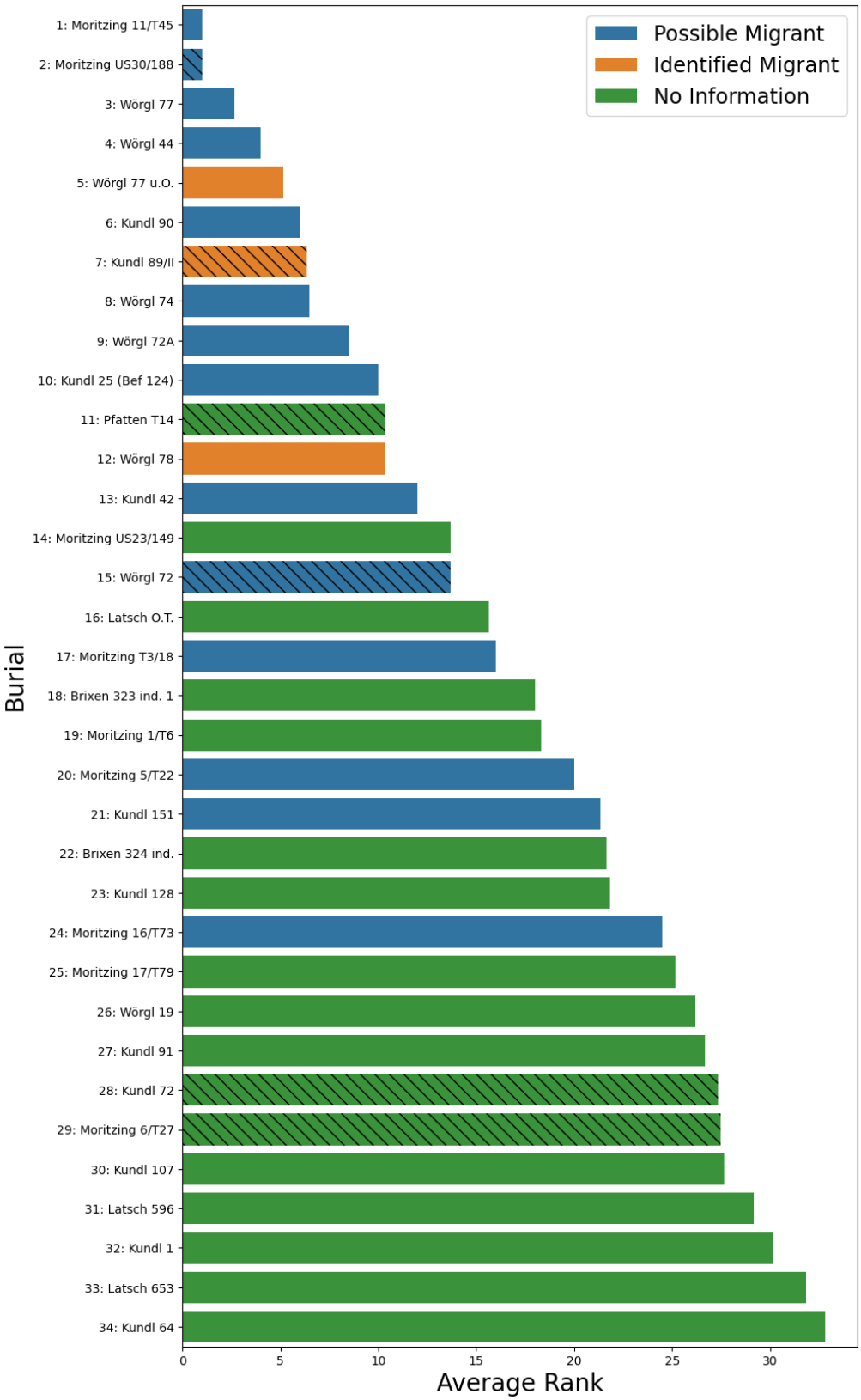


Figure 4.17: Migrant Ranking of the Fritzens-Sanzeno Culture Dataset using the two best Models per Kernel as shown in Table 4.7

4.4.2 Results and Discussion of the Migrant Detection on the Fritzens-Sanzeno Culture Dataset

We present the result of our migrant detection on the Fritzens-Sanzeno culture dataset with the ten best models selected in Figure 4.16 and with the two best models per kernel in Figure 4.17. The x-axis displays the average rank among all models for each individual sample. A description of the calculation and motivation of the average rank can be found in Section 3.8.2. The y-axis displays the different samples by their burial site and burial ID and is enumerated by their ranking. In order to enhance readability, we have included only the samples that, for at least one configuration, are predicted as migrant candidates. Every sample that we did not include within Figures 4.16 and 4.17 exhibited an average rank of 30.9 when using the ten best models and 34.0 when using the two best models per kernel. Each of these samples was labeled as non-migrant for every value for $T1_m$ within the range of 0 to 3.
Further information from the research of Grupe et al. (2020) led to the creation of three groups of samples, which were color-coded within the plot according to their group. The group labeled "No Information" contains all samples without additional context information beyond their burial site. The group labeled "Identified Migrant" contains all samples that were identified as migrant candidates by Grupe et al. (2020). This group contains six samples with archaeological context, such as grave goods, which indicate a migration. All 18 samples that are grouped within "Possible Migrant" exhibit some characteristics that might indicate a migration, such as being statistical outliers, a weak cluster assignment, or being located within the overlapping region of the two locations (Grupe et al., 2020).

Correspondence with Lang (2024) revealed additional archaeological context for the graves where a label as a migrant candidate was provided by the computational model proposed in this work. We obtained no additional archaeological context for the graves that were not identified as migrant candidates except for the Kundl 160 and Kundl 40 graves. Each grave with additional archaeological context is marked as hatched within our result plots. Unmarked graves indicate an archaeological unobtrusive grave that adheres to the cultural burial traditions. Nevertheless, if the grave in question adheres to the burial traditions of the Fritzens-Sanzeno culture, it does not provide sufficient evidence to definitively identify the grave as non-migrant.

It appears that ranks increase uniformly along the y-axis. We are most confident in samples with a low rank being mixed isotope signals and, thus, migrants. As the average rank increases, our confidence in this label decreases. It should be noted that a higher average rank could also be due to a migration that happened longer ago. This ranking cannot be interpreted as a probability. Rather, we present candidate samples that might be migrants and advise further investigation into the archaeological context of these samples.
When using the best ten models, a total of 31 of the 92 samples were identified as potential migrants, as they had been classified as such within the experiment. This represents 34.1% of the total population. Using a more heterogeneous choice of models, which is the result of selecting the best two models per kernel as seen in Figure 4.17, we identified 34 samples,

37.0% of the total population, as potential migrants. The samples Moritzing 6/T27, Latsch 653, and Kundl 72 are additionally identified as migrant candidates. While the order of the ranking of the individual samples differs within both experiments, only these three samples were labeled as migrant candidates in one experiment and not in the other. Within the labeled migrant candidates, the archaeological context of six different graves indicates a deviating burial ritual from the information provided by Lang (2024). The grave Moritzing US30/188 at rank 2 was conspicuous as no grave goods could be found in contradiction to the Fritzens-Sanzeno burial rituals. Kundl 89/II contained ceramics that might be originally from southwestern Germany. Within the grave of Pfatten T14, a larger than normal amount of grave goods was found; however, these grave goods match the Fritzens-Sanzeno culture. Foreign ceramics, possibly originating from the Alpine foothills, were found in the burial Wörgl 72. Kundl 72 contained southern Alpine ceramics. Moritzing 6/T27 is a special case within our dataset, as we are unsure of the origin of the isotope ratio measurements. The grave contained no cremated bone, but the associated isotope measurements were inventoried as cremated bone. Not shown within our plots, as it never got labeled as a migrant candidate, is Kundl 160 with an unusual amulet, probably from the southern Alps. Every other grave contained no grave goods or context that indicated a foreign origin. However, the absence of archaeological conspicuous context does not necessarily imply that the contained individual did not migrate in the past.

Discussion In the following section, we will present the findings from our experiments on the Fritzens-Sanzeno culture dataset. We will utilize the findings of Grupe et al. (2020) and the archaeological context provided within this research and by the correspondence with Lang (2024). We will discuss the individual samples that were previously identified as possible migrants and compare our results with these findings.
Immediately, we observe that our algorithm does not identify individuals Kundl 160, Kundl 89/I, and Kundl 40 as potential migrants. Together with Kundl 89/I, another individual (Kundl 89/II) was buried in the same urn, denoted as grave Kundl 89. The samples Kundl 160, Kundl 40, and Kundl 89/II formed a distinct cluster in the study by Grupe et al. (2020). This cluster was assumed to be identified as they are all dissimilar to the other individuals but do not necessarily share similar features (Grupe et al., 2020). The clustering results of Grupe et al. (2020) are shown within Figure 4.18. The distinct cluster of the three Kundl individuals can be seen in the upper left.

All three burials are notable from an archaeological perspective. Kundl 89 was particularly noteworthy due to the presence of a dish decorated in a style characteristic of southwestern Germany and grave goods that are geographically related to the south of the Alps. The grave goods of Kundl 160, particularly an amulet resembling southern Alpine jewelry, indicate a connection to the south, and the grave of Kundl 40 contains completely unique ceramics. This archaeological context suggests a migration from possibly multiple different locations (Grupe et al., 2020; Lang, 2024).
Our result, namely the missing migration candidate labeling of Kundl 160, Kundl 89/I, and

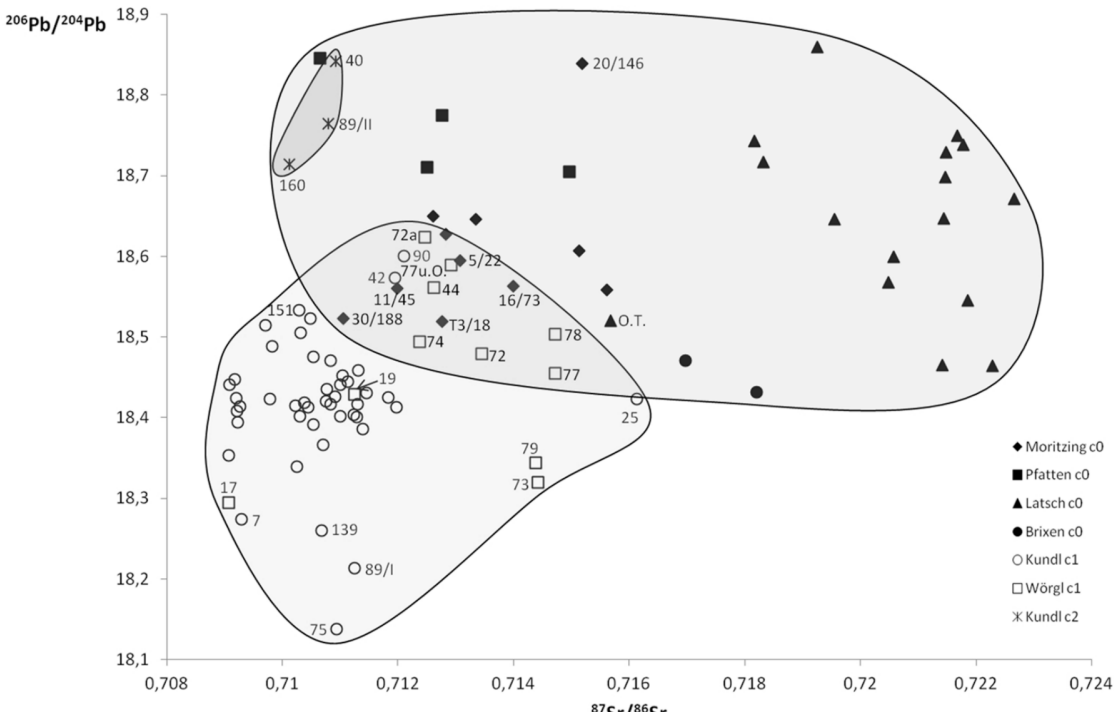


Figure 4.18: Bivariate $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$ Plot with the resulting Clusters of the research by Grupe et al. (2020)

Kundl 40, indicates that these individuals did not migrate from the southern Alpine region. However, a southern Alpine origin should not be definitively excluded based on our results. As a migration from an unknown southern Alpine region might be possible. Furthermore, our result separates the reference regions based on multiple locations. Further experiments that use a finer selection of burial sites might change the outcome of the labeling and include these individuals as migrant candidates.

Kundl 89/II is ranked fourth and seventh in the experiment. Consequently, it was identified as a migrant candidate at an early $T1_m$ value, thus a smaller IBI space, and represents a sample with a high degree of confidence that it exhibits a mixed isotopic signal due to migration. Our ranking and the grave goods associated with this burial indicate that Kundl 89/II migrated from the Southern Alpine area to the Inneralpine region of Kundl. Thus, the grave goods in the southern Alpine style could be associated with Kundl 89/II, and the dish decorated in a southwest German style could be associated with Kundl 89/I. Furthermore, our results indicate that Kundl 160 did not migrate from the south, from an unknown location, or more than 20 years ago, although the grave goods might suggest otherwise. However, Grupe et al. (2020) firmly identified Kundl 160 as non-local based on its grave goods and distinct cluster assignment. Thus, a migration source different than the investigated region south of the Alps seems more plausible than a migration that happened more than 20 years ago. The objective of our approach is not to detect all migrants within a single location but

rather migrants between two reference locations. Consequently, the absence of identification of samples Kundl 160, Kundl 89/I, and Kundl 40 as migrants does not invalidate our findings. At this point, we assume that these individuals may have migrated from a different location. The results of this study align with the findings of Grupe et al. (2020), who proposed that the three individuals grouped within a cluster do not share a common origin and are not defined by their similarities but rather by their differences compared to the rest of the dataset.

Grupe et al. (2020) also identified a group of individuals within an overlap region of both clusters, as seen in Figure 4.18. The individuals contained in the overlap are assumed to be candidates for a mixed isotope signal. This concerns the burials 11/T45, 16/T73, US30/188, 5/T22, T3/18 from Moritzing, 90, 42 from Kundl, and 44, 74, 77, 77 u.O., and 78 from Wörgl. Our algorithm captures this and labels every sample of this group as a migrant candidate. This supports the hypothesis that this region may contain mixed isotope signals, and the samples within this region should be interpreted as migration candidates. Most of these samples are found within the first ranks of our results and, thus, the samples in which we have the highest confidence.
In both experiments, Moritzing 11/T45 and Moritzing US30/188 samples were identified as the most promising candidates. Moritzing US30/188 was noticeable from an archaeological view as it contained no grave goods in difference to the Fritzens-Sanzeno burial tradition. While this alone does not prove a migratory background, it further supports the assumption that this individual migrated from the Inneralpine region to southern Tyrol. For each model, these samples were first identified as potential migrants. We propose a further investigation into their archaeological context to verify whether these individuals migrated from the Inneralpine region to the southern region of Moritzing.

In the previous research of Grupe et al. (2020), a group of 11 samples was identified as statistical outliers. This group was included in the "Possible Migrant" category until further information indicated an immigration background, at which point it was reclassified as "Identified Migrant." Seven out of the 11 samples were identified as possible migrants. The algorithm fails to identify Kundl 139, Kundl 75, and Kundl 7. The three individuals only exhibited outlying lead isotope ratios. Grupe et al. (2020) demonstrated that provenancing individuals by lead isotopic fingerprints alone is not possible within this region. It is, therefore, unsurprising that these samples were not identified as the labeling as possible migrants is insecure and only based on lead isotopes. The archaeological context of all statistical outliers is inconspicuous (Lang, 2024).

For both model selections, as shown in Table 4.7 and 4.6, the first ten ranks consist of samples that already were identified as suspicious by Grupe et al. (2020). Within the first 20 ranks, only five samples exhibited no conspicuous isotope signals or archaeological context in earlier research. Therefore, our findings largely align with those of previous research on the Fritzens-Sanzeno culture. Furthermore, we present previously inconspicuous samples, such as Pfatten T14 and Moritzing US23/149, as potential migrant candidates. Despite

their lack of previous identification, our results indicate that they are ranked closely with Wörgl 78, which was identified as a migrant candidate due to its associated grave goods, its location within the overlap region of the clusters, and its statistical outlier status within the strontium isotope signature. The individual of Pfatten T14 was archaeologically suspicious as the grave did contain rich, albeit local, grave goods (Lang, 2024). Even if the interpretation of this archaeological context is not unambiguous, our results support further research into the origin of the individuals of Pfatten T14. If we assume that the individual Wörgl 78 has a migratory background and that the ranking of our migrant candidates reflects how confident we are in a migratory background, we propose the samples Pfatten T14 and Moritzing US23/149 as previous unknown migrant candidates. In total, we identified 34 migrant candidates within the Fritzens-Sanzeno Dataset.

We approximate the years that have passed since migration for the identified migrant candidates using the method proposed in Section 3.3. Figure 4.19 shows on the x-axis the migrant candidates ordered by their rank within the first experiment, and the y-axis denotes the years that are predicted to have passed since the migration of the individual. A linear regression was fitted and plotted as the blue line surrounded by the shaded 95% confidence interval for that regression. It is not advisable to interpret these results as precise measurements of the time elapsed since the migration. Rather, they should be regarded as approximations with a potential for significant error.

We observe that the linear regression line shows an increasing trend, indicating that as the rank of the migrant candidates increases, the predicted years since migration also tend to increase. The positive slope of the regression line implies a positive correlation between the rank of the migrant candidates and the predicted years since migration. Higher-ranked individuals tend to have migrated more recently. We expected this behavior as the migrants that migrated fewer years ago should be easier to detect. We noted earlier that our confidence in the migrant prediction of the earlier ranks is higher as they are found to be migrant candidates with a smaller IBI space than the samples on later ranks. With these results, we support this thesis as we identify the samples on the first ranks as the samples, which should be easier to identify. The confidence interval around the regression line suggests reasonable certainty about the trend. Although, there is a noticeable variability among the predicted years of the individuals.

The predictions per migrant candidate are presented in Table 4.8. Kundl 89/II is the sample with the shortest time since migration. Every other candidate has migrated more than 7.8 years ago. This is noteworthy because we anticipate that individuals who have recently arrived at the new location would be the easiest to detect, as their isotopic fingerprint has not yet adjusted to the new location. Thus, with reasonable certainty, we found no very early migrants within this dataset except for Kundl 89/II. Furthermore, we observe that the last ranks are predicted to have migrated close to 20 years ago. After 20 years, the isotopic fingerprint should have fully adapted to the new location, rendering the individual unidentifiable as a migrant using isotope analysis. Therefore, samples with predicted years close to 20 years should be treated with caution. Nevertheless, we also conclude that an

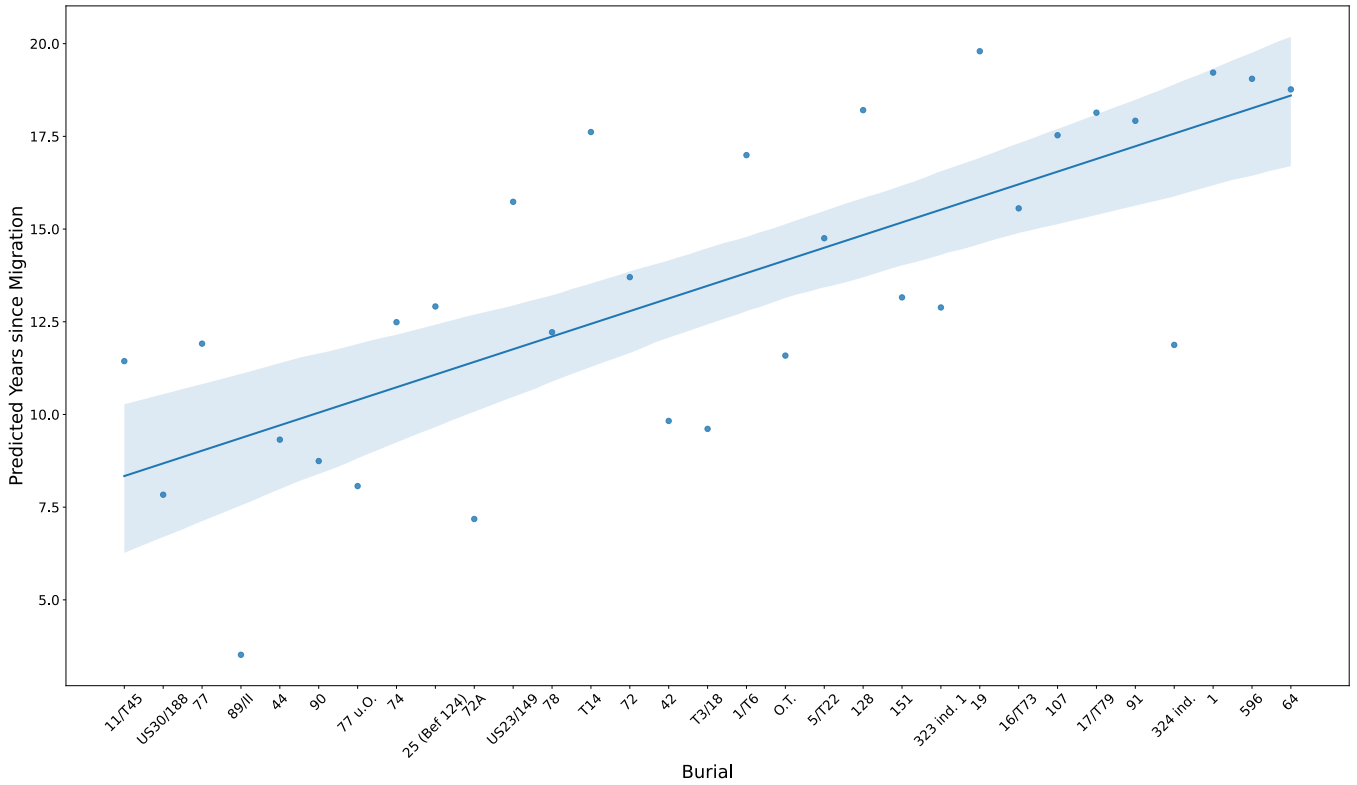


Figure 4.19: Predicted Years since Migration for Fritzens-Sanzeno Migration Candidates

increase in $T1_m$ and a consequent shift of the threshold for the IBI space in the direction of the clusters is not advisable. As we have identified migrants who migrated close to 20 years ago, we expect to have identified all reasonable possible migrant candidates when using our approach. We see that our algorithm is able to find even the hard-to-identify individuals who migrated almost 20 years ago.

Table 4.8: Table of the predicted Years since Migration for the Migrant Candidates

Burial Site	Burial ID	Predicted Years since Migration
Moritzing	11/T45	11.43
Moritzing	US30/188	7.84
Wörgl	77	11.91
Kundl	89/II	3.52
Wörgl	44	9.32
Kundl	90	8.74
Wörgl	77 u.O.	8.07
Wörgl	74	12.49
Kundl	25 (Bef 124)	12.91
Wörgl	72A	7.18
Moritzing	US23/149	15.74
Wörgl	78	12.22
Pfatten	T14	17.62
Wörgl	72	13.70
Kundl	42	9.83
Moritzing	T3/18	9.61
Moritzing	1/T6	16.99
Latsch	O.T.	11.59
Moritzing	5/T22	14.75
Kundl	128	18.21
Kundl	151	13.16
Brixen	323 ind. 1	12.89
Wörgl	19	19.80
Moritzing	16/T73	15.56
Kundl	107	17.53
Moritzing	17/T79	18.14
Kundl	91	17.92
Brixen	324 ind.	11.88
Kundl	1	19.22
Latsch	596	19.05
Kundl	64	18.77

Figure 4.20 and Figure 4.21 contain bivariate plots of $^{87}Sr/^{86}Sr$ and $^{208}Pb/^{204}Pb$. The coloring of each point describes whether the sample belongs to the south Tyrolean or the inner Alpine region. Additionally, the marker corresponds to the exact site of the burial. Each sample identified as a migrant candidate within our migrant detection on the Fritzens-

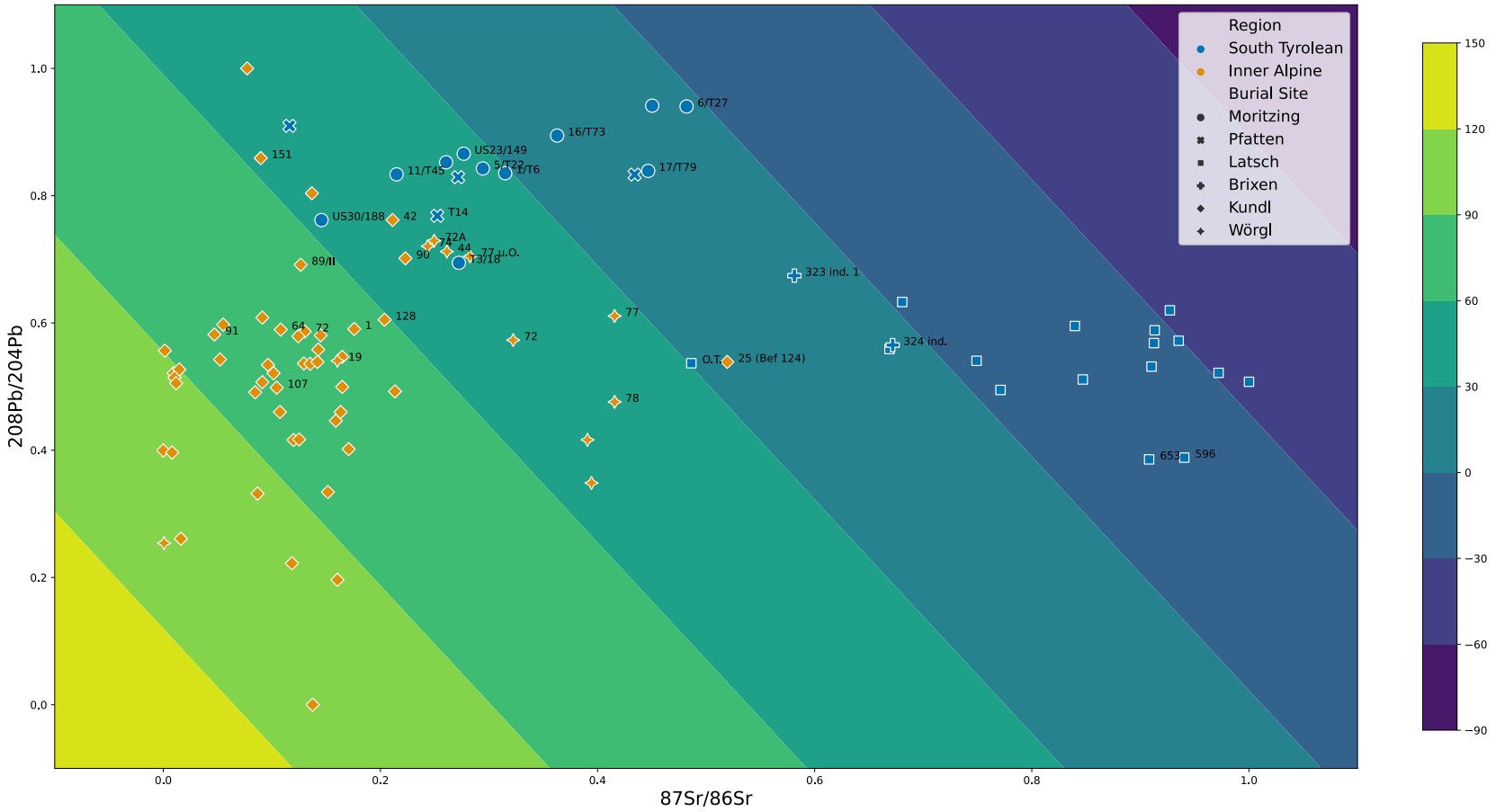


Figure 4.20: Bivariate Plot of $^{87}Sr/^{86}Sr$ versus $^{208}Pb/^{204}Pb$ with the Contour approximating the Decision Boundary of a SVM trained with an rbf Kernel, C=2.656E8, and Gamma=3.511E-06

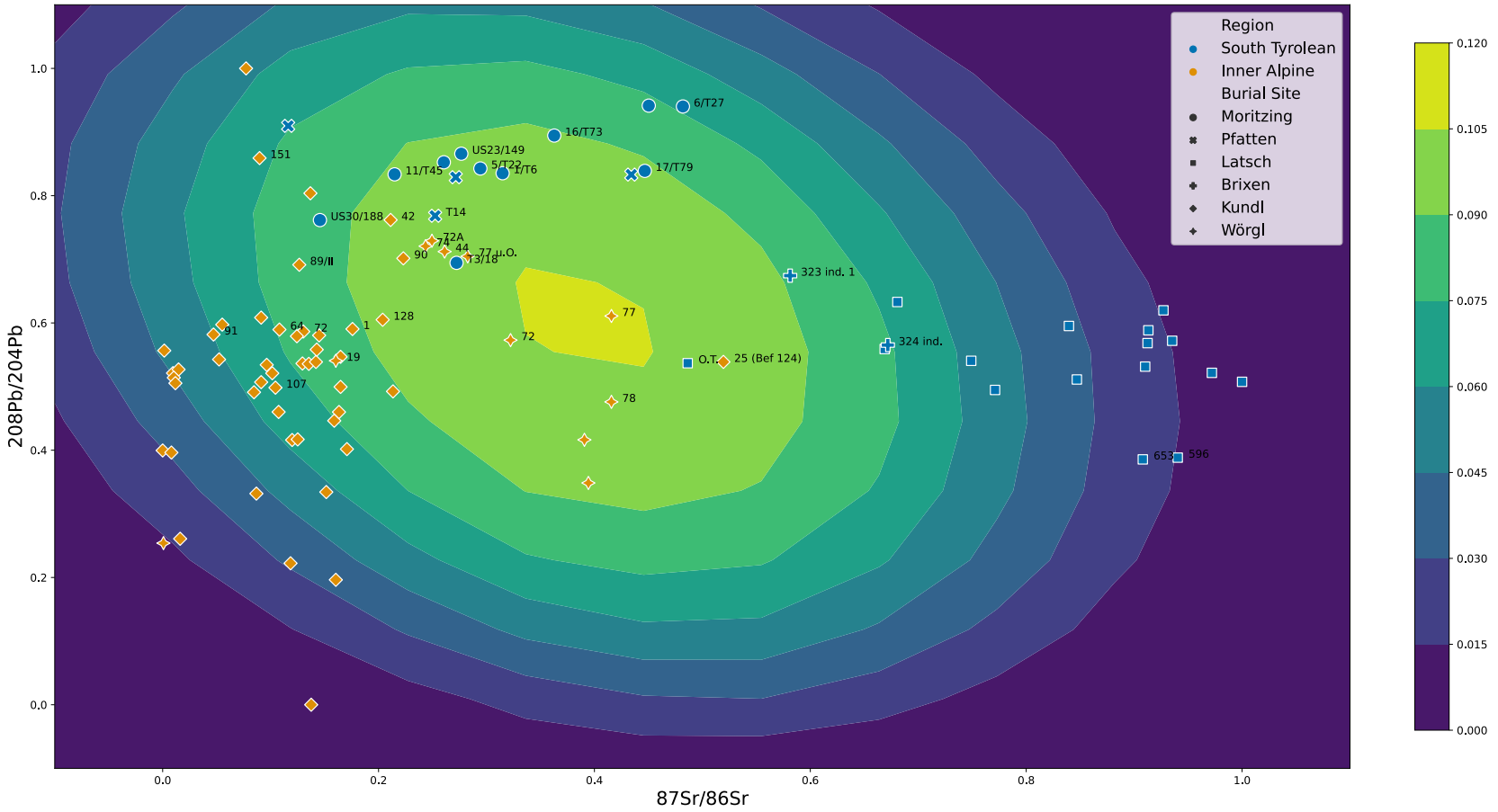


Figure 4.21: Bivariate Plot of $^{87}Sr/^{86}Sr$ versus $^{208}Pb/^{204}Pb$ with the Contour approximating the IBI space with an SVM trained with an rbf Kernel, C=2.656E8, and Gamma=3.511E-06

Sanzeno culture dataset is labeled with its burial. Within Figure 4.20, the contour shows a 2-dimensional approximation of the decision boundary. This approximation is calculated as described within Section 3.4. The contour in Figure 4.21 visualizes an approximation of the IBI space, calculated as described within Section 3.4. As this is only a two-dimensional plot of a six-dimensional feature space, we can not capture the exact patterns of the dataset but can provide an approximation.We observe that the Inneralpine region seems to cluster more densely, while the south Tyrolean region displays a wider spread along the $^{87}Sr/^{86}Sr$. From this plot, we observe that the site Latsch displays a larger $^{87}Sr/^{86}Sr$ than the rest of the dataset, including the other sites from southern Tyrol, while the sites of Moritzing and Pfatten cluster closer to the inner Alpine region.

Within Figure 4.21, we see that the IBI space is located firmly between both clusters, even in this two-dimensional approximation. When comparing our IBI space with the overlap from the research of Grupe et al. (2020) as seen in Figure 4.18, we see a shift towards larger values along the $^{87}Sr/^{86}Sr$ axis. This might be due to using the mean to calculate the CCV. As the Latsch group is located at higher $^{87}Sr/^{86}Sr$ values, it might drag the mean, and thus the IBI space, toward larger $^{87}Sr/^{86}Sr$ values. We propose that future research should include an evaluation of different cluster centroids, such as the median. Even with the difference in IBI space we still managed to capture the individuals within the overlap identified by Grupe et al. (2020).

Within Figure 4.20, we observe that the decision boundary approximation has a linear shape and separates the feature space oblique.

Most of the samples we label migrant candidates appear at the center of the IBI space. Interestingly, the individuals Latsch 653 and Latsch 596 appear at the border of the IBI space and far from the Inneralpine cluster but still get labeled as migrant candidates. In $^{87}Sr/^{86}Sr$, these individuals fall well within the Latsch group. However, in $^{208}Pb/^{204}Pb$, they show a lower measurement than the rest of the Latsch individuals. We suspect that they are labeled as migrant candidates but not shown within the IBI space because they are closer to the Inneralpine cluster within the remaining lead isotope ratios, which are not included within this two-dimensional visualization.

4.5 Results and Discussion of Applying our Approach on the Extended Dataset

Before the presentation and discussion of the results of the migrant detection applied to the extended dataset, it is necessary to consider the validity of the results. The results presented here are more experimental than those reported for the Fritzens-Sanzeno dataset. It is beyond the scope of our expertise to provide a comprehensive interpretation of these results. Furthermore, as we do not have detailed domain knowledge, it was not possible to preprocess the data appropriately. The data is more heterogeneous than the Fritzens-Sanzeno dataset and contains samples from multiple cultures and timeframes.
The dataset includes ratios sampled from dental enamel and cremated and uncremated bone.

As previously discussed in Section 1.3.1, dental enamel and cremated bone are more resistant to diagenetic changes after burial than uncremated bone. It is not possible for us to ascertain the extent to which this potential contamination is addressed. Consequently, the isotope ratios sourced from uncremated bone might be biased toward the geological isotope ratios of the burial environment. Furthermore, factors such as the nearest metal processing place often influence lead isotope ratios. These factors might change over time, and thus, the lead isotope ratios of the sampled humans might not be comparable across different time periods. We also lack information about grave goods or similar archaeological contexts, which might indicate a migration.
Due to these limitations, we will refrain from analyzing the full results and will concentrate on the samples we analyzed from the Fritzens-Sanzeno culture.

4.5.1 Results and Discussion of the Parameter Tuning on the Extended Dataset

The best-performing models of the parameter tuning with the extended dataset are presented within Table 4.9. A degree of none indicates that this kernel does not apply the parameter degree. We divided the extended dataset into three subsets. Each subset contains two regions. The first subset, denoted as 1-2, contains the southern German, denoted as 1, burials, and the Inneralpine, denoted as 2, burials. The second subset, denoted as 1-3, contains the burials of the southern German region and the southern Tyrolean region, denoted as 3. Lastly, the dataset 2-3, contains burials from the Inneralpine and southern Tyrolean regions.
For the dataset containing the southern German and Inneralpine samples (1-2), we observe that all the 10 best models utilize the rbf kernel. Furthermore, eight out of ten models combine high C values with low gamma values. The remaining two models apply a low value with an increased gamma value, albeit the gamma value is still below 1. The F1-Scores range between 0.856162 and 0.810473.
The SVM training on the samples from southern Germany and southern Tyrol (1-3) produced multiple models with a perfect score for each kernel. Thus, to obtain a more heterogeneous choice of models, we selected the best 2 models per kernel for the migrant prediction. The rbf kernels again utilize a high C value in combination with a low gamma value. Conversely, the poly kernel applied medium C values with either extremely high, 1E4, or medium, 1.833E1, gamma values and degrees of 3 and 4. Similarly, no clear trend for the linear kernel is visible as one model utilizes a medium-high C value of 8.697E3 while the other utilizes a high C value of 5.337E6. It stands to note that several other models could have been chosen as many models perfectly separated the samples from both regions.
When trained on the Inneralpine and southern Tyrolean dataset (2-3), seven of the ten best-performing models used the rbf kernel, while three applied a linear kernel. Again, the models with the Gaussian kernel utilize a high C value and a low gamma value. The linear kernel performs best with a medium C value between 6.579E3 and 3.765E3 and low gamma values between 1.097E-3 and 5.99E-4. The best F1-Score is 0.891139, followed by 0.887473,

Table 4.9: Best performing Models of the Parameter Tuning on the Extended Dataset

Dataset	Kernel	C	gamma	degree	Average F1-Score
1-2	rbf	2.656E8	6.734E-3	None	0.856162
1-2	rbf	2.009E8	6.734E-3	None	0.843392
1-2	rbf	1.150E8	6.734E-3	None	0.842170
1-2	rbf	4.642E8	6.734E-3	None	0.837427
1-2	rbf	3.511E8	6.734E-3	None	0.836269
1-2	rbf	8.111E8	6.734E-3	None	0.834060
1-2	rbf	1.520E8	6.734E-3	None	0.831482
1-2	rbf	8.697E7	6.734E-3	None	0.812352
1-2	rbf	6.136E0	2.535E-1	None	0.810473
1-2	rbf	4.642E0	2.535E-1	None	0.810473
1-3	poly	4.833E3	1E4	4	1
1-3	poly	8.859E4	1.833E1	3	1
1-3	linear	8.697E3	None	None	1
1-3	linear	5.337E6	None	None	1
1-3	rbf	1.233E7	1.874E-1	None	1
1-3	rbf	1.322E6	6e-6	None	1
2-3	rbf	1.630E7	8.11E-4	None	0.891139
2-3	rbf	7.055E6	4.43E-4	None	0.887473
2-3	rbf	4.037E6	3.27E-4	None	0.886434
2-3	rbf	3.054E6	3.27E-4	None	0.886434
2-3	rbf	1.748E6	1.098E-3	None	0.886434
2-3	rbf	1.748E6	5.99E-4	None	0.886434
2-3	rbf	1E6	1.097E-3	None	0.886434
2-3	linear	6.579E3	1.097E-3	None	0.886434
2-3	linear	4.977E3	3.27E-4	None	0.886434
2-3	linear	3.765E3	5.99E-4	None	0.886434

and all remaining models produce the same score of 0.886434.

Discussion Overall, the southern German and southern Tyrolean burials can be perfectly separated by applying an SVM while it struggles with separating the southern German and Inneralpine, and, albeit to a lesser extent, the Inneralpine and southern Tyrolean burials. Similar to the Fritzens-Sanzeno dataset, Gaussian kernels produced the best results, which is not unexpected as we assumed the isotopic signature of a location to be Gaussian distributed. When comparing it with the parameter tuning results on the Fritzens-Sanzeno dataset, we see that introducing new samples did not increase the SVM's performance but decreased it. As noted before, multiple factors, such as sampling the isotopic signature from different parts of the individuals, might introduce noise and unknown patterns into this dataset. These results confirm that introducing new sample sources did not increase the ability of the SVM to separate the classes. As we have seen in Section 4.3.4, increased spread and overlapping clusters might decrease the performance of our migrant detection approach. Thus, in addition to the factors named at the beginning of this chapter, the decreased performance of the SVM leads us to be more confident in the results of the Fritzens-Sanzeno dataset than in the results of the extended dataset.

4.5.2 Results and Discussion of the Migrant Detection on the Extended Dataset

In the following, we will present and discuss the results of applying the migrant detection on the extended dataset. Figures 4.22, 4.23, and 4.24 present these results. As we will focus on the Fritzens-Sanzeno culture, we omit all samples not included within the Fritzens-Sanzeno dataset. As in Section 4.4.1, the x-axis corresponds to the average rank of each sample on all used models. The y-axis contains the samples of the Fritzens-Sanzeno dataset and is ranked with the smallest rank assigned to the samples that got labeled, on average, the earliest. This rank is calculated with all samples from the selected regions, thus also including samples not present in the Fritzens-Sanzeno dataset. This ranking helps us evaluate how confident we are in the labeling as migrant candidates.

The southern German and Inneralpine regions Figure 4.22 presents the results of the migrant detection on the subset of the extended dataset containing the southern German and Inneralpine regions. Overall, this dataset contains 270 samples, of which 163, or 60.0%, were identified as migrant candidates. Of the 163 migrant candidates, 29 samples from the Fritzens-Sanzeno dataset were labeled. The 42 other individuals from the Fritzens-Sanzeno dataset were not labeled as migrants. Out of the 29 detected migrant candidates, only 3 were labeled as migrant candidates when our approach was applied to the Fritzens-Sanzeno dataset. 27 of the 29 detected migrant candidates were not detected as migrants between

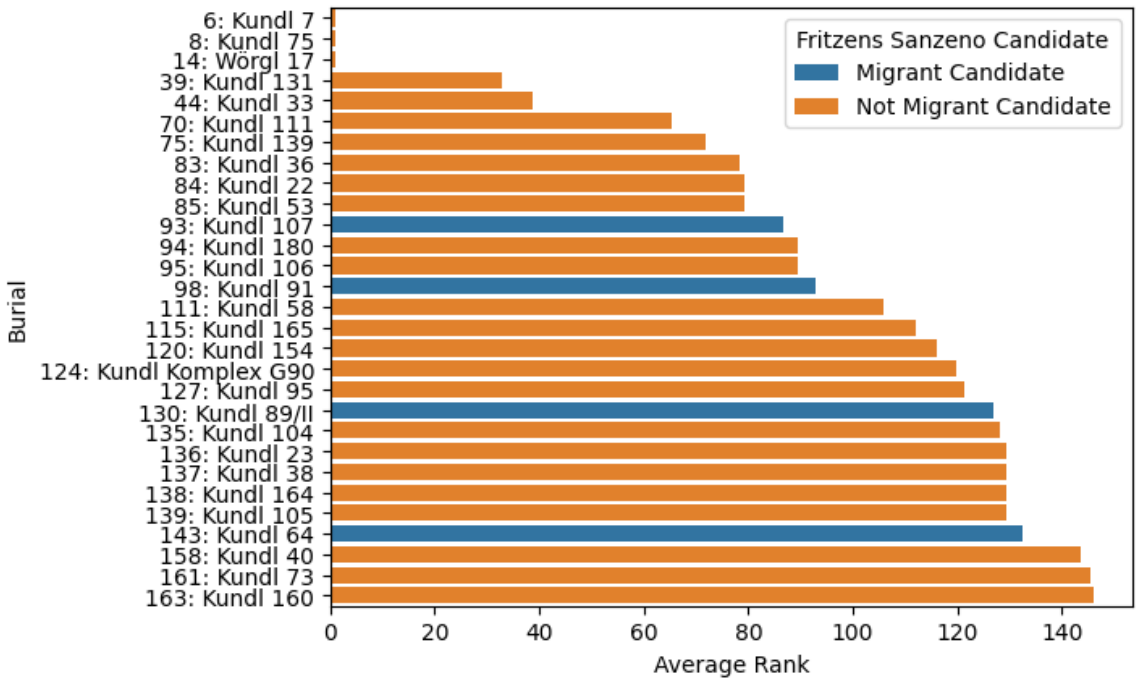


Figure 4.22: Migrant Ranking of the Extended Dataset containing the southern German and Inneralpine Regions (1-2)

the Inneralpine and southern Tyrolean region. This subset of the extended dataset contains 146 samples from the southern German region and 122 Inneralpine samples. Thus, it is roughly balanced. However, both regions contain many different sites and are thus very heterogeneous. The southern German region contains 22 sites, and the Inneralpine region contains 17 different sites.

Discussion When looking at the ranking results for the southern German and Inneralpine region, the individuals Kundl 7, Kundl 75, and Wörgl 17 are the migrant candidates in whom we have the most confidence. They are all ranked below rank 15, and compared to the next individual of the Fritzens-Sanzeno candidate, Kundl 131, at rank 39, they show a substantially lower average rank. We see a strong increase in average rank between Kundl 131 and Kundl 33. All three individuals were among the first-ranked individuals for all 10 models. Thus, it can be seen as sharing a rank. The next individual, Kundl 131, has an average rank of 32.9.

With a large percentage of 60% of all samples labeled as migrant candidates, the lower ranks should not be weighted too heavily. This may be due to strongly overlapping clusters and the SVM model not perfectly capturing the underlying data structure. We have seen within Section 4.3.4 that the performance of our approach degrades with a higher overlap.

Within these lower ranks, the previously identified migrant candidates Kundl 107, Kundl 91, Kundl 89/II, and Kundl 64 are to be found. These individuals were identified within

the results of the Fritzens-Sanzeno dataset as candidates for a migration from the southern Tyrolean to the Inneralpine region. As they are ranked within the lower ranks between 93 and 145, these new results do not change our previous assessment. The confidence in this labeling is low, and because of the low confidence in this labeling, this does not contradict the results of the Fritzens-Sanzeno migrant detections. Finding many earlier identified migrant candidates between the Inneralpine and southern Tyrolean region would lead us to suspect that our approach could not differentiate between multiple regions. This is not the case.

Only 5 out of the first 50 ranks consist of individuals from the Inenralpine region. As both datasets contain similar samples, we would have expected the ranking to include closer to 50% of both regions within the first ranks. This might be due to a larger cluster spread containing the samples from the southern German region.

Within Section 4.4.1, we formulated the idea that the individuals Kundl 40 and Kundl 160 could be migrants from southern Germany to the Inneralpine region. Kundl 40 was assigned rank 158, and Kundl 160 was assigned 163. Thus, even though we detect these individuals as migrant candidates, their ranks indicate low confidence in this ranking. With this experiment, we can not validate or invalidate the hypothesis that Kundl 40 and Kundl 160 migrated from southern Germany. Within the research of Grupe et al. (2020), these individuals formed a special cluster with Kundl 89/II and were identified as candidates for a migratory background. Unfortunately, we cannot provide a more decisive result and can not provide the provenance of these individuals. However, this experiment adds Kundl 7, Kundl 75, and Wörgl 17 to the list of individuals from the Fritzens-Sanzeno culture for which we suspect a migration to have happened within the last 20 years.

The southern German and southern Tyrolean regions The results of the migrant detection on the extended dataset with the southern German and southern Tyrolean regions are presented in Figure 4.23. This subset of the extended dataset contained 179 samples, of which 146 samples are from southern Germany and 33 from southern Tyrol. 101 samples are predicted as migrant candidates, which corresponds to 56.42% of all samples. Out of these 101 migrant candidates, 15 individuals are from the Fritzens-Sanzeno dataset. The other 18 samples of the Fritzens-Sanzeno dataset were not labeled as migrant candidates. The southern Tyrolean region only contains 4 different sites, while the southern German region contains 22 different sites.

Discussion In comparison with the results of the southern German and Inneralpine dataset, as shown in Figure 4.22, we observe a higher relative number of migrant candidates from the Fritzens-Sanzeno dataset in these results. Out of the first 15 ranks, five individuals from the Fritzens-Sanzeno dataset are contained. While in total, not substantially higher than the four found between southern Germany and the Inneralpine region, in relation to the smaller sample size of 33 for southern Tyrol and 122 for the Inneralpine region, this shows a clear difference between both results. Additionally, we do not observe a point in our

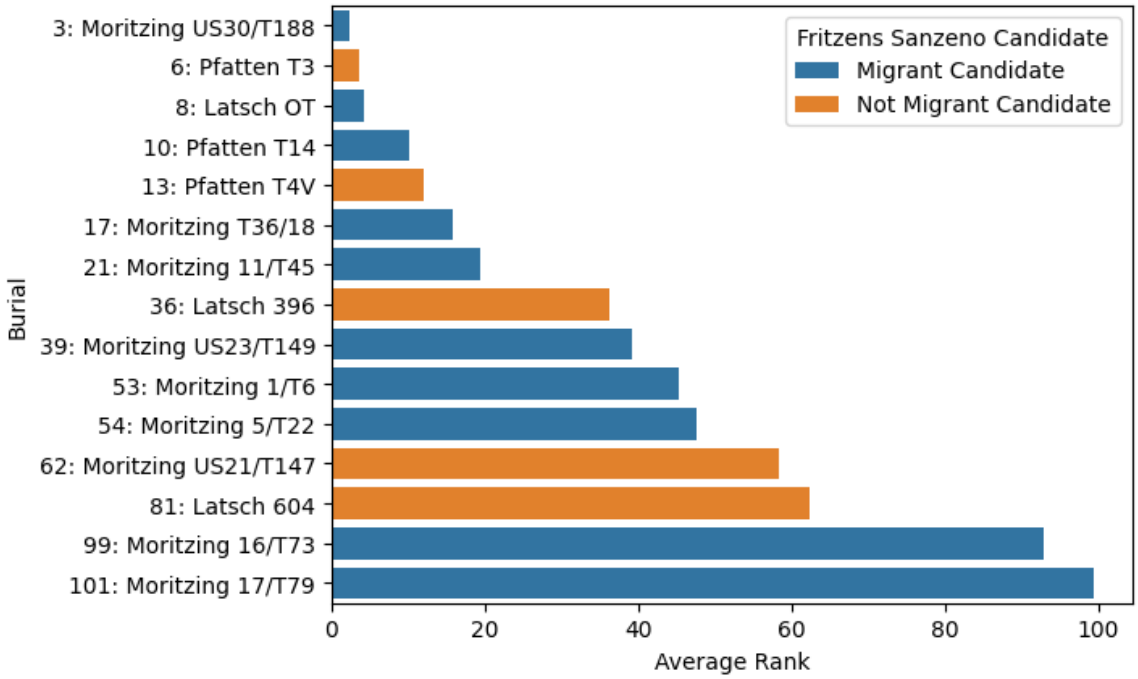


Figure 4.23: Migrant Ranking of the Extended Dataset containing the southern German and southern Tyrolean Regions (1-3)

ranking where the average rank increases rapidly. However, a slight jump in average rank between the sample Moritzing 11/T45 and Latsch 396 can be observed.

We observe that many different SVM settings could perfectly separate the southern German and southern Tyrolean regions. Thus, we conclude that the chosen dataset is suitable for creating an isoscape through the use of SVMs to differentiate between southern German and southern Tyrolean isotopic fingerprints.

Three out of four individuals from Pfatten were ranked very high as migrant candidates. Pfatten T3 was assigned rank 6, Pfatten T14 was assigned rank 10, and Pfatten T4V was assigned rank 13. Either the Pfatten group consists of migrants from southern Germany, or the isoscape of Pfatten is close to that of southern Germany and thus located within the IBI space. The same argument could be made for Moritzing, with nine out of 11 individuals labeled migrant candidates.

Within the results on the Fritzens-Sanzeno dataset, presented in Section 4.4.1, Moritzing US30/T188, Latsch OT, Pfatten T14, Moritzing T36/18 and Moritzing 11/T45 were earlier labeled as migrant candidate, at rank 2, 18, 13, 16, 1 respectively, from the Inneralpine region but now hold early ranks and are labeled as migrant candidates from southern Germany. It is noteworthy that Moritzing T36/18 is known as Moritzing T3/18 within the Fritzens-Sanzeno dataset due to a deviating burial label from the publication of Grupe et al. (2020). With our current knowledge, we can not resolve this ambiguity. Rather, archaeological context or other evidence could help resolve whether these individuals migrated from southern Germany

or the Inneralpine region. The low-ranking individuals from the Fritzens-Sanzeno dataset, Moritzing US30/T188, and Moritzing 11/T45, were captured rather securely as migrants, at rank 1 and rank2, from the Inneralpine region within the Fritzens-Sanzeno dataset results, and thus, we are confident in this labeling and would not dispute it based on these new results. Furthermore, this could indicate that the other Moritzing individuals are mislabeled, as it might be unlikely that 9 out of 11 individuals from a single location would have a migratory background. Without additional domain knowledge, we can not give a decisive answer if it is unlikely that most of the Moritzing individuals have migrated from southern Germany.

Overall, if we disregard the group from Moritzing, our results indicate a migration of the three individuals from Pfatten (T3, T14, T4V) and the individual Latsch OT.

The Inneralpine and southern Tyrolean regions The migrant detection applied on the samples of the Inneralpine and southern Tyrolean regions can be seen in Figure 4.24. This dataset contains 155 samples, and 58 of them, 37.4% of the samples, are predicted as migrant candidates. The Inneralpine subset contains 122 samples, while the southern Tyrolean subset only contains 33. Thus, this dataset is again heavily unbalanced. Out of the 58 samples predicted as migrant candidates, 27 samples were already labeled as migrant candidates within the Fritzens-Sanzeno dataset.

Discussion The set of migrant candidates from this experiment contains 27 of 35 migrant candidates from the migrant detection on the Fritzens-Sanzeno dataset. Thus, it supports our earlier results and does not contradict them. Ten additional samples from the Fritzens-Sanzeno dataset are labeled as migrant candidates, increasing the total number of migrant candidates from the Fritzens-Sanzeno dataset to 45 out of 92. If we consider that only migrations that happened in the last 20 years can be detected, a total of 48.91% migrant candidates seems very high. Our approach seems to be very sensitive and might produce an unknown number of false positives. However, without domain knowledge, no definitive statement can be made if this percentage of migrants is realistic.

Within the early ranks, previously undetected Pfatten T3 (rank 2) and Moritzing US21/T147 (rank 4) emerge as new migrant candidates. The individuals Pfatten T4U and Pfatten T4V also appear at ranks 36 and 37. We also labeled Pfatten T3, earlier ranked 6, and Pfatten T4V, earlier ranked 13, as migrant candidates when comparing the southern German and southern Tyrolean regions. Thus, we can not draw any decisive conclusion about the origin of these individuals but rather produce ambiguous results. They could either be migrants from southern Germany or the Inneralpine region. Especially interesting is the case of Pfatten T3, as it is ranked second and sixth in both experiments while not being ranked when only using the Fritzens-Sanzeno dataset. If we assume, based on the low rank in both experiments, that Pfatten T3 is a migrant, we would expect it to have migrated from southern Germany or one of the Inneralpine locations, which were not included in the Fritzens-Sanzeno dataset.

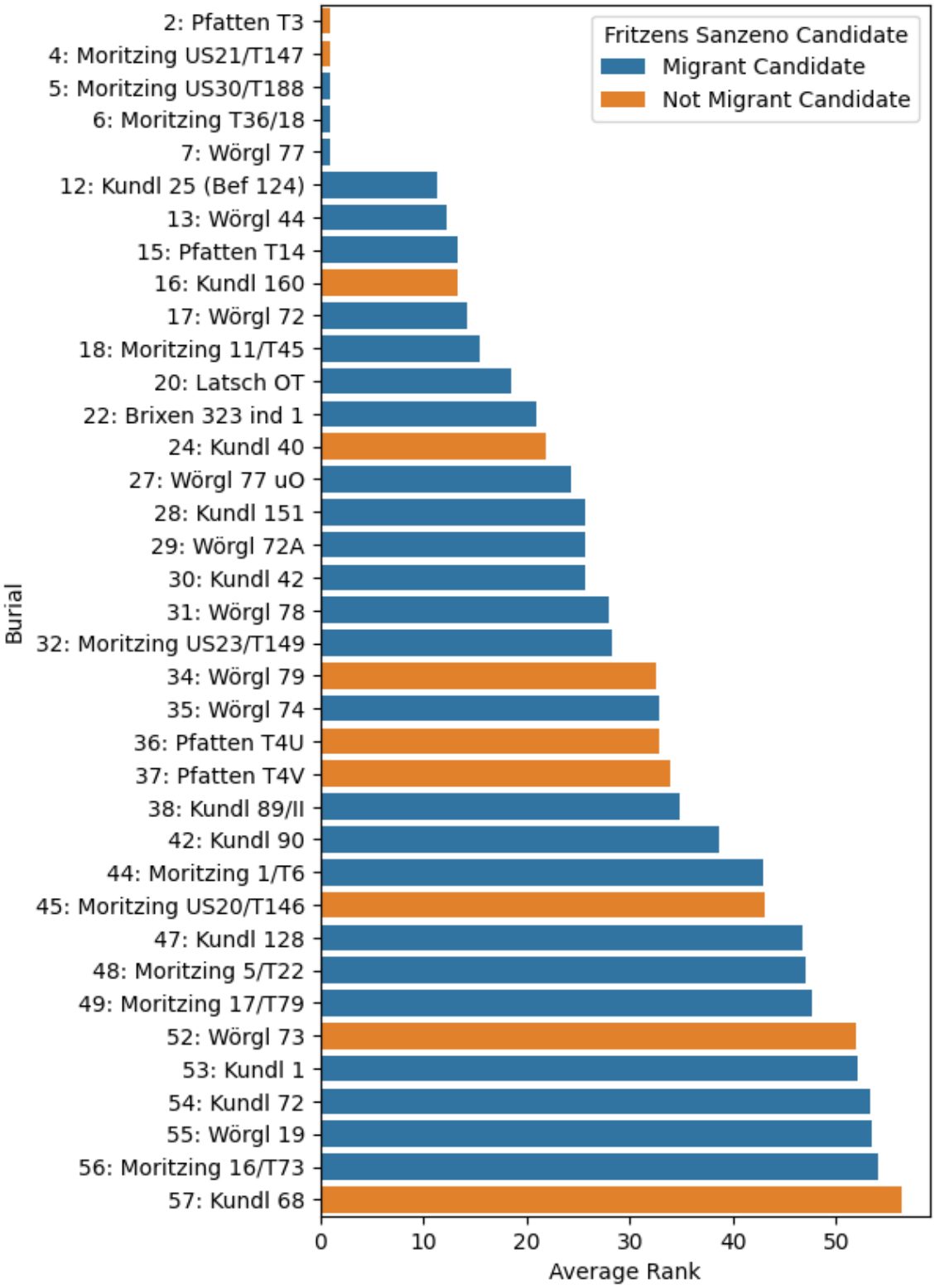


Figure 4.24: Migrant Ranking of the Extended Dataset containing the Inneralpine and southern Tyrolean Regions (2-3)

While we could not validate the idea that Kundl 40 and Kundl 160 originated from southern Germany, we observe that they, within this experiment, got labeled as migrant candidates. Kundl 40 appears at rank 24, and Kundl 160 at rank 16. Although not certain due to the rather high ranks, this could indicate that Kundl 40 and Kundl 160 also originated from one of the Inneralpine regions, which were not included within the Fritzens-Sanzeno dataset. This also provides further evidence that they did not migrate from southern Germany.

Seven individuals were labeled as migrant candidates when using the Fritzens-Sanzeno dataset but were not identified with the extended dataset. This concerns Brixen 324 ind, Moritzing 6/T27, Latsch 653, Latsch 596, Kundl 107, Kundl 91, and Kundl 64. In Section 4.4.1 where we applied our approach on the Fritzens-Sanzeno dataset, they were at ranks 28/22, not identified/29, not identified/33, 30/31, 25/30, 27/27, and 31/34, respectively, for the configurations with the ten overall best models and the best two models per kernel. Thus, all of them either got assigned a high rank for both configurations or were not even captured within the configuration with the ten best models and assigned a high rank when using the two best models per kernel. On the one hand, this highlights that the lower ranks of our migrant candidate ranking consist of insecurely captured migrant candidates. On the other hand, it does lessen the impact of not labeling them as migrant candidates when using the extended dataset.

Overall, we conclude that Pfatten T3, Moritzing US21/T147, Kundl 160, and Kundl 40 pose as promising new migrant candidates, which might originate from a site that was not included within the Fritzens-Sanzeno dataset. Our results indicate that the idea that Kundl 160 and Kundl 40 might originate from southern Germany might not hold true. Lastly, we mostly confirm the findings from applying the migrant detection on the Fritzens-Sanzeno dataset as the extended dataset mostly captured already known migrant candidates and only missed migrant candidates, which were insecurely captured by the Fritzens-Sanzeno dataset.

4.6 Limitations

Definition of IBI and Outliers Our predicate to define an IBI for our use case, as defined in Section 3.1, includes that each IBI instance should be an outlier. However, the natural process of isotopic signature mixing after migration does not guarantee that each data point corresponding to a migrant will be detected as an outlier. Migrants who lived close to 20 years in the new location might not be detectable as outliers from the new location. Their isotopic signal might have adjusted mostly and is not discernible from the distribution of the new location. Our generation of the positive instances for the test set for the evaluation of our algorithm models this linear mixing process. Thus, we will generate points that may have also been generated by the Gaussian models corresponding to the clusters. This will lead to our algorithm being unable to differentiate between points corresponding to either cluster and migrants at the very early or late section of the mixing process and thus will decrease the performance of our algorithm. Within our experiment on the impact of the migration progress, we calculated the Mahalanobis distance of the samples to the closest distribution,

as seen in Figure 4.10. We observed that after 18 years, the Mahalanobis distance of the IBI to its reference cluster is less than the mean plus three times the standard deviation. Thus, one could argue that these points should not be classified as outliers. Section 4.3.3 shows that samples close to the reference cluster, which might not be classified as outliers, are the hardest to predict for our algorithm.

Missing Outliers in Synthethic Test Set As we are not able to generate outliers, which are not IBIs, without using our own algorithm to detect whether they are outliers or not, our evaluation of our algorithm on the synthetic dataset does not cover every aspect of our algorithm. In particular, the boundary perpendicular to the CCV is not sufficiently evaluated. When looking at the results of the first sub-experiment of the preparatory experiments on synthetic data, as described within Section 4.1, we see the impact of the multiplier of the threshold to the CCV. We observe that the F1-Scores increase when we increase the multiplier until we reach the multiplier 3.0. Afterwards, the results remain the same. We additionally execute this experiment with a multiplier of 100. We obtain exactly the same results as with a multiplier of 5. Thus, we derive that the width around the CCV already captures all IBIs with a multiplier of 3.0. Increasing the threshold should, in a real-world scenario, lead to us including outliers that are not IBIs. As our test set does not contain this kind of point, we do not see an increase in false positive samples. Thus, a larger threshold for the distance to the CCV for our test set does not come with the tradeoff of mislabeling outliers as IBIs.

5 Conclusion and Future Work

In this chapter, we will draw the final conclusions on our work and evaluate whether we achieved our goals. Additionally, we will provide an outlook on future work on the topic of identifying IBIs with the help of SVMs.

5.1 Conclusion

In this master's thesis, we set out to achieve several goals in the context of identifying migrants within isotope datasets using machine learning methods. This use-case was modeled after the real-world application of identifying mixed multi-isotope signals, which can be observed after an individual migrates to a new location and the individual's isotope signal begins aligning with the new location's isotope signal. To our knowledge, we propose the first algorithm that combines the concept of cluster in-betweenness with multi-isotope provenance research. In Section 1.4, we set three goals for this thesis.
The first goal was to determine if SVMs are suitable for detecting instances in-between two known clusters. To answer this question, we developed an approach that identifies the space in-between two clusters based on the model provided by the SVM and the concept of the cluster connecting vector. This algorithm determines if a sample contains a mixture of the characteristics of both reference clusters. The algorithm discerns between in-between instances and samples belonging to the reference clusters by identifying the position of the sample in relation to the separating boundary. We introduced the cluster connecting vector as a representative of a perfect linear mixing process between the reference clusters to exclude outliers and samples that show a mixture of characteristics from unknown clusters. Our algorithm identifies whether a sample deviates too far from the assumed mixing process, thereby excluding noise from our predictions. We conducted a series of experiments on a synthetic dataset to evaluate the algorithm's performance and robustness. In these experiments, our approach demonstrated its ability to detect in-between instances in various settings.

Our second goal was to develop a method to evaluate the extent to which the linear mixing process of an isotope signal resulting from migration has advanced. To achieve this, we compared the sample's current position with the cluster connecting vector, which represents a perfect linear remodeling process. When evaluating this approach, we achieved a Mean Squared Error of 0.996, which proved its efficacy.

While the first two goals contained the data science aspects of this thesis, the last goal was dedicated to the domain application of our algorithm. We applied our algorithm to a real-world multi-isotope dataset containing isotope ratios sampled from archaeological human

remains from the Fritzens-Sanzeno culture. An entirely data-driven multi-step approach was used to obtain a ranking of migrant candidates. We provided 34 migrant candidates and ranked them according to our confidence in their labeling. Our results reproduced earlier research on the influence of migration on the genesis of the Fritzens-Sanzeno culture while additionally providing previously unidentified candidates for migration between the inner Alpine and southern Tyrolean research areas. We identified the most challenging migrant candidates whose isotopic signatures had already adjusted to the new location, as they are predicted to have migrated close to 20 years ago.

Our work provides a novel algorithm for provenance research in multi-isotope datasets. It offers an entirely data-driven approach that can be leveraged to obtain a ranking of migrant candidates or provides an individual choice on the trade-off between capturing hard-to-identify migrants and more secure labeling by manually setting the algorithm's parameters. This thesis has demonstrated the potential of SVMs combined with the concept of cluster in-betweenness for identifying migrants within multi-isotope datasets, while achieving our set goals.

5.2 Future Work

Despite the promising results, there are several areas where future research could enhance the detection of IBIs using SVMs.

To determine the orientation of the reference clusters, we calculate the CCV between both clusters. Using the distance to the CCV, we define the boundaries for the IBI space. However, in high-dimensional feature spaces, the interpretability of distances deteriorates due to the curse of dimensionality. To address this issue, instead of relying on the CCV and a distance measure, we can calculate the angles between the samples of both clusters and set a threshold for the angle of a sample to the cluster centers accordingly.

We evaluated our method with a wide range of experiments on a synthetic dataset. However, this can not cover all real-world applications of our approach. Therefore, we recommend applying our method to different real-world isotope datasets to assess the transferability of our results. Analyzing real-world data with a known migrant or IBI label could give new insights into the applicability of our approach in different settings. While our method was designed specifically for the use case of migrant detection within isotope datasets, its underlying concept of in-betweenness may be relevant to other domains, potentially extending the applicability of our approach beyond its original context.

Within the synthetic dataset, we generated a multidimensional Gaussian distribution where each feature was uncorrelated. Consequently, we did not evaluate the effectiveness of our algorithm on correlated datasets. To address this limitation, we propose further experiments on a synthetic dataset with correlated features.

The IBI predicates introduced in the work of Kazempour et al. (2024) and discussed in Section 1.3.3 could also be applied to this problem. A comparison between the results of our

approach and those obtained using these predicates could provide valuable insights and serve as another evaluation of the applicability of the IBI concept on isotopic migrant detection.

When we applied our approach to the isotope datasets, we faced challenges due to limited preprocessing capabilities. Our lack of domain knowledge prevented us from effectively excluding outlier samples, which is particularly difficult since these outliers might be potential migrants, which we want to keep in the dataset. We were unable to distinguish between outliers that are not migrants and those that could be migrants. Nevertheless, outliers and noise could still affect the position of the SVM's decision boundary and, consequently, the IBI space derived from our approach. Therefore, further research in collaboration with archaeologists is necessary to address this issue.

In our analysis of the isotope datasets, we used multiple burial sites for each region. To gain more precise insights into the exact origins of ancient humans, it would be beneficial to compare our results with datasets containing fewer burial sites, or even datasets with a single site per region. This approach could help isolate specific regional isotopic signatures and improve our understanding of human migration patterns. Further research with these focused datasets could provide more detailed insights into the geographic origins and movements of ancient populations.

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.1 Appendix Fritzens-Sanzeno Dataset

Table 1: Fritzens-Sanzeno Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Moritzing	11/T45	0.71199	38.8047	15.6949	18.56	2.4725	1.1826
Moritzing	16/T73	0.714	38.8643	15.708	18.5634	2.4743	1.1818
Moritzing	17/T79	0.71514	38.81	15.6993	18.6072	2.4722	1.1852
Moritzing	US20/T146	0.71519	38.9103	15.7156	18.8394	2.4759	1.1987
Moritzing	US21/T147	0.71261	38.8232	15.6732	18.6501	2.477	1.1899
Moritzing	US23/149	0.71283	38.8364	15.7028	18.6273	2.4732	1.1862
Moritzing	US30/188	0.71105	38.7344	15.6723	18.5233	2.4715	1.1819
Moritzing	5/T22	0.71307	38.8137	15.6933	18.5943	2.4733	1.1849
Moritzing	6/T27	0.71562	38.909	15.723	18.5579	2.4747	1.1803
Moritzing	T3/18	0.71277	38.6687	15.6544	18.5191	2.4701	1.183
Moritzing	1/T6	0.71335	38.8064	15.694	18.6464	2.4727	1.1881
Pfatten	T14	0.7125	38.741	15.6724	18.71	2.4721	1.1938
Pfatten	T3	0.71065	38.8792	15.6926	18.846	2.4774	1.2009
Pfatten	T4U	0.71497	38.8043	15.668	18.7047	2.4767	1.1938
Pfatten	T4V	0.71276	38.8003	15.682	18.7744	2.4742	1.1972
Latsch	396	0.71816	38.5362	15.6493	18.7432	2.4625	1.1977
Latsch	447	0.72058	38.4894	15.6455	18.5992	2.4601	1.1888
Latsch	454	0.72167	38.5957	15.659	18.7497	2.4647	1.1974
Latsch	537	0.72144	38.5092	15.6497	18.6467	2.4607	1.1915
Latsch	596	0.72185	38.3693	15.6255	18.5451	2.4556	1.1869
Latsch	604	0.71925	38.5181	15.6512	18.8594	2.4611	1.205
Latsch	642	0.71955	38.4733	15.6427	18.6462	2.4595	1.192
Latsch	651	0.72048	38.5716	15.6451	18.5674	2.4654	1.1868
Latsch	652f	0.72147	38.5453	15.6535	18.6984	2.4624	1.1946
Latsch	653	0.72141	38.3666	15.6185	18.465	2.4566	1.1822
Latsch	655	0.72228	38.4996	15.6512	18.4639	2.4599	1.1797
Latsch	657	0.71832	38.6085	15.6658	18.717	2.4645	1.1948
Latsch	O.T.	0.71568	38.5144	15.631	18.5198	2.464	1.1848
Latsch	592	0.72178	38.5489	15.6442	18.7382	2.4641	1.1978
Latsch	652D	0.72148	38.565	15.6357	18.7286	2.4664	1.1978
Latsch	652C	0.72266	38.4858	15.6332	18.6716	2.4618	1.1944
Brixen	323 ind. 1	0.71697	38.649	15.6749	18.4703	2.46568	1.1783
Brixen	324 ind.	0.7182	38.5423	15.6499	18.4316	2.46279	1.1777
Kundl	33	0.70907	38.3803	15.6387	18.3528	2.4542	1.1735
Kundl	48	0.71197	38.4711	15.6542	18.4126	2.4576	1.1762
Kundl	53	0.7092	38.4992	15.6619	18.4243	2.4581	1.1764
Kundl	37	0.7107	38.3964	15.6396	18.3664	2.4551	1.1743
Kundl	73	0.71083	38.514	15.6581	18.4705	2.4598	1.1796
Kundl	7	0.70929	38.2446	15.6349	18.274	2.4461	1.1688
Kundl	64	0.71054	38.5661	15.6695	18.475	2.4612	1.179
Kundl	151	0.71029	38.8297	15.7269	18.5329	2.4691	1.1784
Kundl	75	0.71094	37.9895	15.5847	18.1376	2.4376	1.1638
Kundl	85	0.71139	38.3824	15.6396	18.3857	2.4542	1.1756
Kundl	58	0.70982	38.5737	15.6698	18.4882	2.4616	1.1799
Kundl	89/I	0.71125	38.1815	15.6248	18.2135	2.4437	1.1656

Table 1: Fritzens-Sanzeno Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Kundl	85	0.71131	38.5242	15.6697	18.459	2.4585	1.178
Kundl	72	0.71084	38.5631	15.6581	18.417	2.4628	1.1762
Kundl	27	0.71123	38.426	15.6504	18.4032	2.4553	1.1759
Kundl	90	0.7121	38.6756	15.6699	18.6006	2.4682	1.1871
Kundl	95	0.71113	38.3164	15.6147	18.4447	2.454	1.1812
Kundl	91	0.70971	38.5585	15.6532	18.5142	2.4634	1.1828
Kundl	111	0.71025	38.314	15.6316	18.3392	2.4513	1.1733
Kundl	115	0.71104	38.5572	15.6784	18.4518	2.4593	1.1769
Kundl	131	0.70918	38.3771	15.645	18.4471	2.453	1.1791
Kundl	139	0.71068	38.2069	15.6241	18.2601	2.4453	1.1687
Kundl	104	0.71077	38.3973	15.6436	18.4348	2.4545	1.1784
Kundl	23	0.71045	38.499	15.6558	18.4131	2.45924	1.1761
Kundl	36	0.70909	38.5338	15.66	18.4409	2.46078	1.1776
Kundl	42	0.71194	38.7346	15.6973	18.5736	2.46791	1.1833
Kundl	180	0.70927	38.5048	15.6579	18.4137	2.45905	1.176
Kundl	52	0.71091	38.5134	15.6642	18.4262	2.45873	1.1764
Kundl	105	0.71038	38.5119	15.6619	18.4189	2.45892	1.1761
Kundl	113	0.711	38.5164	15.6592	18.4018	2.45974	1.1752
Kundl	114	0.71101	38.5353	15.6721	18.4405	2.45892	1.1764
Kundl	165	0.70978	38.5202	15.6634	18.4227	2.45933	1.1762
Kundl	128	0.71184	38.5812	15.667	18.425	2.46256	1.176
Kundl	22	0.70921	38.4922	15.6564	18.4085	2.45863	1.1758
Kundl	Komplex G90	0.71022	38.4698	15.6475	18.4145	2.4588	1.1768
Kundl	107	0.71049	38.4768	15.6349	18.5226	2.46093	1.1847
Kundl	106	0.70923	38.4837	15.6535	18.3939	2.45845	1.1751
Kundl	19	0.71131	38.478	15.654	18.4167	2.45802	1.1764
Kundl	25 (Bef 124)	0.71613	38.5164	15.665	18.4232	2.45904	1.1761
Kundl	38	0.71031	38.4853	15.6533	18.4018	2.45854	1.1756
Kundl	1	0.71146	38.567	15.6731	18.4302	2.46065	1.1759
Kundl	154	0.71031	38.5845	15.6775	18.5049	2.46124	1.1804
Kundl	25 (Bef L10)	0.71129	38.4394	15.6421	18.4005	2.4574	1.1763
Kundl	68	0.71076	38.5559	15.6771	18.42	2.45929	1.1749
Kundl	164	0.71053	38.4395	15.6412	18.3915	2.45763	1.1758
Wörgl	17	0.70908	38.2379	15.6132	18.2944	2.44912	1.1717
Wörgl	19	0.71125	38.5182	15.6518	18.4284	2.46096	1.1774
Wörgl	44	0.71262	38.686	15.6711	18.5611	2.46863	1.1844
Wörgl	72	0.71345	38.5499	15.6545	18.4792	2.46255	1.1805
Wörgl	72A	0.71246	38.7026	15.6819	18.6233	2.468	1.1877
Wörgl	73	0.71443	38.3304	15.6379	18.3196	2.45113	1.1714
Wörgl	74	0.71238	38.6943	15.6791	18.4943	2.4679	1.1795
Wörgl	77	0.71472	38.587	15.6566	18.455	2.46454	1.1787
Wörgl	77 u.O.	0.71291	38.6787	15.6739	18.5889	2.46772	1.186
Wörgl	78	0.71472	38.455	15.6358	18.5031	2.45953	1.1833
Wörgl	79	0.71438	38.3967	15.6504	18.3436	2.4534	1.1721
Kundl	160	0.71012	38.9675	15.7393	18.7144	2.4758	1.189
Kundl	89/II	0.71079	38.6657	15.6452	18.7644	2.4714	1.1994
Kundl	40	0.71093	38.7756	15.6589	18.8423	2.4763	1.2033

.2 Appendix Extended Dataset

Table 2: Extended Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Eching	26	0.70873	38.38	15.65	18.61	2.45	1.19
Eching	49	0.7085	38.54	15.68	18.88	2.46	1.2
Eching	130	0.70872	38.7	15.71	19.89	2.46	1.27
Eching	140	0.70822	38.4	15.64	18.51	2.46	1.18
Eching	148	0.70858	38.52	15.67	19.06	2.46	1.22
Eching	151	0.70864	38.72	15.74	19.3	2.46	1.23
Eching	156	0.70863	38.38	15.63	18.7	2.46	1.2
Eching	183	0.70932	38.38	15.65	18.91	2.45	1.21
Eching	205	0.70932	38.51	15.67	19.09	2.46	1.22
Eching	219	0.70933	38.64	15.71	20.01	2.46	1.27
Eching	222	0.7088	38.64	15.7	19.64	2.46	1.25
Eching	250	0.70862	38.6	15.71	19.76	2.46	1.26
Eching	254	0.70854	38.59	15.69	19.39	2.46	1.24
Eching	258	0.7096	38.47	15.64	18.71	2.46	1.2
Eching	267	0.70852	38.6	15.69	19.73	2.46	1.26
Eching	346	0.70861	38.38	15.73	19.57	2.44	1.24
Eching	412	0.7087	38.26	15.65	18.68	2.44	1.19
Eching	626	0.71018	38.48	15.69	19.19	2.45	1.22
Eching	721	0.70895	38.42	15.69	19.45	2.45	1.24
Eching	727	0.70872	38.68	15.73	20.03	2.46	1.27
Eching	5105	0.70874	38.78	15.75	19.9	2.46	1.26
Englschalking	14	0.7086	38.57	15.69	19.35	2.46	1.23
Englschalking	15	0.70909	38.76	15.68	18.86	2.47	1.2
Grünwald	U11	0.70926	38.76	15.71	19.66	2.47	1.25
Grünwald	Gw2F G2	0.70902	38.57	15.7	19.3	2.46	1.23
Grünwald	FA	0.70898	38.63	15.68	19.21	2.46	1.23
Grünwald	J31	0.70962	38.69	15.69	19.4	2.47	1.24
Grünwald	Stelle 46	0.70899	38.78	15.72	18.78	2.47	1.2
Grünwald	Stelle 63	0.70991	38.7	15.68	19.15	2.47	1.22
Grünwald	Stelle 101	0.70872	38.82	15.77	19.13	2.46	1.21
Grünwald	293	0.70849	38.5	15.64	18.82	2.46	1.2
Grünwald	382	0.71083	38.75	15.7	19.47	2.47	1.24
Grünwald	503	0.70886	38.61	15.67	18.83	2.46	1.2
Hofoldinger Forst	11	0.71153	38.68	15.7	19.59	2.46	1.25
Hofoldinger Forst	17	0.7089	38.49	15.68	19.3	2.46	1.23
Hofoldinger Forst	4	0.71009	38.59	15.68	19.27	2.46	1.23
Langengeisling	4	0.7098	38.25	15.61	18.42	2.45	1.18
Langengeisling	6	0.71016	38.78	15.68	19.04	2.47	1.21
Obermenzing	1/1951	0.7096	38.61	15.69	19.14	2.46	1.22
Obermenzing	1/1983	0.7085	38.59	15.68	19.07	2.46	1.22
Obermenzing	1/1987	0.70869	38.49	15.66	18.88	2.46	1.21
Obermenzing	19/1985	0.70909	38.4	15.62	18.52	2.46	1.19
Obermenzing	22/1984	0.70881	38.69	15.69	19.37	2.47	1.23
Obermenzing	23/1984	0.70849	38.47	15.67	19.04	2.45	1.21
Obermenzing	26/1984	0.70832	38.43	15.65	18.82	2.45	1.2
Obermenzing	27/1984	0.70827	38.48	15.67	18.71	2.46	1.19
Obermenzing	29/1984	0.70844	38.5	15.66	19.13	2.46	1.22
Obermenzing	31/1984	0.70847	38.39	15.66	18.94	2.45	1.21
Obermenzing	34/1984	0.70844	38.52	15.67	19.22	2.46	1.23
Obermenzing	38/1984	0.7084	38.57	15.67	18.55	2.46	1.18
Obermenzing	39/1984	0.70845	38.61	15.66	18.92	2.47	1.21
Obermenzing	4/1982	0.70852	38.61	15.69	19.15	2.46	1.22
Obermenzing	45/1984	0.70798	38.48	15.66	18.87	2.46	1.2
Obermenzing	5/1951	0.70873	38.34	15.63	18.65	2.45	1.19
Obermenzing	5/1982	0.70877	38.57	15.68	19.26	2.46	1.23
Obermenzing	5/1985	0.70862	38.17	15.6	18.65	2.45	1.2
Obermenzing	50/1984	0.70839	38.45	15.65	18.72	2.46	1.2

Table 2: Extended Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Obermenzing	55/1984	0.70867	38.59	15.73	19.21	2.45	1.22
Obermenzing	7/1985	0.70863	38.54	15.71	18.54	2.45	1.18
Obermenzing	14/1985	0.70914	38.27	15.61	18.66	2.45	1.2
Obermenzing	9/1985	0.70852	38.46	15.63	18.57	2.46	1.19
Poing	B6/15	0.70846	38.65	15.71	19.37	2.46	1.23
Poing	B8	0.7122	38.63	15.67	19.2	2.46	1.22
Poing	B5	0.71126	38.76	15.71	19.25	2.47	1.22
Trudering	153	0.70837	38.55	15.7	19.48	2.46	1.24
Trudering	156	0.71166	38.68	15.7	19.58	2.46	1.25
Trudering	1300	0.70863	38.57	15.69	19.3	2.46	1.23
Unterhaching	92	0.70841	38.26	15.65	18.68	2.45	1.19
Unterhaching	117	0.70998	38.69	15.75	19.45	2.46	1.24
Unterhaching	125	0.7086	38.29	15.59	18.69	2.46	1.2
Waging am See	U1	0.70851	38.76	15.67	19	2.47	1.21
Waging am See	U1/04	0.70846	38.7	15.68	19.15	2.47	1.22
Waging am See	U13	0.70868	38.47	15.63	18.69	2.46	1.2
Gernlinden	60	0.70874	38.65	15.7	19.24	2.46	1.23
Gernlinden	141	0.70937	38.71	15.73	19.3	2.46	1.23
Gernlinden	170	0.70868	38.53	15.67	18.69	2.46	1.19
Aubing	130	0.70914	38.73	15.69	19.3	2.47	1.23
Aubing	22	0.70897	38.8	15.7	19.4	2.47	1.24
Aubing	24	0.71062	38.62	15.67	19.07	2.46	1.22
Aubing	521	0.70908	38.7	15.69	19.09	2.47	1.22
Forstinning	2	0.71045	38.8	15.7	19.29	2.47	1.23
Forstinning	6	0.70992	38.71	15.69	19.24	2.47	1.23
Aschheim	A Block	0.7085	38.76	15.68	18.79	2.47	1.2
Kirchheim	4	0.70872	38.56	15.66	18.94	2.46	1.21
Kleinaitingen	4	0.70876	38.58	15.75	19.99	2.45	1.27
Kleinaitingen	15	0.70844	38.35	15.65	18.74	2.45	1.2
Kleinaitingen	18	0.70851	38.33	15.64	18.58	2.45	1.19
Kleinaitingen	24	0.71011	38.45	15.68	19.37	2.45	1.24
Kleinaitingen	25	0.70826	38.56	15.69	19.69	2.46	1.26
Kleinaitingen	28	0.70903	38.32	15.65	18.76	2.45	1.2
Kleinaitingen	62	0.70938	38.44	15.65	18.79	2.46	1.2
Kleinaitingen	63	0.70917	38.34	15.68	19.19	2.45	1.22
Kleinaitingen	64	0.70907	38.62	15.68	19.4	2.46	1.24
Kleinaitingen	65	0.70912	38.44	15.71	19.56	2.45	1.25
Kleinaitingen	66	0.70912	38.42	15.68	19.35	2.45	1.23
Kleinaitingen	67	0.7087	38.41	15.65	19.09	2.45	1.22
Kleinaitingen	68	0.70948	38.31	15.63	18.36	2.45	1.17
Kleinaitingen	71	0.70905	38.69	15.69	19.12	2.47	1.22
Kleinaitingen	161	0.70985	38.47	15.73	19.67	2.45	1.25
Kleinaitingen	162	0.70898	38.48	15.68	18.93	2.46	1.21
Kleinaitingen	164	0.70898	38.48	15.68	19.19	2.45	1.22
Kleinaitingen	167	0.71001	38.42	15.66	18.91	2.45	1.21
Kleinaitingen	179	0.70902	38.56	15.69	18.94	2.46	1.21
Kleinaitingen	186	0.70846	38.15	15.73	19.79	2.42	1.26
Kleinaitingen	189	0.7084	38.38	15.62	18.75	2.46	1.2
Kleinaitingen	190	0.70836	38.54	15.68	19.17	2.46	1.22
Kleinaitingen	168A	0.70872	38.23	15.66	19.05	2.44	1.22
Königsbrunn-Zeller	6	0.7088	38.4	15.67	19.06	2.45	1.22
München		0.70872	38.25	15.66	18.84	2.44	1.2
Zuchering	137	0.70893	38.54	15.66	19.02	2.46	1.21
Zuchering	145	0.70838	38.68	15.68	18.84	2.47	1.2
Zuchering	146	0.70842	38.76	15.69	19.02	2.47	1.21
Zuchering	177	0.70916	38.83	15.68	19.05	2.48	1.22
Zuchering	336	0.70886	38.69	15.67	18.78	2.47	1.2
Zuchering	371	0.70903	38.57	15.67	18.91	2.46	1.21
Zuchering	380	0.70889	38.43	15.63	18.77	2.46	1.2
Zuchering	384	0.70864	38.43	15.63	18.65	2.46	1.19
Zuchering	392	0.70872	38.5	15.64	18.9	2.46	1.21

Table 2: Extended Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Zuchering	404	0.70844	38.53	15.67	18.85	2.46	1.2
Zuchering	411	0.70922	38.52	15.66	18.98	2.46	1.21
Zuchering	503	0.70919	38.42	15.63	18.76	2.46	1.2
Zuchering	506	0.70933	38.54	15.68	18.64	2.46	1.19
Zuchering	538	0.70898	38.65	15.65	18.63	2.47	1.19
Zuchering	545	0.70957	38.57	15.65	19.02	2.47	1.22
Zuchering	551	0.70968	38.42	15.65	18.47	2.45	1.18
Garching an der Alz	9	0.71002	38.47	15.69	19.29	2.45	1.23
Garching an der Alz	15	0.70878	38.16	15.69	19.11	2.43	1.22
Flintsbach am Inn	A	0.70848	38.77	15.68	18.79	2.47	1.2
Flintsbach am Inn	B	0.70861	38.78	15.68	18.73	2.47	1.19
Flintsbach am Inn	C	0.70848	38.23	15.63	18.31	2.45	1.17
Flintsbach am Inn	E	0.70832	38.6	15.65	18.65	2.47	1.19
Kelheim	19	0.7095	38.58	15.67	18.81	2.46	1.2
Kelheim	26	0.70896	38.46	15.65	18.65	2.46	1.19
Kelheim	29	0.70932	38.66	15.69	18.94	2.46	1.21
Kelheim	33	0.7096	38.11	15.59	18.29	2.44	1.17
Kelheim	38	0.70936	38.26	15.62	18.46	2.45	1.18
Kelheim	54	0.70974	38.55	15.67	18.83	2.46	1.2
Kelheim	urn 56	0.70936	38.72	15.69	18.95	2.47	1.21
Künzing	19	0.70931	38.75	15.71	19.1	2.46	1.2
Künzing	20	0.70929	38.6	15.68	18.9	2.46	1.21
Künzing	3 (B1820)	0.7094	38.62	15.67	18.92	2.46	1.21
Künzing	7 (B1381)	0.70928	38.65	15.71	18.86	2.46	1.2
Künzing	88 (B1917)	0.70895	38.61	15.67	18.95	2.46	1.21
Künzing	A (Obj-Nr 2155)	0.70932	38.56	15.67	18.39	2.46	1.17
Künzing	B (Obj-Nr 2264)	0.70903	38.68	15.69	18.84	2.47	1.2
Künzing	D (Obj-Nr 2448)	0.7091	38.76	15.73	19.15	2.46	1.22
Wörgl	17	0.70908	38.24	15.61	18.29	2.45	1.17
Wörgl	19	0.71125	38.52	15.65	18.43	2.46	1.18
Wörgl	44	0.71262	38.69	15.67	18.56	2.47	1.18
Wörgl	72	0.71345	38.55	15.65	18.48	2.46	1.18
Wörgl	72A	0.71246	38.7	15.68	18.62	2.47	1.19
Wörgl	73	0.71443	38.33	15.64	18.32	2.45	1.17
Wörgl	74	0.71238	38.69	15.68	18.49	2.47	1.18
Wörgl	77	0.71472	38.59	15.66	18.46	2.46	1.18
Wörgl	77 uO	0.71291	38.68	15.67	18.59	2.47	1.19
Wörgl	78	0.71472	38.46	15.64	18.5	2.46	1.18
Wörgl	79	0.71438	38.4	15.65	18.34	2.45	1.17
Hötting	20	0.71788	38.48	15.67	18.28	2.46	1.17
Hötting	69	0.70875	38.44	15.66	18.74	2.45	1.2
Hötting	73	0.70877	38.35	15.64	18.78	2.45	1.2
Hötting	79	0.70891	38.58	15.66	18.64	2.46	1.19
Hötting	62	0.70889	37.92	15.99	22.82	2.37	1.43
Hötting	67	0.7087	38.2	15.63	18.31	2.44	1.17
Wilten	16	0.71094	38.68	15.68	18.68	2.47	1.19
Wilten	8	0.71024	38.51	15.65	18.5	2.46	1.18
Mühlau	15	0.71189	38.45	15.66	18.29	2.46	1.17
Kundl	33	0.70907	38.38	15.64	18.35	2.45	1.17
Kundl	48	0.71197	38.47	15.65	18.41	2.46	1.18
Kundl	53	0.7092	38.5	15.66	18.42	2.46	1.18
Kundl	37	0.7107	38.4	15.64	18.37	2.46	1.17
Kundl	73	0.71083	38.51	15.66	18.47	2.46	1.18
Kundl	7	0.70929	38.24	15.63	18.27	2.45	1.17
Kundl	64	0.71054	38.57	15.67	18.48	2.46	1.18
Kundl	151	0.71029	38.83	15.73	18.53	2.47	1.18
Kundl	160	0.71012	38.97	15.74	18.71	2.48	1.19
Kundl	75	0.71094	37.99	15.58	18.14	2.44	1.16
Kundl	85	0.71139	38.38	15.64	18.39	2.45	1.18
Kundl	58	0.70982	38.57	15.67	18.49	2.46	1.18
Kundl	89/I	0.71125	38.18	15.62	18.21	2.44	1.17

Table 2: Extended Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Kundl	85	0.71131	38.52	15.67	18.46	2.46	1.18
Kundl	72	0.71084	38.56	15.66	18.42	2.46	1.18
Kundl	27	0.71123	38.43	15.65	18.4	2.46	1.18
Kundl	90	0.7121	38.68	15.67	18.6	2.47	1.19
Kundl	95	0.71113	38.32	15.61	18.44	2.45	1.18
Kundl	L79	0.71092	38.39	15.63	18.43	2.46	1.18
Kundl	91	0.70971	38.56	15.65	18.51	2.46	1.18
Kundl	L428	0.71111	38.64	15.68	18.56	2.46	1.18
Kundl	L76	0.70975	38.76	15.69	18.63	2.47	1.19
Kundl	111	0.71025	38.31	15.63	18.34	2.45	1.17
Kundl	115	0.71104	38.56	15.68	18.45	2.46	1.18
Kundl	554	0.70921	38.49	15.66	18.49	2.46	1.18
Kundl	131	0.70918	38.38	15.65	18.45	2.45	1.18
Kundl	139	0.71068	38.21	15.62	18.26	2.45	1.17
Kundl	733	0.71212	38.53	15.66	18.49	2.46	1.18
Kundl	89/II	0.71079	38.67	15.65	18.76	2.47	1.2
Kundl	839	0.7099	38.46	15.64	18.45	2.46	1.18
Kundl	104	0.71077	38.4	15.64	18.43	2.45	1.18
Kundl	23	0.71045	38.5	15.66	18.41	2.46	1.18
Kundl	36	0.70909	38.53	15.66	18.44	2.46	1.18
Kundl	40	0.71093	38.78	15.66	18.84	2.48	1.2
Kundl	42	0.71194	38.73	15.7	18.57	2.47	1.18
Kundl	180	0.70927	38.5	15.66	18.41	2.46	1.18
Kundl	52	0.71091	38.51	15.66	18.43	2.46	1.18
Kundl	105	0.71038	38.51	15.66	18.42	2.46	1.18
Kundl	113	0.711	38.52	15.66	18.4	2.46	1.18
Kundl	114	0.71101	38.54	15.67	18.44	2.46	1.18
Kundl	165	0.70978	38.52	15.66	18.42	2.46	1.18
Kundl	128	0.71184	38.58	15.67	18.42	2.46	1.18
Kundl	22	0.70921	38.49	15.66	18.41	2.46	1.18
Kundl	Komplex G90	0.71022	38.47	15.65	18.41	2.46	1.18
Kundl	107	0.71049	38.48	15.63	18.52	2.46	1.18
Kundl	106	0.70923	38.48	15.65	18.39	2.46	1.18
Kundl	19	0.71131	38.48	15.65	18.42	2.46	1.18
Kundl	25 (Bef 124)	0.71613	38.52	15.66	18.42	2.46	1.18
Kundl	38	0.71031	38.49	15.65	18.4	2.46	1.18
Kundl	1	0.71146	38.57	15.67	18.43	2.46	1.18
Kundl	154	0.71031	38.58	15.68	18.5	2.46	1.18
Kundl	25 (Bef L10)	0.71129	38.44	15.64	18.4	2.46	1.18
Kundl	68	0.71076	38.56	15.68	18.42	2.46	1.17
Kundl	164	0.71053	38.44	15.64	18.39	2.46	1.18
Mühlbachl-Matrei	36	0.70971	38.53	15.62	18.81	2.46	1.2
Mühlbachl-Matrei	83	0.70982	38.45	15.63	18.54	2.46	1.19
Vomp	29	0.71042	38.62	15.68	18.57	2.46	1.18
Vomp	19230/332 9/	0.71191	38.71	15.69	19.01	2.47	1.21
Vomp	248	0.70887	38.58	15.7	18.49	2.46	1.18
Vomp	250	0.70956	38.48	15.67	18.71	2.45	1.19
Vomp	305	0.7093	38.76	15.71	18.95	2.47	1.21
Vomp	304	0.70946	38.71	15.69	18.68	2.47	1.19
Vomp	454	0.7096	38.67	15.72	19	2.46	1.21
Ampass	15	0.71353	38.48	15.67	18.51	2.46	1.18
Kitzbühel	18/08	0.7109	38.61	15.68	18.67	2.46	1.19
Kitzbühel	19/08	0.7116	38.68	15.71	18.55	2.46	1.18
Kitzbühel	20/08	0.70914	38.67	15.67	18.56	2.47	1.18
Kitzbühel	21/08	0.71113	38.76	15.71	18.41	2.47	1.17
Kitzbühel	23/08	0.71129	38.71	15.67	18.66	2.47	1.19
Kitzbühel	7/07	0.71022	38.74	15.68	18.71	2.47	1.19
Fügen-Kapfing	18	0.71593	38.7	15.67	18.55	2.47	1.18
Fügen-Kapfing	19	0.71734	38.74	15.68	18.51	2.47	1.18
Ellbögen	1993	0.71272	38.73	15.69	18.56	2.47	1.18
Ellbögen	4	0.71591	38.61	15.65	18.71	2.47	1.2

Table 2: Extended Dataset

Site	Burial	87Sr/ 86Sr	208Pb/ 204Pb	207Pb/ 204Pb	206Pb/ 204Pb	208Pb/ 207Pb	206Pb/ 207Pb
Karrösten	2	0.71747	38.4	15.64	18.39	2.46	1.18
Telfs	3	0.70916	38.45	15.63	18.72	2.46	1.2
Ambras	19	0.71126	38.83	15.69	18.86	2.48	1.2
Obereching	5	0.70856	38.69	15.69	18.84	2.47	1.2
Obereching	7	0.70873	38.72	15.67	18.89	2.47	1.21
Obereching	8	0.70877	38.73	15.68	18.94	2.47	1.21
Obereching	18	0.70887	38.86	15.69	18.94	2.48	1.21
Obereching	21	0.70918	38.76	15.65	18.89	2.48	1.21
Obereching	27	0.70885	38.77	15.68	18.92	2.47	1.21
Obereching	29	0.70896	38.7	15.65	18.94	2.47	1.21
Obereching	33	0.70889	38.74	15.67	18.92	2.47	1.21
Obereching	42	0.70889	38.73	15.67	18.89	2.47	1.21
Obereching	57	0.70866	38.66	15.65	18.66	2.47	1.19
Obereching	59	0.7086	38.76	15.66	18.87	2.48	1.21
Obereching	70	0.70872	38.43	15.64	18.37	2.46	1.17
Obereching	74	0.70907	38.74	15.67	18.82	2.47	1.2
Obereching	80	0.70854	38.79	15.68	18.91	2.47	1.21
Obereching	129	0.70895	38.76	15.67	18.92	2.47	1.21
Obereching	139	0.7086	38.79	15.68	18.7	2.47	1.19
Obereching	152	0.70881	38.72	15.65	18.83	2.47	1.2
Obereching	157	0.70846	38.72	15.66	18.67	2.47	1.19
Bischofshofen	291	0.71126	38.87	15.69	18.77	2.48	1.2
Bischofshofen	347	0.70891	38.87	15.69	18.84	2.48	1.2
Bischofshofen	452	0.71246	39.15	15.71	19.02	2.49	1.21
Bischofshofen	532	0.70968	38.9	15.67	18.81	2.48	1.2
Saalfelden-Taxau	10	0.71048	38.73	15.69	18.67	2.47	1.19
Saalfelden-Taxau	15	0.71056	38.75	15.68	18.72	2.47	1.19
Saalfelden-Taxau	34	0.71344	38.74	15.68	18.68	2.47	1.19
Brixen	323 ind 1	0.71697	38.65	15.67	18.47	2.47	1.18
Brixen	324 ind 2	0.7182	38.54	15.65	18.43	2.46	1.18
Latsch	396	0.71816	38.54	15.65	18.74	2.46	1.2
Latsch	447	0.72058	38.49	15.65	18.6	2.46	1.19
Latsch	454	0.72167	38.6	15.66	18.75	2.46	1.2
Latsch	537	0.72144	38.51	15.65	18.65	2.46	1.19
Latsch	596	0.72185	38.37	15.63	18.55	2.46	1.19
Latsch	604	0.71925	38.52	15.65	18.86	2.46	1.21
Latsch	642	0.71955	38.47	15.64	18.65	2.46	1.19
Latsch	651	0.72048	38.57	15.65	18.57	2.47	1.19
Latsch	652f	0.72147	38.55	15.65	18.7	2.46	1.19
Latsch	653	0.72141	38.37	15.62	18.47	2.46	1.18
Latsch	655	0.72228	38.5	15.65	18.46	2.46	1.18
Latsch	657	0.71832	38.61	15.67	18.72	2.46	1.19
Latsch	OT	0.71568	38.51	15.63	18.52	2.46	1.18
Latsch	592	0.72178	38.55	15.64	18.74	2.46	1.2
Latsch	652D	0.72148	38.57	15.64	18.73	2.47	1.2
Latsch	652C	0.72266	38.49	15.63	18.67	2.46	1.19
Moritzing	11/T45	0.71199	38.8	15.69	18.56	2.47	1.18
Moritzing	16/T73	0.714	38.86	15.71	18.56	2.47	1.18
Moritzing	17/T79	0.71514	38.81	15.7	18.61	2.47	1.19
Moritzing	US20/T146	0.71519	38.91	15.72	18.84	2.48	1.2
Moritzing	US21/T147	0.71261	38.82	15.67	18.65	2.48	1.19
Moritzing	US23/T149	0.71283	38.84	15.7	18.63	2.47	1.19
Moritzing	US30/T188	0.71105	38.73	15.67	18.52	2.47	1.18
Moritzing	5/T22	0.71307	38.81	15.69	18.59	2.47	1.18
Moritzing	6/T27	0.71562	38.91	15.72	18.56	2.47	1.18
Moritzing	T36/18	0.71277	38.67	15.65	18.52	2.47	1.18
Moritzing	1/T6	0.71335	38.81	15.69	18.65	2.47	1.19
Pfatten	T14	0.7125	38.74	15.67	18.71	2.47	1.19
Pfatten	T3	0.71065	38.88	15.69	18.85	2.48	1.2
Pfatten	T4U	0.71497	38.8	15.67	18.7	2.48	1.19
Pfatten	T4V	0.71276	38.8	15.68	18.77	2.47	1.2